

Belief Disagreement, Instability, and Uncertainty under Stochastic Recall*

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Abstract

Expectations are typically modeled as functions of an agent’s information set. Using pre-registered evidence, we show that beliefs depend instead on the information agents retrieve when forming them. Even after observing identical histories of inflation or stock returns, individuals recall different observations, and their beliefs track what they retrieve. Selective recall therefore generates disagreement, within-person belief instability, and heterogeneity in subjective uncertainty without any new information. We formalize these patterns in a memory-based model in which agents store observed signals, retrieve each only probabilistically, and update optimally using recalled rather than objectively available information. The model yields sharp predictions for how beliefs vary with the composition of stored information, recall probabilities, and contextual cues. Large-scale experiments that hold information fixed and intervene only after exposure confirm these predictions, while survey evidence on macroeconomic and financial expectations supports their external relevance. Disagreement, instability, and uncertainty are thus structural features of belief formation rather than mere measurement error.

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1 Introduction

Two people exposed to the same news headlines about inflation or stock returns can recall different pieces of information and form sharply different beliefs; the same person, asked again the next day under no new information, can give a meaningfully different answer. We document this pattern systematically in four pre-registered two-wave field studies on stock returns, inflation, tariff policy, and an electric-vehicle company. Such disagreement and instability are difficult to reconcile with the canonical Bayesian benchmark, under which agents who share common priors and observe identical signals reach posterior consensus and hold stable beliefs. We argue that they reflect a primitive stage of belief formation that is usually taken for granted: retrieval. Agents store all observed signals but retrieve only some of them, at random, when forming a belief, a friction we call *stochastic recall*. The broader implication is that expectations are functions not of the information an agent has stored but of the information that comes to mind: having information is not, by itself, a stable input to belief formation. Our central result is that this friction alone, with common information, priors, and updating rules, generates belief disagreement, belief instability, and heterogeneity in subjective uncertainty.

Our field studies elicit not only beliefs but also detailed recall of the information presented, holding the information environment fixed across two waves. Faced with the same sequence, individuals recall strikingly different pieces of it, and the same individual recalls different pieces across the two waves. What an individual believes closely tracks what that individual recalls: recalled content predicts beliefs in levels and in revisions. Variation in retrieval thus accounts for a large share of the variation in beliefs under an unchanged information environment: the decision-maker (DM) has already been shown the relevant information, but only some of it comes to mind when a belief is formed.

Motivated by these facts, we develop a memory-based model of belief formation with *stochastic recall*. Agents observe the same signals, store them all in their memory database, share a common prior, and apply Bayes' rule conditional on the signals they recall. The only friction is retrieval: when forming a belief, each stored signal is recalled with some probability. This minimal departure from standard learning isolates imperfect recall from heterogeneous information acquisition, selective attention, and biased updating, and it coincides with the Bayesian benchmark when recall is complete. Holding the prior and information fixed, stochastic recall alone

breaks both posterior consensus and belief stability among ex ante identical agents: different agents retrieve different pieces of the same history, generating cross-sectional disagreement, and the same agent retrieves different pieces at different points in time, generating belief instability with no new information.¹ Because posterior confidence depends on the balance of recalled evidence, recall heterogeneity also produces dispersion in subjective uncertainty.

The model yields three testable predictions (Predictions 1–3). First, disagreement, instability, and subjective uncertainty are greatest when the information history is *ambivalent* (positive and negative evidence are closely balanced); disagreement, instability, and the heterogeneity in subjective uncertainty vanish only under perfectly one-sided evidence or perfect recall. Second, raising the recall probability (through effort, incentives, repetition, or salience) reduces disagreement and stabilizes beliefs in the absence of new information. Third, contextual cues have non-monotonic effects: cues that reinforce dominant signals reduce dispersion and uncertainty, whereas cues that prime minority, that is, non-dominant, signals can first raise them and, once sufficiently strong, reduce them.

We test these predictions in a controlled environment. Because all participants within a group see the same fixed information, and our retrieval-effort and cue manipulations operate only after exposure, our design separates stochastic recall from alternative explanations based on heterogeneous attention or information acquisition. In Experiment 1, about 4,800 participants infer whether a hypothetical company is good or bad from 20 news items, each either positive or negative about the company. Holding the news sequence fixed within each treatment group, we vary three margins: the balance of positive and negative news (informational ambivalence); retrieval effort, through Stop-and-Think prompts and a surprise monetary incentive for accurate recall; and contextual cues that prime recall of majority or minority items.

Experiment 2 replicates the disagreement design in a fresh sample of about 2,000 participants, and Experiment 3 re-elicits their beliefs and recall 24 hours later, so that any belief instability across the two elicitations is driven primarily by retrieval. Experiments 4 and 5 extend these qualitative designs to a dynamic, quantitative environment: about 2,300 participants observe

¹Because beliefs can move without new information or any systematic reduction in uncertainty, stochastic recall provides a mechanism for the excess belief movement documented in [Augenblick and Rabin \(2021\)](#). It similarly generates the individual-level belief overreaction documented in [Bordalo, Gennaioli, Ma and Shleifer \(2020b\)](#), in which forecast revisions negatively predict subsequent forecast errors: the recalled sample is redrawn at each forecast, so the recall shock that moves the current revision enters the subsequent forecast error with the opposite sign, inducing a negative revision–error covariance.

a realization of a persistent stochastic process, report continuous beliefs about its mean, and return one day later for re-elicitation under an unchanged information set.

In the qualitative experiments we elicit binary and probabilistic beliefs alongside recall, allowing us to measure belief levels, cross-sectional dispersion, and subjective uncertainty within the same design. The data confirm these predictions: (1) disagreement increases sharply with informational ambivalence, rising more than thirteen-fold from one-sided to balanced information environments, and falls by nearly 80 percent when retrieval effort is increased; (2) subjective uncertainty is heterogeneous across individuals, highest when recalled evidence is balanced and rising with informational ambivalence; (3) cueing dominant signals lowers both disagreement and subjective uncertainty, while cueing minority signals nearly triples disagreement and raises uncertainty; (4) in the 24-hour re-elicitation, belief revisions occur with no new information and track within-individual changes in recall, with revision rates rising from 7 percent in the most one-sided environment to 30 percent in the most ambivalent and falling to 19 percent with greater retrieval effort; and (5) in the quantitative Experiments 4 and 5, which elicit continuous beliefs, disagreement and instability in mean estimates rise with the volatility and persistence of the underlying process.

The same patterns appear in naturally occurring expectations data. In the Michigan Survey of Consumers, disagreement, instability, and subjective uncertainty in households' stock-return and inflation expectations are all higher after more mixed return and inflation histories, and salient recent returns shift disagreement asymmetrically depending on whether they reinforce or run against the dominant sign of recent history. We read this evidence as external validity for the experimental comparative statics rather than as an independent test of the mechanism.

Taken together, the evidence traces all three patterns to a single friction. In every design, the information environment is fixed and the interventions arrive only after exposure, so which stored signals come to mind is the margin the designs leave free to adjust; measured recall then accounts for the belief variation on each margin we manipulate, and interventions that raise recall effort (Stop-and-Think prompts and incentives for accurate recall) reduce disagreement and instability. The patterns are thus structural features of how beliefs are formed rather than reporting noise or forgetting: reporting noise has only a limited channel through which our post-exposure interventions could move beliefs, and deterministic forgetting is ruled out because signals absent from one elicitation re-emerge at the next.

By measuring and manipulating recall directly, the evidence does more than document that beliefs are stochastic: it locates an important source of the stochasticity at the retrieval stage. An influential alternative locates it instead in noisy processing: the mind encodes information into an imprecise internal impression, and beliefs are an optimal but deterministic function of that impression (Khaw, Li and Woodford, 2021; Woodford, 2020; Azeredo da Silveira, Sung and Woodford, 2024). Under noisy processing, the stored record itself is corrupted and reading it is deterministic; under stochastic recall, the record is intact and reading it is stochastic. Several of our findings favor retrieval: belief revisions move with the measured content of recall; content-free cues shift beliefs by changing which signals are recalled; signals absent from one elicitation re-emerge at the next, while information lost from a corrupted record cannot return; and subjective uncertainty varies with the balance of the evidence an individual happens to recall.

We build on and relate to several strands of the literature. First, the literature on memory and beliefs (Bordalo, Conlon, Gennaioli, Kwon and Shleifer, 2023, 2026; Bordalo, Burro, Coffman, Gennaioli and Shleifer, 2024a; Bordalo, Gennaioli, La Porta and Shleifer, 2024b; Gennaioli, Leva, Schoenle and Shleifer, 2024; Conlon and Kwon, 2025; Wachter and Kahana, 2024; Taubinsky, Butera, Saccarola and Lian, 2026; D’Acunto and Weber, 2022; Augenblick, Jack, Kaur, Masiye and Swanson, 2026) studies how recalled signals are weighted in belief formation, emphasizing similarity-based and cue-driven retrieval. In frameworks that explicitly specify a memory database, the central object is the probability that a stored signal is recalled; applications have focused on the first moment of beliefs, constructing them from many samples of the database, so that realized retrieval variation averages out.² Graeber, Roth and Zimmermann (2024) and Musolff, Roth and Zimmermann (2026) provide experimental evidence that memory shapes how beliefs form and evolve over time. We document and model an idiosyncratic component of retrieval: random variation in which signals come to mind at any moment, alongside the systematic, cue-driven composition that this literature emphasizes. Relatedly, Augenblick, Backus, Little and Moore (2025) show that disagreement and overprecision arise when agents condition on one of several plausible models of fully visible data; our stochastic draw is instead over the recalled sample, under a known model.

²Section 3.1 provides a detailed discussion of the relationship between our approach and recent memory-based models.

Two further strands focus on the information environment itself. The informational-friction literature (Lucas, 1972; Woodford, 2003; Sims, 2003; Mankiw, Reis and Wolfers, 2003; Mackowiak and Wiederholt, 2009; Coibion and Gorodnichenko, 2015) attributes dispersion to heterogeneous acquisition or processing of information and typically implies dynamic consistency under a fixed information set: absent new signals, beliefs do not systematically change. The exposure-based literature (Malmendier and Nagel, 2016; Kuchler and Zafar, 2019; Malmendier and Wachter, 2024; D’Acunto, Malmendier, Ospina and Weber, 2021; D’Acunto, Malmendier and Weber, 2021; D’Acunto, Weber and Yin, 2026) attributes dispersion to differences in exposed information histories. We instead isolate a channel that operates even when information histories coincide and no new signals arrive: the same agent may retrieve different pieces of stored information at different times, so beliefs vary both across agents and within an agent over time.

More broadly, the posterior consensus and belief stability of canonical Bayesian learning (Aumann, 1976; Geanakoplos and Polemarchakis, 1982) rest on an implicit assumption of perfect recall; relaxing that single assumption is enough to overturn both, even under common priors and identical histories.

Section 2 presents the motivating field evidence. Section 3 develops the model and derives its predictions. Section 4 presents the main experiment, Section 5 the belief-instability experiments, and Section 6 the quantitative extension.³ Section 7 combines the experimental results to distinguish stochasticity in retrieval from noise in processing. Section 8 presents external-validity evidence from the Michigan Survey of Consumers. Section 9 concludes.

2 Motivating Evidence from Four Field Studies

Standard macroeconomic models assume perfect recall. To assess this assumption empirically, we conduct four pre-registered online field survey studies spanning financial, macroeconomic, policy, and firm-level settings.

2.1 Design of Field Studies

Overview The four field studies share a common two-wave panel structure. In Part 1 (Day 1), participants are exposed to a fixed sequence of real-world information and then report both

³Appendix Table A.1 gives a complete overview of the paper’s data collections, summarizing the four field studies and five experiments together with their domains, samples, elicitations, incentives, and pre-registration numbers.

their beliefs and recall of the information. In Part 2 (Day 2), conducted approximately 24 hours later, the same participants report their beliefs again and complete an analogous recall task. No new information is provided between the two parts.

Information and Measurement In Part 1 of each study, information is presented one item per screen in a fixed sequence, mirroring the gradual arrival of information in real life. Participants cannot return to previous screens.

Study 1 (Stock Market Returns). Participants view the twelve monthly U.S. stock market returns for calendar year 2025 (January–December), each displayed as a percentage. They are then asked:

“Based on the information you just received, what is your best estimate of the average monthly stock market return in 2025?”

Responses are numerical percentages. The recall task provides a blank entry field for each month, and participants report as many monthly returns as they remember.

The remaining studies follow the same format. In Study 2 (Inflation), participants view the ten annual U.S. inflation rates for 2016–2025 and are asked for their best estimate of the average annual inflation rate over the past 10 years; the recall task mirrors Study 1, with one entry field per year. In Study 3 (Tariff Policy Headlines), participants view 11 opinion-article headlines about recent U.S. tariff policy (five positive, six negative) and are asked to describe the overall sentiment of these articles. In Study 4 (Electric Vehicle Company Headlines), participants view 11 recent news headlines about a major electric vehicle company (six positive, five negative) and are asked to describe the overall sentiment of the news coverage. In Studies 3 and 4, responses are recorded on a four-point scale from *Very negative* to *Very positive*, with the order of options randomly flipped across participants, and the recall task asks for the number of positive and the number of negative headlines observed.

Sample and Incentives Participants are adult U.S. respondents recruited through the online survey platform CloudResearch using its Connect participant pool.⁴ Approximately 2,700 respondents were recruited across the four studies, with roughly 600 valid participants per study

⁴Connect is a proprietary, managed panel of U.S. adults who enroll directly on the CloudResearch platform; the pool is large, diverse, and broadly representative of the U.S. adult population in terms of age, gender, income, and education. We use CloudResearch’s built-in quality assurance system (Sentry) to pre-screen respondents prior to survey entry based on historical response behavior, device-level checks, and automated bot-detection signals.

in Part 1 (conducted on January 29, 2026) and each participant restricted to a single study; 84 percent returned for Part 2 one day later.⁵ Appendix Table A.2 reports demographic characteristics by study. Participants received fixed participation payments of \$2 for Part 1 and \$1 for Part 2. In Studies 1 and 2, belief elicitations in both parts were incentivized for accuracy: the 10 percent of responses closest to the true average of the displayed series received a \$1 bonus. In Studies 3 and 4, bonuses were tied to the accuracy of the recall task (correct counts of positive and negative headlines).⁶

2.2 Empirical Patterns

Immediately after exposure, individuals retrieve different information from the same history. Figure 1 plots the Day 1 and Day 2 recall distributions across the four studies; dispersion is wide and similar on both days. The interquartile range (IQR) of recalled stock returns is about 1.6 percentage points (p.p.) around a true mean of 1.33 percent, and the IQR of recalled inflation is close to 1 p.p. around a true mean of 3.3 percent. In Studies 3 and 4, the IQR of the recalled positive share is 25 and 16 p.p. against true positive shares of 45 and 55 percent, with support spanning nearly the full unit interval.

Beliefs inherit this dispersion (Appendix Figure A.1). Appendix Table A.4 shows that beliefs closely track contemporaneous recall. In Studies 1–2 the elasticity of beliefs with respect to recalled averages is close to unity (about 0.90 on Day 1 and 0.88–0.94 on Day 2); in Studies 3–4, a 10 p.p. increase in the recalled positive share predicts a 0.26–0.42 increase in a binary evaluation index on Day 1 and a 0.19–0.27 increase on Day 2. The estimates are robust to demographic controls.⁷

Recall is not only heterogeneous across individuals but also unstable within them. Figure 2 plots within-individual changes in recalled information between the two elicitations, 24 hours

⁵Pre-specified data-quality screens (an attention check and completion-time thresholds, excluding responses shorter than 2 minutes or longer than 20 minutes, approximately the bottom and top 1 percent of completion times) exclude approximately 8 percent of respondents. We additionally exclude the roughly 2 percent who report taking notes during the information screens. Return to Part 2 is not selective: regressing an indicator for Part-2 participation on demographic characteristics study by study, joint *F*-tests of the demographic coefficients are insignificant in all four studies (Appendix Table A.3).

⁶While these studies also have a correct overall sentiment, belief accuracy is less naturally scored on the four-point evaluation scale than in Studies 1 and 2, where responses can be compared directly to the true numerical average of the displayed series.

⁷Andre, Pizzinelli, Roth and Wohlfart (2022) and Binder, Kuang and Tang (2026) document analogous patterns showing that heterogeneity in recalled mechanisms is a first-order determinant of belief heterogeneity.

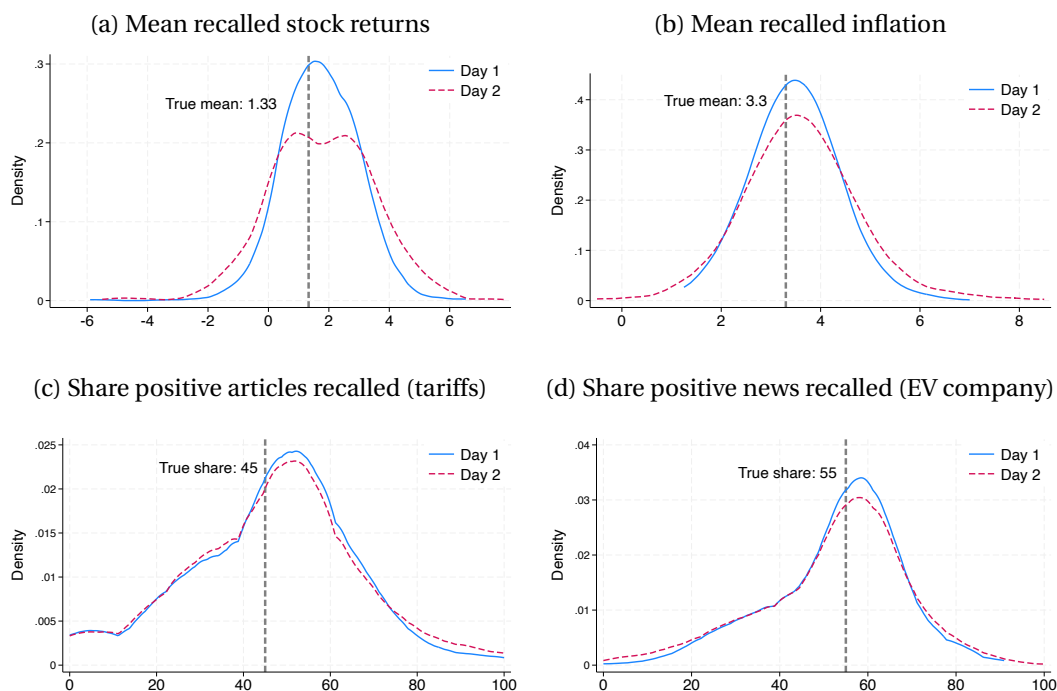


Figure 1: MEMORY RECALL OF THE SAME INFORMATION IS HIGHLY HETEROGENEOUS

Notes: The figure plots the cross-sectional distribution of retrieved information (memory recall) elicited immediately after exposure (Day 1, solid) and one day later (Day 2, dashed), holding the information history fixed. Panels (a) and (b) display the distribution of the mean recalled monthly stock returns over the past 12 months and the mean recalled annual inflation rates over the past 10 years. Panels (c) and (d) display the distribution of the percentage share of positive opinion headlines/news items recalled (out of all recalled items) for tariff policy and for the electric-vehicle company.

apart: many participants recall higher or lower average returns and inflation on Day 2 than on Day 1, and the recalled positive share often shifts by tens of p.p. Beliefs are similarly unstable. Appendix Figure A.2 shows the within-individual belief revisions: the IQR of revisions is 2.3 p.p. for stock returns and 1.6 p.p. for inflation, and 43 percent (41 percent) of respondents reverse the sign of their tariff (EV) evaluation between elicitations.

Crucially, the two instabilities are linked at the individual level. Appendix Table A.5 shows that belief revisions track recall revisions with a slope close to one in Studies 1–2 (0.93 for stock returns, 0.89 for inflation); in Studies 3–4, a 10 p.p. change in the recalled positive share predicts a 0.21–0.27 change in the evaluation index. The estimates are robust to demographic

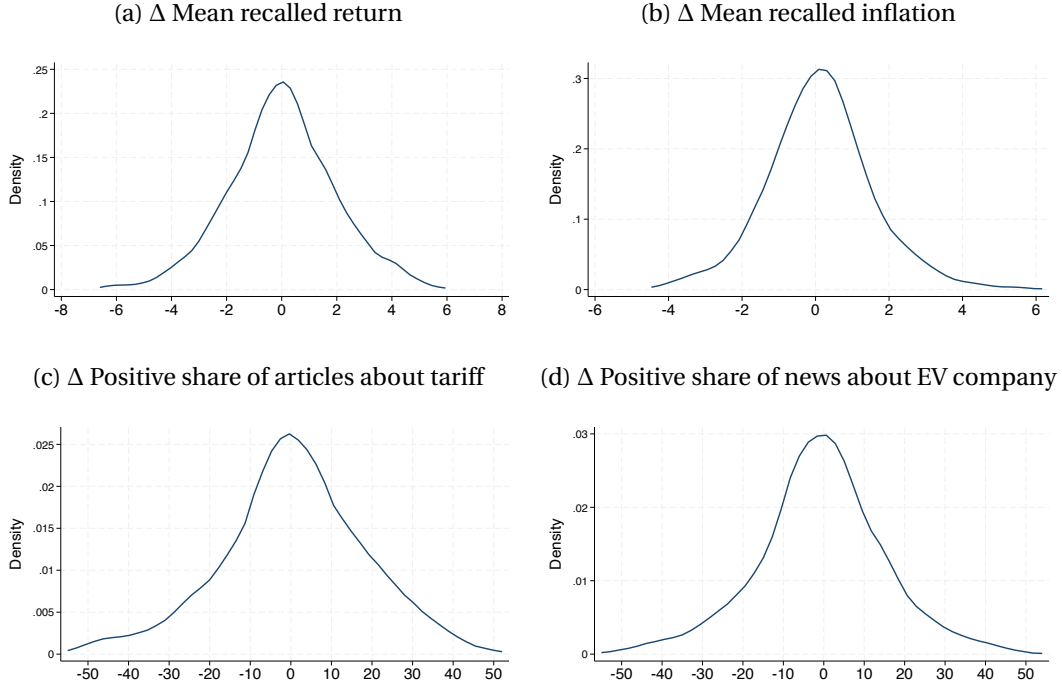


Figure 2: WITHIN INDIVIDUALS, MEMORY RETRIEVAL OF THE SAME INFORMATION VARIES SUBSTANTIALLY OVER TIME

Notes: The figure plots the cross-sectional distribution of within-individual changes in memory retrieval between Day 1 and Day 2 across the four field studies. Panels (a) and (b) display the distribution of changes in mean recalled stock returns and mean recalled inflation rates between Day 2 and Day 1, respectively. Panels (c) and (d) summarize the distribution of changes in the share of positive recalled headlines between the two elicitations. No new information is provided between the two waves.

controls.⁸ Cleaner evidence that stochastic recall drives disagreement and instability comes from the controlled experiments presented in Sections 4–6, in which we directly manipulate the information environment, recall effort, and contextual cues.

Result 1 (Recall and beliefs in the field). *Under identical information histories and no arrival of new information: (i) across individuals, recalled information is highly heterogeneous, and beliefs track recalled information approximately one for one; (ii) within individuals, recall varies substantially across elicitations 24 hours apart, and belief revisions closely track recall revisions.*

⁸We cannot fully exclude the possibility that some participants draw on prior experience or preexisting beliefs, or seek outside information between elicitations, particularly in Studies 1 and 2; heterogeneity in numerical skills may also contribute to dispersion in quantitative beliefs in those studies. However, the substantial heterogeneity observed in *recalled* information itself emerges immediately after exposure and does not require advanced numerical ability, and in Studies 3 and 4 participants evaluate *qualitative* headlines constructed for the study and base their responses on the presented content. Importantly, our controlled experiments are designed to shut down these channels by construction: the qualitative experiments concern a hypothetical company and the quantitative experiment an abstract, synthetically generated data series, so participants have no prior experience or preexisting beliefs to draw on and no informative outside source to consult; we nonetheless obtain the same empirical patterns as Result 1.

Existing belief formation models Canonical information-friction models (Sims, 2003; Mackowiak and Wiederholt, 2009; Mankiw, Reis and Wolfers, 2003; Lucas, 1972; Woodford, 2003) generate disagreement through heterogeneous information acquisition or processing, and typically imply dynamic consistency of beliefs under a fixed information set. Models of noisy cognitive processing (Khaw, Li and Woodford, 2021; Woodford, 2020; Azeredo da Silveira, Sung and Woodford, 2024) generate stochastic beliefs through noise realized at the encoding stage, as an imprecise internal impression of the information: beliefs are deterministic given the impression, no fresh stochasticity arises at the moment a belief is formed, and no prediction ties belief movements to the content of measured recall. Result 1 documents exactly that: recall is heterogeneous across individuals and unstable within them under a shared, fixed information history, and beliefs move with it. Memory-based models (Gennaioli, Leva, Schoenle and Shleifer, 2024; Conlon and Kwon, 2025; Taubinsky, Butera, Saccarola and Lian, 2026) supply the natural primitive, the probability that a stored signal is recalled; existing work has focused on its first-moment implications, with cues and similarity shifting the level of recall and beliefs. Our focus is the second moments: a memory-based model with stochastic recall generates retrieval heterogeneity under shared information and joint instability of recall and beliefs without new information as direct implications of a single friction, locating the stochasticity in beliefs at the retrieval stage rather than at encoding. The Supplementary Appendix (Kuang, Tang and Weber, 2026) provides a systematic comparison. Motivated by these patterns, the next section develops such a model, in which beliefs depend on the stored information retrieved at the moment of belief formation: stochastic recall.

3 Stochastic Recall and Belief Formation

We preserve Bayesian updating conditional on recalled information, but relax the benchmark assumption that stored information is perfectly retrieved when beliefs are formed. This single friction generates belief disagreement, within-individual instability, and heterogeneous subjective uncertainty among ex ante identical DMs, with sharp predictions across informational ambivalence, recall probability, and contextual cues.

3.1 Model Setup

Decision problem We study settings in which a DM forms beliefs about an economic variable (e.g., inflation, house prices, or asset returns) on the basis of previously observed information. To isolate the core mechanism transparently, we first study a binary-state environment in which the DM forms a qualitative belief about whether an asset is in a good or bad state, capturing the logic of the qualitative experiments and providing a reduced-form representation of many directional-forecasting settings. Section 6 extends the model to quantitative expectation formation.

Baseline environment The model environment builds on and is closely related to Enke, Schwerter and Zimmermann (2024). Consider a continuum of DMs, indexed by i , who form beliefs about the value of an asset Y . The asset can be in one of two states, G (good) or B (bad), which are not directly observable. All DMs share a common prior $P(G) = P(B) = 0.5$.⁹

Each DM observes the same realized history of independently and identically distributed binary signals $S = \{s_1, s_2, \dots, s_N\}$, where each signal is either positive (*pos*) or negative (*neg*). We code signals as $s_k = 1$ for positive and $s_k = -1$ for negative signals. Let N^+ and N^- denote the numbers of positive and negative signals, respectively, with total signal count $N = N^+ + N^-$. The full collection of observed signals is stored in the DM’s memory database; the friction studied in this paper operates only at the retrieval stage when beliefs are formed. Full storage is a modeling benchmark: empirically, the object our designs identify is an effective accessibility probability of previously presented signals, and the evidence favors a retrieval interpretation of it (see Section 4.3).

Signals are informative about the true state. Let $b \in (0.5, 1)$ denote signal diagnosticity, defined by $P(\text{pos} \mid G) = P(\text{neg} \mid B) = b$, so that a signal is more likely to match the true state than to contradict it. Each DM i forms a qualitative belief about the asset value, denoted by $\hat{Y}_i \in \{1, -1\}$, where 1 corresponds to state G and -1 to state B .

Recall stage Motivated by Result 1, we separate Bayesian learning into two components, perfect retrieval of stored information and Bayesian updating conditional on the information that comes to mind, and relax the first while preserving the second. DM i recalls a random subset of the

⁹To isolate the role of imperfect memory in belief formation, the prior is deliberately assumed to be uninformative as in Enke, Schwerter and Zimmermann (2024); we also induce this prior in the main experiments presented later. The Supplementary Appendix examines an extension of the model with a biased prior.

stored signals through a probabilistic sampling process: let $\tilde{S}_i$ denote the *recalled information set*, the signals that come to mind when DM i forms a belief. Beliefs are based on $\tilde{S}_i$ rather than on S : the recalled information set need not coincide with the objective one, and it varies across DMs and, for the same DM, across elicitations. As a tractable baseline, we assume that each previously observed signal is recalled independently with probability $p \in (0, 1)$.¹⁰ Let X_i^+ and X_i^- denote the numbers of positive and negative signals in $\tilde{S}_i$.¹¹

Information-processing stage The DM applies Bayes' rule to the recalled signals. Under the symmetric prior and signal structure described above, posterior log odds are proportional to the excess of recalled positive over recalled negative signals. A derivation in the Supplementary Appendix shows that optimal updating therefore implies a majority rule: if the DM recalls at least as many positive as negative signals, the DM believes the asset is good; otherwise, the DM believes the asset is bad.¹² Formally,

$$\hat{Y}_i = \begin{cases} 1 & \text{if } X_i^+ - X_i^- \geq 0, \\ -1 & \text{otherwise.} \end{cases} \quad (3.1)$$

This retrieval-stage friction is orthogonal to, and can coexist with, mechanisms that distort signal acquisition or signal weighting, such as inattention or diagnostic expectations.

The distribution of beliefs Due to probabilistic recall, DMs retrieve different subsets of the same signal history: $X_i^+ \sim \text{Binomial}(N^+, p)$ and $X_i^- \sim \text{Binomial}(N^-, p)$, which are approximately normal when N^+ and N^- are large.¹⁴ Let $q = P(\hat{Y}_i = 1) = P(X_i^+ \geq X_i^-)$ denote the probability that

¹⁰Memory can fail at three stages: when information is first acquired (encoding, with probability e), when information is stored over time (retention, r), and when it is recalled (retrieval, p). What a DM recalls at any single moment depends only on the product erp , so all cross-sectional predictions and the estimation in Section 4.4 apply with p read as this combined probability. Only comparisons of the same DM's beliefs at two points in time require the stochasticity at the retrieval stage, because what was encoded and retained does not change between elicitations (see supporting evidence in Section 4.3).

¹¹The independence assumption is merely a benchmark that allows us to cleanly isolate retrieval stochasticity rather than a literal interpretation about the psychological processes governing memory. In Section 3.2, we allow the recall probability to depend on contextual cues, and the Supplementary Appendix relaxes the baseline further, including biased priors, recency effects, repeated sampling from memory, and correlated recall across elicitations.

¹²When $X_i^+ = X_i^-$, the DM is exactly indifferent between the two states; for concreteness, equation (3.1) classifies the asset as good in this case. When N is large enough, ties are negligible.

¹³When no signal is recalled ($X_i^+ = X_i^- = 0$), the DM forms beliefs from the prior alone; with N stored signals each recalled independently with probability p , this occurs with probability $(1 - p)^N$, which is negligible in our experiments (for example, $0.5^{20} \approx 10^{-6}$ when $N = 20$ and $p = 0.5$). We therefore ignore this case in the theoretical analysis.

¹⁴Bordalo, Conlon, Gennaioli, Kwon and Shleifer (2023) employs the same normal approximation to binomial recall in studying probability judgments; Lemma 1 applies it to the cross-sectional distribution of beliefs.

a DM recalls at least as many positive as negative signals.

Lemma 1. 1. *Approximate distribution of beliefs.*

$$\hat{Y}_i = \begin{cases} 1 & \text{with probability } q, \\ -1 & \text{with probability } 1 - q, \end{cases}$$

where

$$q = \Phi\left(\frac{(N^+ - N^-)p}{\sqrt{p(1-p)N}}\right).$$

2. *Cross-sectional moments.*

$$E(\hat{Y}_i) = 2q - 1, \quad \text{Var}(\hat{Y}_i) = 4q(1 - q).$$

The proof is provided in [Appendix C](#). When DMs form binary beliefs and the population is large, a fraction q believes the asset is good, while a fraction $1 - q$ believes it is bad. The numerator inside $\Phi(\cdot)$ captures the imbalance in stored positive versus negative signals; the denominator captures recall-induced sampling noise. Disagreement, $4q(1 - q)$, is maximal when $q = \frac{1}{2}$ and declines as q moves away from $\frac{1}{2}$.¹⁵

Relationship with existing memory models In the memory-based models of [Gennaioli, Leva, Schoenle and Shleifer \(2024\)](#), [Conlon and Kwon \(2025\)](#), [Conlon \(2025\)](#), and [Taubinsky, Butera, Saccarola and Lian \(2026\)](#), DMs are typically assumed to sample from memory sufficiently often to recover the cued distribution, so realized sampling variation averages out: cues alter the relative weights placed on stored elements, and beliefs are stable conditional on the cues. Our DMs instead sample memory a limited number of times (once, in the main text), so the recalled subset itself is random; in the limit of many memory draws, beliefs approach the perfect-recall Bayesian benchmark. The systematic, cue-driven recall these models emphasize is complementary to the idiosyncratic stochasticity we isolate: cues shape the probabilities with which stored signals come to mind, and our analysis concerns the realized draws given those probabilities. In this sense, stochastic recall requires no different model of memory: the primitive is the same recall probability that defines this family of models, and we study the distributional implications of basing beliefs on a limited number of realized draws from memory rather than

¹⁵Holding p and signal proportions fixed, disagreement converges to zero as $N \rightarrow \infty$: as the memory database grows large, recalled samples almost surely reflect the underlying signal imbalance (this follows directly from Lemma 1). In practice, disagreement may be more persistent once additional memory mechanisms (e.g., recency effects and heterogeneous rehearsal) are taken into account.

on the average over many draws.¹⁶

Expectation and opinion formation The framework applies broadly to expectation, opinion, and judgment formation. An individual forecasting stock returns may retrieve realized returns, earnings news, or broader economic narratives. The model thus formalizes the empirical framework in Andre, Pizzinelli, Roth and Wohlfart (2022), in which heterogeneity in beliefs arises because different mechanisms or narratives are retrieved from memory. The same logic applies to political and policy judgments. A voter evaluating a policy, candidate, or salient issue such as tariffs may have stored many relevant considerations (opinion pieces, statistics, personal experiences, and partisan arguments) but retrieves only a subset when responding. In the language of Zaller and Feldman (1992), stochastic recall microfoundations variation in which stored considerations come to mind.

3.2 Predictions: Disagreement, Instability, and Uncertainty

We measure belief disagreement as the cross-sectional variance of realized binary beliefs, $\text{Var}(\hat{Y}_i)$, and summarize signal composition using the degree of informational ambivalence,

$$A = 1 - \left| \frac{N^+ - N^-}{N} \right|, \quad (3.2)$$

which is maximal when signals are evenly split and minimal when all signals point in the same direction.

For belief instability, a single DM's belief is elicited at two dates, t and $t + 1$, with no new information arriving in between; each elicitation draws a fresh, independent recall realization. The probability that beliefs differ across the two dates is $\Pi(N^+, N^-, p) = 2q(1 - q)$, which in a large population also equals the proportion of DMs who revise their beliefs between elicitations.¹⁷

For subjective uncertainty, Bayes' rule implies that the posterior probability that the asset is good is $\mu_i = [1 + ((1 - b)/b)^{D_i}]^{-1}$, where $D_i \equiv X_i^+ - X_i^-$ denotes recalled net evidence. We measure subjective uncertainty as the posterior variance of the state indicator, $U_i = \mu_i(1 - \mu_i)$, which is

¹⁶Within this family, Bordalo, Conlon, Gennaioli, Kwon and Shleifer (2026) also study belief disagreement, with a different microfoundation.

¹⁷Independence across elicitations is a benchmark rather than a necessary ingredient. The Supplementary Appendix allows correlated recall, with each signal's second retrieval repeating the first with probability κ and drawn fresh otherwise: every cross-sectional prediction is unchanged for any κ , because each elicitation's recalled evidence keeps the same distribution, and the switching probability becomes $2q(1 - q)(1 - \kappa_B)$, where κ_B is the belief correlation induced by κ . Instability therefore remains positive with the same comparative statics for all $\kappa < 1$, and $2q(1 - q)$ is the $\kappa = 0$ benchmark, so the close match between predicted and observed switching in Section 5 indicates that retrieval is approximately fresh at the belief margin.

highest when the posterior is closest to 1/2. [Augenblick and Rabin \(2021\)](#) use this measure of uncertainty; it captures uncertainty about the state rather than confidence in the accuracy of a stated belief.

Lemma 2. *At the individual level, subjective uncertainty U_i is a strictly decreasing function of the absolute recalled net evidence $|D_i| = |X_i^+ - X_i^-|$.*

The proof is provided in [Appendix D](#): beliefs formed from balanced recalled evidence are held with low uncertainty, while one-sided recalled evidence produces certain beliefs.

Prediction 1 (Informational ambivalence). *Holding N and p fixed, an increase in informational ambivalence A raises (i) belief disagreement $\text{Var}(\hat{Y}_i)$, (ii) within-individual belief instability Π , and (iii) expected subjective uncertainty $E[U_i]$.*

Prediction 2 (Recall probability). *Holding (N^+, N^-) fixed, an increase in the recall probability p lowers both belief disagreement and within-individual belief instability.*

Proofs are provided in [Appendix E](#), [Appendix F](#), and [Appendix G](#). When the stored history is ambivalent, recalled evidence is frequently close to balanced, so small recall fluctuations flip the recalled majority, raising disagreement and instability, and posteriors concentrate near 1/2, raising uncertainty. A higher recall probability makes recalled subsets larger and more representative of the same underlying history, so beliefs increasingly coincide across DMs and across dates. Interventions that raise recall through effort, incentives, repetition, or salience therefore compress disagreement and instability.¹⁸

Contextual cues We extend the baseline setting to incorporate associative cue-driven recall, following [Bordalo, Conlon, Gennaioli, Kwon and Shleifer \(2023\)](#); [Conlon and Kwon \(2025\)](#); [Enke, Schwerter and Zimmermann \(2024\)](#). Each signal s_k is associated with a context C_k , such as a story or an image. A contextual cue does not itself constitute a signal and does not provide new information, but matches the context associated with some stored signals. The cue increases the probability that matched signals are recalled.

We assume that all positive signals share one context and all negative signals share another, distinct context.¹⁹ A contextual cue matches the context associated with one signal type (either

¹⁸Expected subjective uncertainty is not globally monotone in p : when the stored history has a clear majority, more complete recall makes recalled evidence more one-sided and lowers uncertainty, but when the history is exactly balanced, more complete recall can raise uncertainty by making recalled evidence more balanced.

¹⁹The analysis extends to settings in which signals of the same sign are associated with multiple contexts, so that a cue matches only a subset of positive or negative signals.

positive or negative) and increases the probability of recalling each matched signal by $a > 0$. As a result, the recall probability for the cued signal type rises from p to $\tilde{p} = p + a$, while the recall probability for the non-cued type remains equal to p . Feasibility requires

$$a \in (0, 1 - p].$$

Let $\lambda \in (0, 1)$ denote the proportion of the cued signal type in the memory database (i.e., $\lambda = N^+ / N$ if positive signals are cued and $\lambda = N^- / N$ if negative signals are cued).

Two model-implied thresholds summarize the comparative statics under minority cueing. The first, Δ , is the cue strength at which cued disagreement returns to its no-cue level.²⁰ The second,

$$\Theta = \frac{|N^+ - N^-|p}{N^m}, \quad N^m = \min\{N^+, N^-\},$$

is the cue strength at which the cued (minority) type is recalled in the same expected number as the dominant type, so that expected recalled net evidence is zero.²¹

Prediction 3 (Contextual cues). *Let $a \in (0, 1 - p]$ denote the strength of a contextual cue that increases the recall probability of one signal type from p to $p + a$.*

1. *If the cue enhances recall of the signal type that constitutes the majority in the memory database, both belief disagreement and expected subjective uncertainty fall below their no-cue levels and are strictly decreasing in the cue strength a .*
2. *If the cue enhances recall of the minority signal type, disagreement exceeds its no-cue level when $a < \Delta$, equals it at $a = \Delta$, and falls below it when $a > \Delta$. Whenever $\Theta \leq 1 - p$, both disagreement and expected subjective uncertainty are single-peaked in cue strength: strictly increasing for $a < \Theta$, maximized at $a = \Theta$, and strictly decreasing thereafter; if $\Theta > 1 - p$, both are increasing in a throughout the feasible range.*

The proof is provided in [Appendix H](#). Cueing the dominant signal type reinforces the evidence that most DMs already recall, increasing overlap in recalled information and making recalled evidence more one-sided. As a result, both disagreement and uncertainty decline monotonically. By contrast, cueing the minority signal type initially makes recalled evidence more balanced on

²⁰In closed form, $\Delta = \frac{(2\lambda-1)^2 p(1-2p) - 2(2\lambda-1)p(1-p)}{\lambda(1-p) + (2\lambda-1)^2 p}$; see [Appendix H](#) for the derivation.

²¹The statements for disagreement are exact under the normal approximation of Lemma 1; those for expected subjective uncertainty rely on an additional mean-based approximation, discussed and verified numerically in the Supplementary Appendix.

average, thereby increasing both disagreement and uncertainty. Only when the cue is sufficiently strong to overcome the underlying signal imbalance, and such a cue strength is feasible, do the outcomes begin to decline.²²

Appendix I collects the underlying result in one formal statement: whenever the stored history contains both signal types and recall is imperfect, posterior consensus, common subjective uncertainty, and belief stability all fail with strictly positive probability, and all three are restored as $p \rightarrow 1$.

Model extension The Supplementary Appendix extends the framework along several dimensions, including: (i) continuous-signal and recency-weighted environments, (ii) probabilistic beliefs, (iii) repeated memory sampling, (iv) correlated recall across elicitations, and (v) biased priors. Across these cases, informational ambivalence interacting with imperfect recall continue to govern the key comparative statics. Sections 4 and 5 test Predictions 1–3 experimentally.

4 The Belief Disagreement Experiment

We test the mechanism in a series of controlled experiments. Our experiments hold the information environment fixed across participants and vary retrieval conditions only after exposure, which allows us to test the model’s comparative statics for cross-sectional disagreement, subjective uncertainty, and within-individual belief instability (Predictions 1–3). We run three pre-registered experiments, summarized in Table 1. All three build on the experimental design of [Enke, Schwerter and Zimmermann \(2024\)](#), in which participants infer the type of a hypothetical company from a sequence of positive and negative news items. Experiment 1 (the Belief Disagreement Experiment) varies informational ambivalence, retrieval effort, and contextual cues; Experiment 2 replicates the ambivalence and retrieval-effort design in a fresh sample; and Experiment 3 (the Belief Instability Experiment) re-elicits the Experiment 2 participants approximately 24 hours later under unchanged information, isolating belief revisions that arise primarily from stochastic recall. This section presents Experiment 1; Section 5 presents Experiments 2 and 3.

²²These reversals need not be feasible for all parameter values. Since the cued recall probability cannot exceed one, cue strength is bounded by $a \leq 1 - p$. If baseline recall p is sufficiently high, the relevant threshold can lie above this bound. In that case, minority cueing may remain in the region in which it raises disagreement and/or subjective uncertainty throughout the feasible range.

4.1 Design of Experiment 1

Experiment 1: Belief Disagreement Experiment Participants form beliefs about the quality of a hypothetical company in a stylized asset-evaluation task. The company is described as objectively either “good” or “bad,” with equal ex-ante probability. Participants are informed of a common signal-generating process: for a good company, positive signals occur with probability 0.65 and negative signals with probability 0.35, while for a bad company the reverse holds. Information is conveyed through a sequence of binary signals (“news”) embedded in short contextual narratives consisting of brief stories and accompanying images.²³

For identification, however, the realized signal history we show to participants is fixed by design within each treatment group. Thus, all participants within a group observe the same realized sequence of signals, even though the instructions describe the general signal-generating environment.²⁴

The experiment uses a between-subject design with nine treatment groups. The groups vary along three dimensions: informational ambivalence, retrieval effort, and contextual cues. Across all groups, participants observe 20 signals. The complete signal sequences are reported in Section 3 of the experimental instructions in the Appendix.

Treatment groups The nine groups of Experiment 1 are organized into three sets summarized in Panel A of Table 1. The first set (Groups 1–5) varies informational ambivalence by varying the signal composition (N^+ , N^-) while holding the total number of signals fixed at $N = 20$. Groups 1 and 5 share the same low ambivalence with opposite majorities; Groups 2 and 4 do the same at intermediate levels of ambivalence; Group 3 is most balanced. This symmetric design helps isolate ambivalence from the direction of the signal stream.

The second set (Groups 6–7) holds the signal composition fixed at the composition of Group 3 (the most ambivalent environment) and manipulates retrieval after exposure. Group 6 implements a *Stop-and-Think* procedure in which participants complete a structured recall module *before* reporting beliefs. Group 7 layers *Incentivized Recall* onto Stop-and-Think by adding a

²³The hypothetical company, binary states and signals, and purely qualitative information minimize the influence of prior experience, domain knowledge, and numerical ability on the results.

²⁴The two are consistent: the instructions describe the signal-generating process without stating that a fresh sequence is drawn for each participant, and each fixed history could have been produced by that process, since any mix of positive and negative news can occur under either company type.

surprise monetary reward for accurate recall.

The third set (Groups 8–9) holds the signal composition fixed at the Group 2 composition (intermediate ambivalence) and introduces contextual cues at the belief-elicitation stage. In Group 8, the image associated with the dominant (positive) signal type is used as a cue; in Group 9, the image associated with the minority (negative) signal type is used. Participants are explicitly told that the images are illustrative and contain no information about the company’s type; this information is reinforced through a comprehension check.

Table 1: TREATMENT GROUPS ACROSS THE THREE QUALITATIVE EXPERIMENTS

Group	(N^+, N^-)	A	Manipulation	Prediction tested
Panel A. Belief Disagreement Experiment (Experiment 1; sample size = 4,758)				
<i>A.1. Informational ambivalence</i>				
1	(18, 2)	0.20	—	Pred. 1
2	(14, 6)	0.60	—	Pred. 1
3	(11, 9)	0.90	—	Pred. 1
4	(6, 14)	0.60	—	Pred. 1
5	(2, 18)	0.20	—	Pred. 1
<i>A.2. Recall effort (Group 3 composition)</i>				
6	(11, 9)	0.90	Stop-and-Think	Pred. 2
7	(11, 9)	0.90	Stop-and-Think + incentivized recall	Pred. 2
<i>A.3. Contextual cues (Group 2 composition)</i>				
8	(14, 6)	0.60	Cue: dominant (positive) signal type	Pred. 3
9	(14, 6)	0.60	Cue: minority (negative) signal type	Pred. 3
Panel B. Belief Disagreement Robustness Experiment (Experiment 2; sample size = 2,039)				
1	(18, 2)	0.20	—	Pred. 1
2	(14, 6)	0.60	—	Pred. 1
3	(11, 9)	0.90	—	Pred. 1
4	(11, 9)	0.90	Stop-and-Think	Pred. 2
Panel C. Belief Instability Experiment (Experiment 3; sample size = 1,744)				
1	(18, 2)	0.20	24-hour re-elicitation	Pred. 1
2	(14, 6)	0.60	24-hour re-elicitation	Pred. 1
3	(11, 9)	0.90	24-hour re-elicitation	Pred. 1
4	(11, 9)	0.90	24-hour re-elicitation (Stop-and-Think at Day 1)	Pred. 2

Notes: All groups receive $N = 20$ signals about a hypothetical company whose type (good or bad) is the object of inference. $A = 1 - |N^+ - N^-|/N$ is the informational ambivalence of the signal history. The manipulations are described in the text; all are introduced after the full signal sequence has been shown. Panel C re-elicits the same participants as Panel B approximately 24 hours later with no new information. All three experiments were conducted on Prolific and pre-registered at the AEA RCT Registry (Experiment 1: #17150; Experiments 2 and 3: #17671).

Procedures Participants first read detailed instructions describing the prior, the signal structure, the independence of draws (sampling with replacement), and the incentive scheme. The instructions impose a common prior, $P(G) = 0.5$, and a common signal structure with diagnosticity $b = 0.65$.

Participants then complete seven multiple-choice comprehension checks, identical across all nine treatment groups and with answer-order randomized across respondents; only respondents who pass all comprehension checks are allowed to proceed to the information-exposure stage.²⁵ Participants then observe the sequence of 20 news items, presented sequentially as short contextual events with illustrative images.

After exposure, participants report both a binary belief and a probabilistic belief about the company's type.²⁶ Participants then complete recall tasks eliciting the number of positive and negative signals remembered as well as the valence of the 1st, 5th, 15th, and last signals. The recalled counts are the aggregate recall measure used throughout the analysis; the probed items provide a separate, item-level measure, which we use to assess whether imperfect recall reflects encoding or retrieval (Section 4.3) and to test for motivated recall and for systematic differences in recall accuracy across signal positions and signs (Section 5.3). We provide full procedural details in the Online Instruction.

Why the design identifies retrieval The identifying logic is timing. All participants within a treatment group observe the same realized signal history under the same presentation conditions, and the retrieval-effort interventions and contextual cues arrive only after the signal sequence has been seen. By construction, the treatments therefore cannot affect which signals participants observe or attend to at exposure; they can only affect which of the previously presented signals are retrieved when beliefs are formed. Belief elicitation is accuracy-incentivized in all groups, and a comprehension check before exposure ensures participants understand the task and signal structure. The evidence thus rests on three sources of identification: post-exposure retrieval interventions, contextual cues at the belief stage, and repeated belief elicitations under a fixed information set (Experiment 3).

²⁵The comprehension checks verify: (i) that more positive (negative) news makes the company more likely to be good (bad); (ii) that news draws are independently and with replacement; (iii) that the same event can occur multiple times and should be counted each time; and (iv) that illustrative images contain no information beyond the text.

²⁶Probabilistic beliefs are elicited as percentage probabilities. Respondents are randomly asked to report either $P(\text{good})$ or $P(\text{bad})$. We place all responses on a common scale by defining $p_i = P(\text{good})$ when $P(\text{good})$ is elicited directly, and $p_i = 1 - P(\text{bad})$ otherwise.

Sample The Belief Disagreement Experiment was conducted online via Prolific.²⁷ Data collection began on 18 November 2025 and lasted eight days. Approximately 6,900 respondents were initially recruited from a broad U.S. general-population sample. About 75 percent passed all comprehension checks, yielding 4,758 valid completed responses.²⁸ The median completion time was approximately 15 minutes. Appendix Table A.6 shows that the treatment groups are balanced on demographics.

Incentives Participants received a fixed participation payment of \$2. In addition, they earned a \$0.50 bonus if their belief about the company’s type was correct. Participants in Group 7 earned an additional \$0.50 bonus for accurate recall.

4.2 Findings: Disagreement

Let D_g denote disagreement in group g , measured as the cross-sectional variance of participants’ reported binary beliefs about the company’s type.

Disagreement increases with informational ambivalence Prediction 1 predicts

$$D_3 > \{D_2, D_4\} > \{D_1, D_5\}.$$

Disagreement in the most ambivalent environment (Group 3; variance 0.640) is more than thirteen times larger than in the least ambivalent environments (Groups 1 and 5; variances 0.039 and 0.047).²⁹ The top panel of Figure 3 shows the complete pattern: disagreement peaks in Group 3, falls sharply in Groups 1 and 5, and takes intermediate values in Groups 2 and 4 (0.199 and 0.187).³⁰

²⁷Prolific maintains a large, diverse participant pool that is broadly representative of the U.S. adult population in terms of age, gender, income, and education. Within each platform, participants in each of our collections are excluded from all others at recruitment. The field studies (Section 2) and the quantitative experiments (Section 6) were run on CloudResearch’s Connect pool instead: after the qualitative experiments had recruited nearly 10,000 respondents on Prolific, recruitment under these exclusions slowed, and we moved the remaining collections to a fresh platform.

²⁸We screen out likely bots flagged by the Qualtrics and Prolific quality checks, respondents who fail an attention check, and responses with completion times shorter than 5 minutes or longer than 30 minutes, thresholds that correspond approximately to the bottom and top 1 percent of completion times; together these screens remove about 5 percent of responses. We also exclude respondents who report taking notes during the information screens, around 2.5 percent of the sample.

²⁹The patterns do not depend on using the binary belief. Classifying beliefs by the direction of the reported probability (good if at least 50 percent) reproduces all three comparative statics (p values < 0.01). In terms of the level of the reported probabilistic beliefs, their cross-sectional variance also rises with ambivalence and falls with retrieval effort (p values < 0.05). See Section S.7 of the Supplementary Appendix.

³⁰We discuss the bottom panel (using recall-generated beliefs in place of reported beliefs) in Section 4.3.

Increased retrieval effort reduces disagreement Prediction 2 predicts

$$D_3 > \{D_6, D_7\}.$$

Stop-and-Think alone (Group 6) reduces the variance of beliefs from 0.640 in Group 3 to 0.370; adding a surprise accuracy incentive on top of Stop-and-Think (Group 7) lowers it further to 0.135. Both reductions are statistically significant (p value < 0.01), and the combined retrieval-effort intervention reduces disagreement by nearly 80 percent relative to the high-ambivalence baseline, consistent with evidence that prompting individuals to retrieve stored information shifts their beliefs (Augenblick, Jack, Kaur, Masiye and Swanson, 2026).

Contextual cues have asymmetric effects on disagreement Prediction 3 predicts that cueing dominant signals reduces disagreement,

$$D_2 > D_8,$$

whereas cueing minority signals can increase disagreement when the cue is weak,

$$D_2 < D_9 \quad \text{if } a < \Delta.$$

Cueing the dominant positive signal type in Group 8 slightly reduces disagreement relative to the no-cue benchmark in Group 2. By contrast, cueing minority negative signals in Group 9 substantially increases disagreement: belief variance increases to 0.565, almost tripling relative to Group 2. The estimation in Section 4.4 puts the cue strength in Group 9 at $a^- = 0.475$, while the corresponding threshold implied by the analytical expressions underlying Prediction 3 is $\Delta = 0.87$. Because $a^- < \Delta$, the cue is weak in the model's sense, and the observed increase in disagreement is consistent with the theory.

4.3 Mechanism: Recall-Generated Beliefs and Retrieval Shifts

The model suggests that agents form beliefs by comparing recalled positive and negative signals; because we observe both stated beliefs and recalled signal counts, we can test this mechanism directly. We establish the recall-belief link with three pieces of evidence: cross-sectional alignment between recall-generated and reported beliefs (this subsection), causal shifts in beliefs from retrieval-effort and contextual-cue interventions (this subsection, Table 2), and within-individual co-movement of recall and belief revisions in Experiment 3 (Section 5).

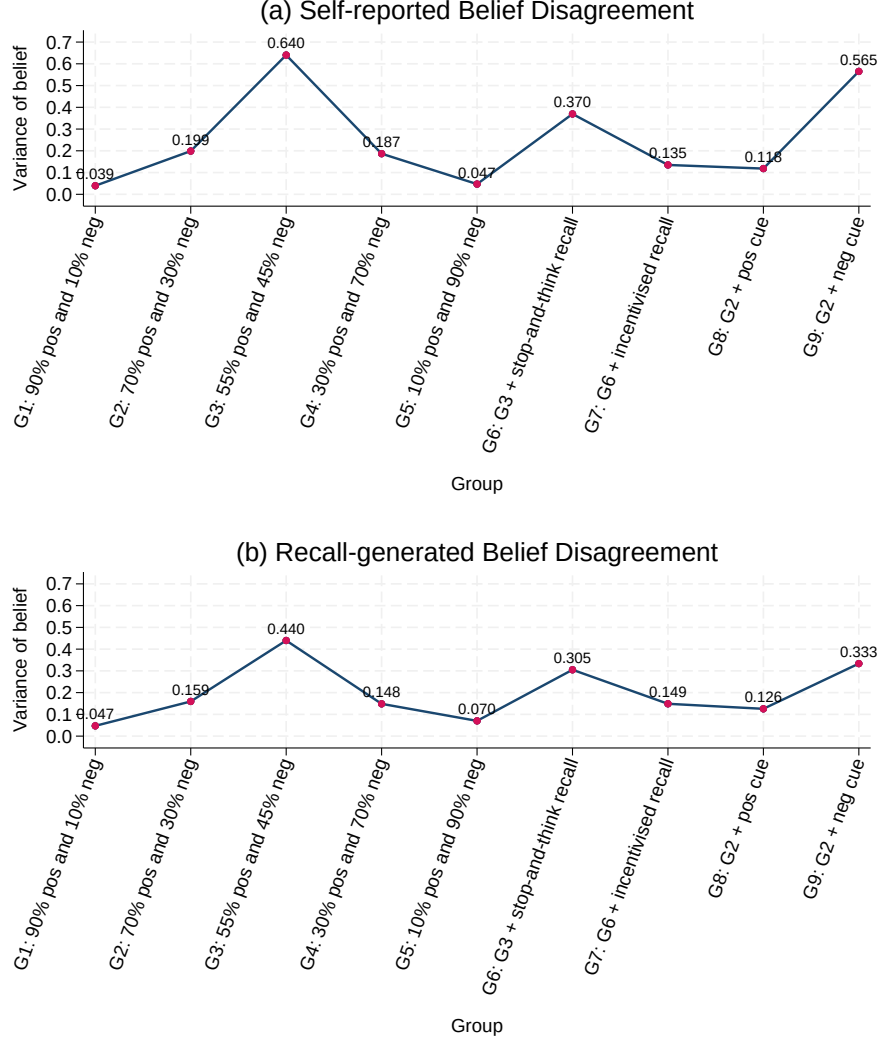


Figure 3: DISAGREEMENT ABOUT THE COMPANY STATE BY GROUP (REPORTED AND RECALL-GENERATED BELIEFS)

Notes: The figure plots disagreement in participants' binary beliefs across the nine experimental groups, measured as the cross-sectional variance within each group. The top panel uses self-reported binary beliefs (good = 1, bad = -1); the bottom panel uses recall-generated beliefs, coded as 1 if recalled positive signals are at least as many as recalled negative signals and -1 otherwise. Treatment groups are described in Table 1.

Recall-generated beliefs closely match reported beliefs For each participant, we substitute the number of recalled positive and negative signals into equation (3.1) to construct an individual-level binary *recall-generated belief* and compare it to the directly reported belief.³¹ Appendix

³¹Tied recalled counts ($X_i^+ = X_i^-$) are generally rare, somewhat more common only in the most ambivalent groups (Groups 3 and 6). Equation (3.1) classifies these cases as good, and we use the same convention here and in the simulated method-of-moments estimation, so the theory, the estimation, and the empirical construction share the same exact finite-sample binomial model. The results are not sensitive to this convention: even in Groups 3 and 6 the observed belief moments lie within the band spanned by the alternative conventions (classifying ties as good, as bad, or splitting them evenly). Under the normal approximation of Lemma 1 the distinction vanishes; the estimation in Section 4.4 simulates the exact binomial model and does not rely on that approximation.

Table A.7 shows that reported beliefs increase monotonically with the share of recalled positive signals and are strongly predicted (p value < 0.01) by an indicator for whether recalled positives exceed recalled negatives. Most notably, recall-generated beliefs match reported beliefs for 98 percent of participants.³²

Table 2: EFFECT OF RECALL EFFORT AND CONTEXTUAL CUES ON RECALL OUTCOMES

	(1)	(2)
<i>Panel A. Effect of recall effort on recall accuracy (Groups 3, 6, 7)</i>		
<i>Dep. Variable: absolute recall error in the share of positive news (p.p.)</i>		
Group 3	3.737*** (0.115)	3.817*** (0.426)
Relative to Group 3		
Group 6: Group 3 + Stop-and-Think	-0.586*** (0.186)	-0.553*** (0.189)
Group 7: Group 6 + incentivized recall	-0.846*** (0.151)	-0.849*** (0.153)
Demographics		✓
Observations	1,576	1,576
R-squared	0.016	0.032
<i>Panel B. Effect of contextual cues on recall composition (Groups 2, 8, 9)</i>		
<i>Dep. Variable: share of recalled positive news (%)</i>		
Group 2	59.395*** (0.205)	59.131*** (0.946)
Relative to Group 2		
Group 8: Group 2 + pos. cue	8.548*** (0.350)	8.503*** (0.352)
Group 9: Group 2 + neg. cue	-3.103*** (0.340)	-3.122*** (0.341)
Demographics		✓
Observations	1,603	1,603
R-squared	0.410	0.419

Notes: Panel A regresses the absolute recall error in the percentage-point share of positive news on group indicators for Groups 3, 6, and 7 (Group 3 is the omitted reference). Panel B regresses the share of recalled positive news on group indicators for Groups 2, 8, and 9 (Group 2 is the reference). Column (2) adds demographic controls in both panels. Robust standard errors in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Interventions shift retrieval in the predicted direction The bottom panel of Figure 3 computes disagreement using recall-generated beliefs only. The same treatment pattern reappears. Disagreement in recall-generated beliefs peaks in the most ambivalent environment (Group 3; variance 0.440), is low in the least ambivalent environments (Groups 1 and 5; 0.047 and 0.070), and is intermediate in Groups 2 and 4 (0.159 and 0.148). Increasing retrieval effort compresses

³²This excludes cases in which participants recall equal numbers of positive and negative signals. Including such cases reduces the consistency rate only slightly. Appendix Figure A.4 replicates the disagreement analysis using only participants for whom recall-generated and reported beliefs coincide and yields similar patterns.

disagreement relative to Group 3, from 0.440 to 0.305 when Stop-and-Think is added (Group 6) and to 0.149 when the accuracy incentive is added as well (Group 7). Associative cues also shift recall-generated disagreement relative to the no-cue baseline Group 2 (0.159), lowering it under the positive cue (Group 8; 0.126) and raising it under the negative cue (Group 9; 0.333). Because these beliefs are computed solely from recalled signals, heterogeneous priors or idiosyncratic noise in reported beliefs cannot drive them.

Table 2 quantifies the retrieval shifts directly. Retrieval effort improves recall precision: Stop-and-Think reduces the absolute recall error by 16 percent relative to Group 3, and adding the accuracy incentive reduces it by 23 percent (Panel A). Cues shift the composition of recalled signals: relative to an uncued recalled-positive share of 59.4 percent, the positive cue raises the share by 8.55 p.p. and the negative cue lowers it by 3.10 p.p. (Panel B), shifts large enough to move a nontrivial fraction of participants across the recall-balance threshold that determines recall-generated beliefs.

Measured recall accounts for most of the treatment effects So far we have documented the two links separately: the treatments change what participants recall, and what participants recall shapes their beliefs. If the treatments move beliefs by changing recall, then measured recall should account for most of their effect on beliefs, leaving little room for channels that bypass memory. Appendix Table A.13 quantifies this direct link, with Panel A covering the retrieval-effort treatments and Panel B the cue treatments. In the cue sample (Panel B), regressing reported binary beliefs on the dominant-recall indicator yields a coefficient of 0.92; adding the cue treatments themselves, which pick up any influence of the cue on reported beliefs outside of measured recall (such as salience at the reporting stage, a different interpretation of the news, or experimenter demand), leaves the effect of recall essentially unchanged (0.90, p value < 0.01) and an order of magnitude larger than the largest residual cue effect (moving reported beliefs by 1.81 versus 0.15 on the two-point scale). Controlling for a full set of recalled net-evidence indicators absorbs about three quarters of the negative-cue effect on beliefs (-0.24 to -0.06) and leaves no significant positive-cue effect; the same controls absorb roughly half of the retrieval-effort effects (Panel A; 0.33 to 0.13 for Group 7). Because recalled counts are themselves a noisy, post-elicitation measure of retrieval, these absorption shares are lower bounds: the effect of recall is understated and the residual treatment effects overstated. The treatments move beliefs

primarily by moving what is recalled.

Encoding differences cannot drive these effects Participants who see the same screens need not store the same information, so one may worry that the patterns above reflect differences in what was stored in memory when the news was shown (encoding) rather than in what is retrieved from memory when beliefs are formed. For most of our results the distinction is immaterial: beliefs depend only on the product of encoding, retention, and retrieval (Section 3). But the interpretation of the treatment effects does depend on where the friction originates, and four patterns suggest it is at the retrieval stage. First, the retrieval-effort and cue treatments arrive only after all signals have been seen, when encoding is complete, yet they move beliefs. Second, recall accuracy is statistically indistinguishable across the five ambivalence environments (Appendix Table A.8), so encoding differences cannot explain why disagreement varies across them. Third, signals absent from a participant’s Day-1 recall frequently re-emerge on Day 2 (Appendix Table A.16); information that was never stored cannot return. Fourth, the Stop-and-Think and Incentivized-Recall interventions make recalled counts more accurate but do not make participants any better at identifying individual signals when probed directly, with accuracy at chance for every position and sign (Appendix Table A.9): they help participants retrieve more of what is stored rather than sharpen memory for particular items, consistent with stochastic access to stored content.

4.4 Quantitative Estimation

The findings so far show that beliefs move in the directions the model predicts. We now ask whether the model fits the data quantitatively: can a single recall probability account for average beliefs in all nine groups at once, and how large is the friction at baseline and under the treatments? We fit the model with a simulation-based minimum-distance estimator targeting within-group mean beliefs. The estimator jointly identifies a baseline recall probability $\hat{p}$ from the informational-ambivalence variation in Groups 1–5, retrieval-effort effects $\hat{p}_6$ and $\hat{p}_7$ from Groups 6 and 7, and asymmetric cue effects $\hat{a}^+$ and $\hat{a}^-$ from Groups 8 and 9. Appendix K details the binomial simulation, the objective function, and the group-specific specifications.

Figure 4 compares experimental moments with model-implied moments evaluated at the estimated parameter vector $(\hat{p}, \hat{p}_6, \hat{p}_7, \hat{a}^+, \hat{a}^-)$. The top panel shows that the model closely

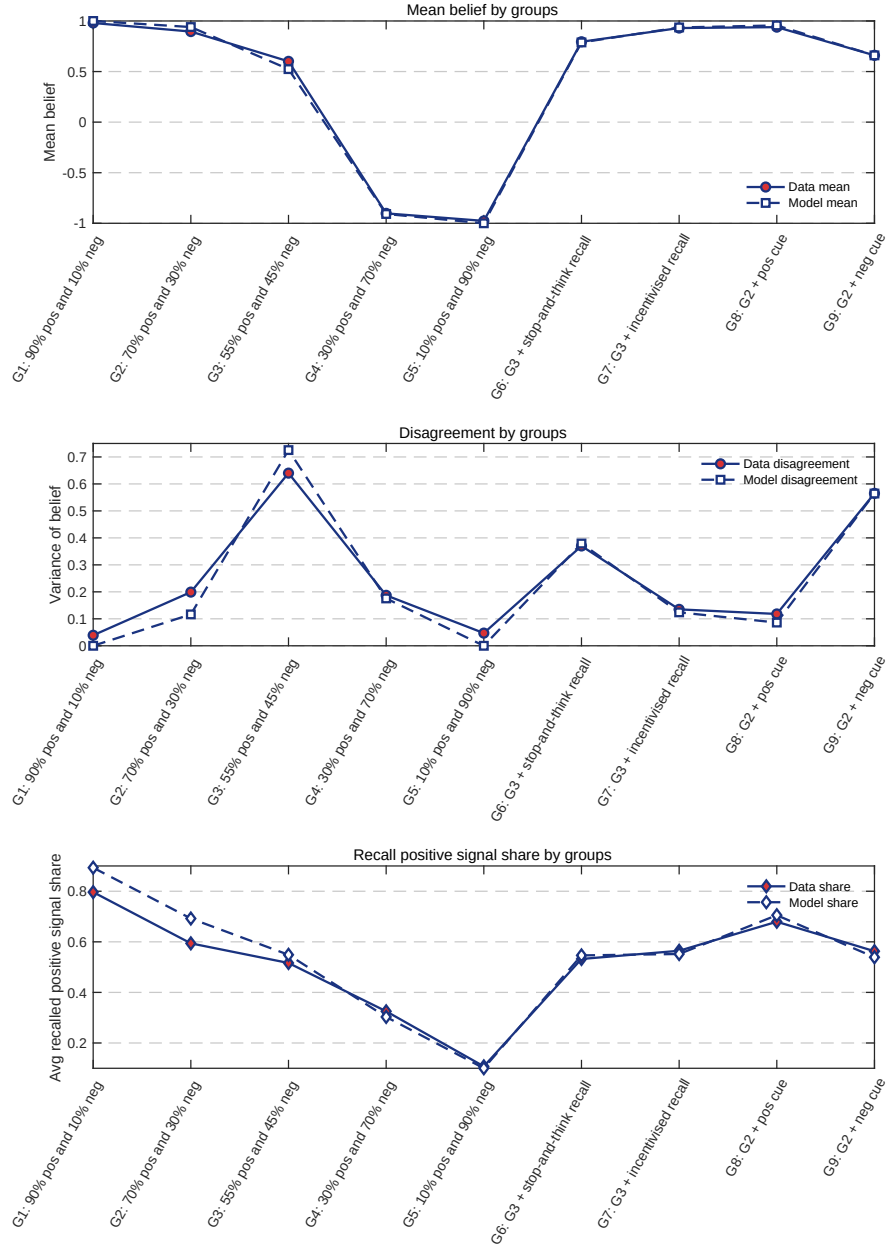


Figure 4: MODEL FIT: BELIEFS, DISAGREEMENT, AND RECALL ACROSS TREATMENTS

Notes: The figure compares experimental moments (solid) with model-implied moments (dashed) across the nine groups. Top panel: within-group mean belief. Middle panel: within-group variance of binary beliefs (disagreement). Bottom panel: average share of recalled positive signals. Groups 1–5 identify the common baseline recall probability $\hat{p} = 0.51$; Groups 6–7 are fitted with elevated recall probabilities $\hat{p}_6 = 0.81$ (Stop-and-Think) and $\hat{p}_7 = 0.90$ (Stop-and-Think + incentivized recall); Groups 8–9 increase the recall probability of the cued signal type by $\hat{a}^+ = 0.02$ (positive cue) and $\hat{a}^- = 0.475$ (negative cue).

matches the level and ordering of mean beliefs across groups. The middle panel shows that the model reproduces the non-monotonic disagreement pattern, the large reductions from retrieval effort, and the divergence between the two cue treatments. The bottom panel shows that the model also closely tracks the recalled share of positive signals across treatments. This

fit is informative because it validates the mechanism directly: the same parameter values that explain belief disagreement also explain retrieval patterns.³³ Throughout, the estimates carry the interpretation given in Section 3: $\hat{p}$ is an effective accessibility probability, indexing how much of the stored history is reflected in retrieved evidence, rather than a literal item-level recall rate. The retrieval-effort estimates also sharpen the comparison with existing memory-based models: Groups 6 and 7 receive no new information, yet beliefs shift and disagreement falls, and the estimation rationalizes both by raising a single parameter, the recall probability, an interpretation shared by Augenblick, Jack, Kaur, Masiye and Swanson (2026), who likewise model a prompting-based recall intervention as increasing the probability that stored items are retrieved. This margin is native to the memory-model family, whose central object is the recall probability of a stored signal.

4.5 Findings: Subjective Uncertainty

Under stochastic recall, the retrieval draw determines not only a DM’s belief but also how uncertain she is about the state. To test the model’s uncertainty predictions, let $U_i = p_i(1 - p_i)$ denote individual subjective uncertainty, as in Augenblick and Rabin (2021), where $p_i \in [0, 1]$ is respondent i ’s reported probability that the company is good. This is the empirical counterpart of the model’s uncertainty measure, the posterior variance $\mu_i(1 - \mu_i)$ of Section 3.2, with the reported probability standing in for the posterior: U_i is maximized at $p_i = 0.5$ and falls as beliefs become more extreme. For the group-level comparisons across treatments, let $U_g = E[U_i \mid i \in g]$ denote average uncertainty in group g .

Realized net imbalance strongly predicts subjective uncertainty Moving from a balanced recalled draw to a net imbalance of 10 reduces belief uncertainty by 0.03, about 12 percent of the full theoretical range $U_i \in [0, 0.25]$. Figure 5 illustrates this micro-level mechanism: subjective uncertainty is highest when recalled positive and negative signals are relatively balanced and declines as recalled evidence becomes more one-sided. Panel A of Appendix Table A.14 shows that the relationship is precisely estimated: the coefficient on realized net imbalance is -0.003 (p value < 0.01), with similar estimates after controlling for demographics, consistent with Lemma 2.

³³As a related check that uses no estimated parameters, Appendix Figure A.5 compares the group moments of the recall-generated beliefs of Section 4.3 with those of stated beliefs: the two agree closely in all nine groups, for both mean beliefs and disagreement, so recall alone reproduces the ambivalence pattern and the belief shifts induced by the retrieval-effort and cue treatments.

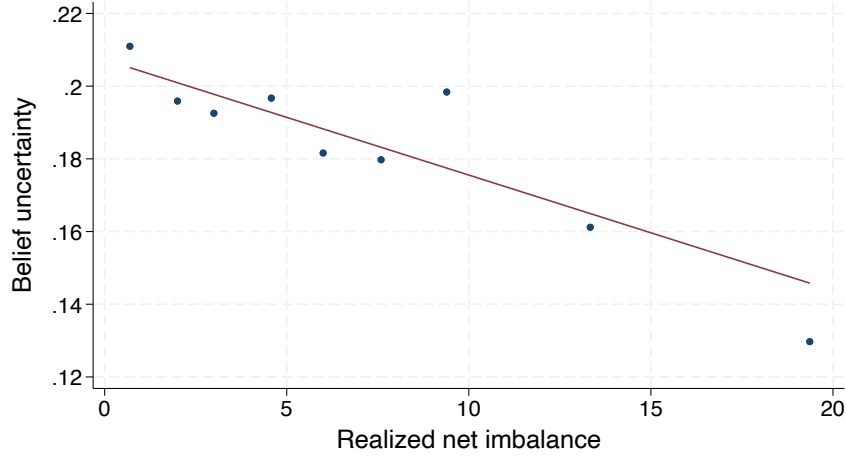


Figure 5: REALIZED NET IMBALANCE AND BELIEF UNCERTAINTY

Notes: The figure plots binned averages of individual belief uncertainty $U_i = p_i(1 - p_i)$ against realized net imbalance, defined as the absolute value of $X_i^+ - X_i^-$, where X_i^+ and X_i^- denote the numbers of positive and negative items recalled. The solid line is the fitted linear relationship. Panel A of Appendix Table A.14 reports the corresponding individual-level regression estimates and inference.

Subjective uncertainty increases with informational ambivalence Prediction 1 predicts

$$U_3 > \{U_2, U_4\} > \{U_1, U_5\}.$$

Belief uncertainty in the most balanced environment (Group 3) exceeds the most one-sided environments by up to 30 percent of the theoretical range $U_i \in [0, 0.25]$. Panel B of Appendix Table A.14 confirms this pattern. Taking Group 3 (55 percent positive) as the reference category, belief uncertainty is significantly lower in Group 2 by 0.019, in Group 4 by 0.022, in Group 1 by 0.053, and in Group 5 by 0.075 (p values < 0.01). The estimates are essentially robust after controlling for demographics.³⁴

Contextual cues have asymmetric effects on subjective uncertainty Prediction 3 predicts

$$U_8 < U_2, \quad U_9 > U_2 \text{ for a sufficiently weak minority cue.}$$

A dominant positive cue lowers belief uncertainty by about 19 percent relative to the no-cue baseline, while a non-dominant negative cue raises it by about 14 percent, consistent with the asymmetric effects predicted by Prediction 3. Panel C of Appendix Table A.14 confirms this pattern. Relative to Group 2 (baseline uncertainty 0.198), the positive cue lowers uncertainty by

³⁴Panel B also shows that the recall interventions in Groups 6 and 7 reduce subjective uncertainty relative to Group 3. Group 6 (Stop-and-Think) lowers uncertainty by 0.021; Group 7 (Stop-and-Think + incentivized recall) lowers it by 0.033 (both p values < 0.01 , robust to demographic controls).

0.037 and the negative cue raises it by 0.028 (both p values < 0.01 ; identical with demographic controls).³⁵

Result 2 (Disagreement and uncertainty under fixed information). *Holding the information history fixed: (i) belief disagreement and subjective uncertainty rise with informational ambivalence; (ii) post-exposure retrieval effort reduces both; (iii) an uninformative contextual cue lowers both when it matches the dominant signal type and raises both when it matches the minority type; (iv) at the individual level, subjective uncertainty declines with the realized net imbalance of recalled evidence. The treatment effects operate primarily through measured recall.*

5 Belief Instability: 24-Hour Re-Elicitation

Experiment 2 replicates the disagreement design of Experiment 1 in a new sample; Experiment 3 re-elicits the same participants approximately 24 hours later under unchanged information, so that any belief revision reflects a fresh retrieval draw (Predictions 1 and 2); Graeber, Roth and Zimmermann (2024) use a similar one-day delayed re-elicitation to isolate the effect of memory on beliefs.

5.1 Design and Cross-Validation

Design Experiment 2 follows the Experiment 1 protocol with two modifications. First, the sample is drawn fresh from the same U.S. general-population participant pool, with respondents who participated in Experiment 1 excluded at recruitment. Second, the experiment uses four treatment groups, summarized in Panel B of Table 1: three replicate the informational-ambivalence manipulation (matching Groups 1–3 of Experiment 1), and one implements a Stop-and-Think intervention analogous to Group 6. Experiment 3 (Panel C of Table 1) re-contacts these participants approximately 24 hours later. Participants are reminded that they previously observed a sequence of news about the company but receive no new information, and then report (i) a binary belief, (ii) a probabilistic belief, and (iii) recalled positive and negative signal counts.³⁶

Sample The study was conducted online via Prolific on 22–23 January 2026. In Part 1, we recruited approximately 2,900 new respondents, yielding 2,039 valid completed responses after

³⁵With $\hat{p} = 0.51$ and cue strength $a^- = 0.475$ as above, the analytical expression implies $\Theta = 0.68 > a^-$: the negative cue is weak, consistent with Prediction 3.

³⁶The order of belief elicitation and recall mirrors the Day-1 structure; Stop-and-Think participants again complete recall before belief elicitation.

comprehension checks.³⁷ Participants who completed Part 1 were re-invited one day later for Part 2; the re-participation rate was 86 percent, yielding 1,744 respondents in Part 2.³⁸ Median completion time was approximately 13 minutes for Part 1 and 3 minutes for Part 2. Appendix Table A.15 reports respondent demographics.

Cross-validation and Day-1 baseline Experiment 2 serves two purposes: a same-design cross-validation of Experiment 1 in a fresh sample, and the Day-1 baseline for Experiment 3’s within-individual evidence. Appendix Figure A.7 shows that disagreement in self-reported beliefs again increases with informational ambivalence (Group 1 < Group 2 < Group 3) and falls under Stop-and-Think (Group 4 < Group 3), for both self-reported and recall-generated beliefs.

5.2 Findings

In Experiment 3, let R_g denote the revision rate in group g , the proportion of participants who switch their binary belief between Day 1 and Day 2.

Belief instability increases with informational ambivalence Prediction 1 predicts $R_3 > R_2 > R_1$. Figure 6 shows that belief revision is rare when information is highly one-sided (Group 1; 7.3 percent for self-reported beliefs, 0.2 percent for recall-generated beliefs), becomes more frequent in intermediate environments (Group 2; 11.3 percent and 4.3 percent), and peaks in the most ambivalent environment (Group 3; 30.3 percent and 26.0 percent). Appendix Table A.17 reports the corresponding regression estimates, with revision rates rising significantly with informational ambivalence (p value < 0.01).

Increased retrieval effort reduces belief instability Prediction 2 predicts $R_3 > R_4$. Consistent with this prediction, revision rates fall sharply: from 30.3 percent to 18.8 percent for self-reported beliefs and from 26.0 percent to 20.1 percent for recall-generated beliefs (p value < 0.01; Appendix

³⁷In Part 1, we screen out likely bots flagged by the Qualtrics and Prolific quality checks, respondents who fail an attention check, and responses with completion times shorter than 5 minutes or longer than 30 minutes, thresholds that correspond approximately to the bottom and top 1 percent of completion times; together these screens remove about 4 percent of responses. We also exclude respondents who report taking notes during the information screens, around 2 percent of the sample.

³⁸Return is not selective on observables: a joint F -test of the demographic coefficients in a regression of Part-2 participation on demographic characteristics is insignificant (Appendix Table A.3).

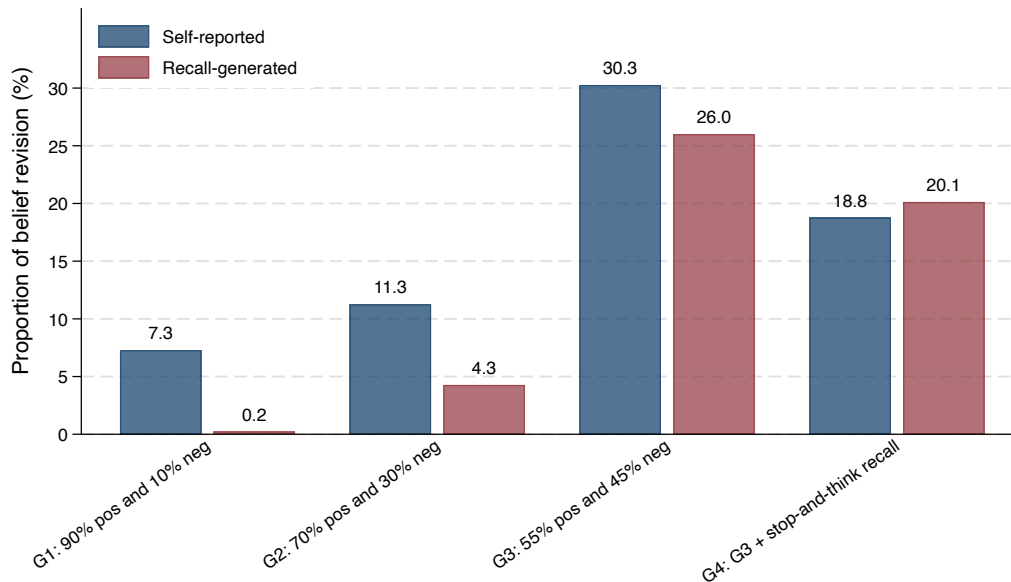


Figure 6: BELIEF INSTABILITY OVER TIME

Notes: The figure shows the proportion of participants whose belief about the company's type changes between Part 1 (Day 1) and Part 2 (Day 2) in Experiment 3. For self-reported beliefs, instability is defined as a change in the reported binary belief from good to bad or vice versa. For recall-generated beliefs, instability is the change in the mechanically generated belief implied by recalled information. Groups 1–3 vary the signal mix while holding the total number of signals fixed; Group 4 shares Group 3's composition but adds a Stop-and-Think retrieval intervention.

Table A.17).³⁹

Mechanism The instability patterns also test the mechanism of Section 4.3 within individual. Appendix Table A.18 shows that, within individuals, changes in the recalled positive-news share between Part 1 and Part 2 strongly predict belief revisions for both self-reported and recall-generated binary beliefs: a 10 p.p. increase in the recalled positive share raises the belief-revision index by 0.29 for self-reported beliefs and 0.39 for recall-generated beliefs (p values < 0.01). The directional specification gives the same picture: a same-sign within-individual shift in recall predicts a same-sign belief revision (coefficients 0.114 and 0.149, p values < 0.01). Conditioning on retrieval makes the location of the instability explicit: among participants whose recalled majority switches between the two days, 83 percent switch their stated belief, against 7 percent among those whose recalled majority is unchanged (Panel C of Appendix Table A.18); the latter share bounds from above the instability attributable to stochasticity downstream of retrieval,

³⁹Revision rates based on the direction of the reported probability show the same pattern: 1.0, 10.6, and 30.0 percent in Groups 1–3, and 9.5 percent with retrieval effort (Group 4; p values < 0.01), as does the variance of within-person revisions in the level of the reported probabilistic beliefs. See Section S.7 of the Supplementary Appendix.

such as reporting noise.⁴⁰ These patterns directly parallel the within-individual co-movement of recall and beliefs in the field studies (Result 1).

Result 3 (Belief instability under fixed information). *Re-eliciting beliefs from the same participants 24 hours later with no new information: (i) the share of participants revising their binary belief rises with informational ambivalence; (ii) retrieval effort reduces revision rates; (iii) within individuals, belief revisions closely track revisions in recalled information.*

Recall alone predicts the revision rates The model also pins down the level of each group's revision rate, not just the ordering. Under stochastic recall, a participant's belief on each day is a fresh retrieval draw that comes up "good" with the same probability, so the probability of switching between the two days is determined by that probability alone. We measure it from recall data: let $\hat{q}$ be the share of a group's returning participants who recall at least as many positive as negative signals on Day 1; the implied Day-2 switching rate is then $2\hat{q}(1 - \hat{q})$, with no stated-belief data and no free parameters. Appendix Figure A.9 shows that this prediction tracks observed switching closely: predicted and observed rates are within 5 p.p. of each other in every group.

5.3 Discussion: Alternative Explanations

We now address the principal alternative explanations for the recall–belief link: reverse causality or a common latent judgment, elicitation artifacts, measurement error, and recall accuracy heterogeneity. Appendix O addresses five further alternatives: experimenter demand, cognitive defaults, external recording, pure forgetting, and stochastic-choice and noisy-aggregation models.

Reverse causality and common latent judgment The recall–belief link could reflect reverse causality, with respondents adjusting recall to match a pre-formed belief, or a common latent judgment Z_i driving both recall and beliefs. Both alternatives are not consistent with the data. Under motivated recall, signals whose sign matches the respondent's belief should be recalled more accurately; pooling the 1st, 5th, 15th, and last signals across Groups 1–5, recall accuracy

⁴⁰Recalled content carries comparable weight in the cross-section: adding the recalled majority to treatment-group indicators raises the R^2 for Day-1 beliefs from 0.05 to 0.61, and stated and recall-generated beliefs coincide for 96 percent of participants on Day 1 and 95 percent on Day 2. Most of the within-group heterogeneity in beliefs is thus variation in what participants retrieve, and the small remainder again bounds the scope for stochasticity downstream of retrieval.

does not depend on whether the signal matches the respondent's belief (marginal effect 0.012, p value > 0.10 , with or without position fixed effects and demographic controls). Under a common-state interpretation, a retrieval-effort intervention should sharpen reported recall without affecting beliefs. The Incentivized Recall treatment (Group 7) rules out this alternative: Appendix Table A.10 uses Group 7 as an instrument for recall, restricted to Groups 3 and 7. The first stage is strong (F -statistics of 443 for the recalled positive share and 121 for realized net imbalance), and the treatment significantly raises the reported probability that the company is good by 15.4 p.p. while reducing subjective uncertainty by 0.033 units; these patterns are hard to reconcile with a common-state model. Because Group 7 also changes the elicitation order relative to Group 3, Panel B of the same table repeats the exercise against Group 6, which holds the order fixed so that only the incentive differs: the incentive alone shifts the recalled positive share (first-stage $F = 126$) and raises the reported probability that the company is good by 5.0 p.p. while reducing uncertainty by 0.012.⁴¹

Elicitation artifacts A related concern is that the elicitation itself creates the recall-belief association: beliefs anchored on the contemporaneous recall task, recall made artificially salient by being elicited first, or a taste for consistent answers (Falk and Zimmermann, 2013). Three features alleviate these concerns. First, in Groups 1–5 and on Day 2 of Experiment 3, beliefs are elicited *before* recall, so stated beliefs can be neither anchored on recall output nor inflated by recall priming; the one treatment that reverses the order (Group 6) raises the coincidence between recall-generated and stated beliefs only from 90 to 95 percent, an upper bound on any anchoring or priming effect since Group 6 also strengthens recall. Second, the consistent answer is not transparent: recall is elicited as raw counts of positive and negative signals, which must be aggregated before they imply a belief. Third, elicitation artifacts cannot explain why recall itself is heterogeneous and unstable (Result 1), why belief revisions track recall revisions under an identical within-session elicitation order (Result 3), or why disagreement, instability, and uncertainty respond to exogenous variation in ambivalence, recall effort, and cues, since these patterns are computed from beliefs alone.

⁴¹Three further patterns reinforce this view: within-individual revisions in the field studies track recall roughly one-for-one (Appendix Table A.5), whereas classical reporting noise would attenuate the slope below one under a common-state model; minority cues in Group 9 amplify both disagreement and subjective uncertainty (Result 2), against the compression a uniform Z_i shift would predict; and Stop-and-Think (Group 6) shifts beliefs although the signal history is identical to Group 3.

Measurement error Recall in our experiments is verifiable: reports are graded against the true signal sequence, so recall error is an objective quantity, and the retrieval-effort interventions make reports objectively more accurate (reductions of 16 and 23 percent in absolute recall error; Table 2), which is possible only if reports draw on genuine memory content rather than being constructed at elicitation. Several features of the results provide additional evidence inconsistent with classical measurement error in recall: the near-perfect alignment between recall-generated and stated beliefs, the systematic co-movement of recall and belief revisions, and the structured comparative statics. The Experiment 3 re-elicitation lets us address the concern even more directly, following Gillen, Snowberg and Yariv (2019). Recall reports on the two days share a persistent individual component of retrieval, while each day adds a fresh recall draw and its own reporting error; instrumenting Day 2 recall with Day 1 recall therefore isolates the persistent component and removes classical reporting noise, any transient state common to the same day's reports (attention, mood, or effort at the time of the survey), and any contamination of the Day 2 recall report by the same day's stated belief. The first stage is strong ($F > 1,000$), documenting the persistent component, and the 2SLS estimates in Appendix Table A.19 are significant with the predicted signs and at least as large as the corresponding OLS estimates: a 10 p.p. increase in the Day 2 recalled positive share raises the Day 2 reported probability that the company is good by 5.06 p.p. and the binary belief by 0.22, while a one-unit increase in realized net imbalance reduces subjective uncertainty by 0.003 (all p values < 0.01).

Recall accuracy heterogeneity Could the observed disagreement reflect systematic heterogeneity in recall accuracy rather than stochastic retrieval? Four pieces of evidence from Experiment 1 suggest no. Recall accuracy is statistically indistinguishable across signal positions and signal signs (Appendix Table A.9), across levels of informational ambivalence in Groups 1–5 (Appendix Table A.8), across participant demographics (Appendix Table A.11), and across demographics within the cue treatments Groups 8 and 9 (Appendix Table A.12). These results are perhaps not very surprising, given that the news items about the company provided in the experiment were simple and qualitative. The one individual characteristic that predicts recall is financial

literacy.⁴² Randomization balances it across arms (the share with high literacy ranges from 42 to 48 percent across groups, $p = 0.76$), so it cannot drive the treatment comparisons, and it operates as the model predicts: pooling the nine groups, participants with high literacy recall the positive share more accurately (absolute error 6.0 versus 6.7 p.p., $p < 0.01$ with group fixed effects) and display less disagreement (variance of the binary belief around its group mean 0.20 versus 0.30, $p < 0.01$), consistent with a higher recall probability (Prediction 2), while the ambivalence, effort, and cue comparative statics hold within each literacy subsample.

6 Quantitative Extension: Beliefs in a Dynamic Environment

Sections 4 and 5 established stochastic recall in a stylized binary environment. This section shows that the same mechanism extends to quantitative expectations about a dynamic process: participants observe the same realized history of a persistent stochastic process and share identical priors and belief rules, yet their mean estimates differ and change over time.

6.1 Setup and Predictions

DMs observe the same realized history $\{Z_1, \dots, Z_T\}$ of a stationary AR(1) process with persistence ρ , shock standard deviation σ_ε , and unknown long-run mean, so the unconditional variance of the process is $\sigma_\varepsilon^2 / (1 - \rho^2)$. As in the binary model, each DM recalls each realization independently with probability p and estimates the mean by averaging the recalled observations; at a second elicitation with no new information, a fresh recall draw generates a revised estimate. Appendix J develops the model formally and proves the following predictions, where disagreement is the cross-sectional variance of mean estimates and instability is the expected squared within-individual revision.

Prediction 4 (Quantitative disagreement). *Cross-sectional disagreement in mean estimates increases with the volatility and the persistence of the process.*

Prediction 5 (Quantitative instability). *Within-individual instability likewise increases with the volatility and the persistence of the process.*

⁴²Financial literacy is measured by the three standard questions of Lusardi and Mitchell (2014): whether \$100 in a savings account at a fixed interest rate of 3 percent per year grows to more than, exactly, or less than \$115 after five years; whether, with an interest rate of 3 percent and inflation of 2 percent per year, the money in the account buys more than, the same as, or less than today after one year; and whether buying a stock mutual fund usually provides a safer return than buying a single company's stock. High literacy denotes answering all three correctly (46 percent of participants).

Both results have a single source: each elicited belief is the sample mean of a random subset of a common history, so dispersion across DMs, and across elicitations within a DM, is proportional, conditional on the realized history, to that history’s sample variance (Appendix J); volatility and persistence raise dispersion by raising the variance of the histories the process generates, $\sigma_\varepsilon^2/(1 - \rho^2)$ in expectation.⁴³ The closed forms in Appendix J additionally imply that both decline with the recall probability and the length of the history, margins we do not vary in this environment.

6.2 Experimental Design

We implement two quantitative belief-elicitation experiments in this environment. In both, participants observe a common sequence of $T = 20$ realizations from a stable AR(1) process with unknown long-run mean, and no new information arrives between Day 1 and Day 2.

Experiment 4: quantitative disagreement Participants are randomly assigned to one of three groups that vary the persistence and volatility of the underlying AR(1) process. Group 1 observes a low-persistence, low-volatility process ($\rho = 0.1, \sigma = 0.5$); Group 2 a high-persistence, low-volatility process ($\rho = 0.9, \sigma = 0.5$); and Group 3 a high-persistence, high-volatility process ($\rho = 0.9, \sigma = 5$). In all groups, the long-run mean is fixed at $\mu = 5$ and unknown to participants. Participants are told that the data points are generated from a stable process, but not the form of the process or its parameters. Each group observes a single realized draw from its process: the three displayed histories share an identical sample mean (5.15) by construction and have sample variances of 0.18, 0.42, and 24.91, so the treatments vary the dispersion of the values on screen while holding their average fixed. Because the treatment is embodied in one realized history per group, cross-group comparisons are comparisons across these realized histories, and we read them through the model’s predictions conditional on the realized history (Appendix J). After viewing the sequence, participants complete unrelated tasks, then report their best estimate of the series average (with accuracy incentives) and complete a free-recall task in which they list any values they remember.

⁴³A simulated minimum distance calibration analogous to that of Section 4.4 reproduces the cross-sectional disagreement across the three treatment groups of Experiment 4; we provide estimation details in the Supplementary Appendix.

Experiment 5: quantitative instability Participants who complete Experiment 4 are re-invited approximately 24 hours later and again report their best estimate of the series average (with accuracy incentives) and complete the same free-recall task, without being shown the data.

Sample Experiments 4 and 5 were conducted online via CloudResearch using its Connect participant pool.⁴⁴ The experiment targeted a U.S. general-population sample. Approximately 2,500 participants were recruited for Experiment 4 on 1 February 2026, yielding 2,301 valid completed responses, of whom 1,867 (about 81 percent) returned to complete Experiment 5 one day later.⁴⁵ Median completion time was approximately 12 minutes in Experiment 4 and 4 minutes in Experiment 5. Appendix Table A.20 reports respondent demographics by group.

6.3 Results

Disagreement increases with volatility and persistence Consistent with Prediction 4, holding persistence fixed at $\rho = 0.9$, increasing volatility raises cross-sectional disagreement in mean estimates by a factor of 4.3 on Day 1 and 5.3 on Day 2 (variances 3.23 versus 13.81, and 4.09 versus 21.74; panel (a) of Appendix Figure A.11). Holding volatility fixed at $\sigma_\varepsilon = 0.5$, increasing persistence from $\rho = 0.1$ to $\rho = 0.9$ raises the variance of mean estimates by a factor of about seven on Day 1 (0.47 to 3.23) and five on Day 2 (0.78 to 4.09). The cross-sectional variance of recall-generated averages moves in parallel (panel (b)), and Appendix Figure A.10 shows the belief distributions widening sharply as the history becomes more volatile. Because each group sees one fixed sequence of values, these are comparisons across the displayed sequences: given the values on screen, the model predicts that belief dispersion scales with how spread out those values are, their sample variance, plus reporting noise (Appendix J), and the estimates follow exactly this ordering. The Supplementary Appendix reports a calibration of the model on these realized histories.

⁴⁴Connect and the Sentry pre-screening are described in footnote 4. We switched from Prolific to CloudResearch in order to draw on a fresh participant pool after the qualitative experiments had recruited a large number of respondents from Prolific. We additionally implemented multiple pre-specified in-survey attention and consistency checks, including an explicit attention check and screening for completion times shorter than 3 minutes (bottom 1 percent) or longer than 20 minutes (top 1 percent); these screens together remove about 6 percent of completed responses.

⁴⁵We also exclude respondents who report taking notes during the information screens, around 1 percent of the sample.

Instability increases with volatility and persistence Consistent with Prediction 5, holding persistence fixed, the variance of within-individual belief revisions over 24 hours rises by a factor of about five with volatility (5.36 to 27.54), and the variance of recall revisions by a factor of about twelve (0.70 to 8.21). Holding volatility fixed, revision variance rises by a factor of about 4.5 with persistence (1.19 to 5.36) and recall-revision variance by about 2.6 (0.27 to 0.70). Appendix Figures A.13 and A.14 plot the revision distributions.

Mechanism: beliefs move closely with recalled averages Appendix Table A.22 quantifies the mapping from recalled information to quantitative beliefs. In the cross section (Panels A–B), coefficients of mean estimates on recalled averages range from 0.78 to 0.95 on Day 1 and from 0.70 to 0.99 on Day 2 (p values < 0.01). Within individuals (Panel C), coefficients on changes range from 0.75 to 0.94. Quantitative beliefs thus move nearly one-for-one with recalled information, across participants and within participants over time. Applying the instrumental variable approach from Section 5.3 to the two-day structure (Appendix Table A.23; first-stage $F > 200$), the 2SLS estimate of the recall-belief slope is 0.460 pooled and 0.509 with group fixed effects (p values < 0.01). The instrumented slopes are smaller than the OLS ones, as expected: beliefs and recall are reported in the same session, so same-day influences on how a participant answers, such as mood or effort, move both reports together and inflate the OLS slope, while the instrument, reported a day earlier, is free of these influences. Under either estimate, beliefs load strongly and significantly on recalled information.

Alternative explanations The alternative mechanisms ruled out in Section 5.3 for the qualitative experiments carry over to the continuous-belief setting. Beliefs are elicited before recall in both Day 1 and Day 2 sessions, ruling out within-session anchoring, and pre-specified screens exclude the small share of respondents (around 1 percent) who report taking notes. Pure forgetting is again rejected: if a value, once forgotten, could not return, every value recalled on Day 2 would already appear in the participant’s Day-1 recall. Instead, 90 percent of participants report at least one Day-2 value absent from their Day-1 recall, even counting values within 2.5 percent of a Day-1 report as repeats (Appendix Table A.21); most of these re-emerging values match data points actually displayed, indicating retrieval rather than noise.

Result 4 (Quantitative beliefs under fixed information). *In the dynamic environment: (i) cross-sectional disagreement in mean estimates and within-individual revisions over 24 hours both*

increase with the volatility and the persistence of the observed history; (ii) beliefs and belief revisions closely track recalled averages, with slopes between one half (instrumented) and one (least squares).

7 Stochasticity in Retrieval versus Noise in Processing

Our model and models of noisy processing share the same discipline, optimal use of whatever the mind delivers, and differ in what the mind delivers. In noisy-processing models, the mind encodes the information into an imprecise internal impression, a noisy but genuinely informative summary, and beliefs are an optimal but deterministic function of that impression (Khaw, Li and Woodford, 2021; Woodford, 2020; Azeredo da Silveira, Sung and Woodford, 2024): the stochasticity enters at the encoding stage, posterior precision is set by the noise parameters, and information the impression misses is permanently lost.⁴⁶ In our model, the record of stored signals is intact and the stochasticity enters at retrieval: each elicitation draws a random subset of the stored signals, and beliefs are Bayesian given the draw. In short, under noise in processing the record is corrupted and reading it is deterministic; under stochastic recall the record is intact and reading it is stochastic. The observable difference is the role of recalled content: a noisy impression assigns no role to which particular signals a respondent can produce, whereas under stochastic recall beliefs and confidence move with the realized recalled sample.

Several pieces of empirical evidence favor the memory-based account with stochastic recall and place the mechanism at the retrieval stage: each follows directly from stochastic sampling of an intact memory store, and none can be generated by noise in the processing of information. (1) *Beliefs track measured recall*: revisions co-move with recall revisions, nearly one-for-one in the qualitative experiments (where stated beliefs switch overnight for 83 percent of participants whose recalled majority switches, versus 7 percent of those whose recall is unchanged) and with slopes between one half and one in the quantitative experiments; under processing noise, which particular signals come to mind plays no role.⁴⁷ (2) *Content-free cues move beliefs through recall* in the qualitative experiments, but an impression that reflects only information has no channel

⁴⁶In the noisy-memory model of Azeredo da Silveira, Sung and Woodford (2024), information not remembered at one date cannot be retrieved at any later date.

⁴⁷Section 5.3 rules out that a common internal impression drives both reports. Moreover, no information is displayed at the re-elicitations, so there is no present stimulus whose noisy encoding could move beliefs (Khaw, Li and Woodford, 2021); beliefs that swing must swing with what comes to mind.

for an uninformative cue. (3) *Forgotten signals return*: a day later, 90 percent of participants in the quantitative experiments produce a value missing from their Day-1 recall, and most re-emerging values match numbers actually shown; a corrupted record cannot return what it has lost. (4) *Confidence moves with the recalled sample* in the qualitative experiments, whereas under processing noise precision is set by the noise parameters, not by the realized draw. The interventions show where the mechanism operates: the effort and cue treatments arrive only after all signals have been seen, yet they move beliefs, and they do so through measured recall, making recalled counts objectively more accurate while probed item-level recall stays at chance, the signature of retrieving more from an intact store rather than of a sharper impression.⁴⁸

8 External Validity: Evidence from the Michigan Survey of Consumers

This section asks whether the comparative statics isolated experimentally also appear in naturally occurring expectations data: greater disagreement, instability, and uncertainty after more mixed histories, and an asymmetric effect on disagreement of salient recent returns acting as cues. We focus on stock price expectations in the Michigan Survey of Consumers (MSC); [Appendix R](#) presents analogous results for quantitative inflation expectations.

Empirical setup We study the MSC stock price expectation question from June 2002 to March 2025. The MSC elicits respondents’ subjective probability $p_{it} \in [0, 1]$ that a \$1,000 investment in a diversified stock fund will be worth more than \$1,000 one year later. We construct three outcomes: a binary directional expectation (+1 if $p_{it} \geq 0.5$, -1 otherwise) whose monthly cross-sectional standard deviation measures disagreement; an indicator for switching direction between consecutive interviews (about six months apart) for within-individual instability; and $U_{it} = p_{it}(1 - p_{it})$ for subjective uncertainty (defined as in [Section 4.5](#)).⁴⁹

The empirical analogue of informational ambivalence is the balance of positive and negative realized S&P 500 returns over a trailing window. For horizon $h \in \{1Y, 5Y, 10Y\}$ and counting

⁴⁸The two sources of noise are not mutually exclusive; indeed, [Woodford \(2020\)](#) notes that for symbolic information, noise in storage and retrieval likely outweighs noise in perception. [Afrouzi, Kwon, Landier, Ma and Thesmar \(2023\)](#) study an intermediate case in which retrieval returns a noisy summary of an intact history; the Supplementary Appendix provides a detailed comparison with noisy-processing models.

⁴⁹Results are robust to defining the binary directional expectation with probability cutoffs other than 0.5.

frequency $f \in \{\text{Monthly, Quarterly}\}$, define

$$A^f(h) = 1 - \frac{|\#Neg^f(h) - \#Pos^f(h)|}{\#Neg^f(h) + \#Pos^f(h)},$$

which is highest when recent returns are evenly balanced and lowest when they are overwhelmingly of one sign.

Ambivalence is associated with disagreement, instability, and uncertainty Table 3 shows that more ambivalent recent return histories are associated with higher disagreement (Panel A), higher belief instability (Panel B), and higher subjective uncertainty (Panel C) at all horizons and sampling frequencies, with the relationship strongest at longer windows. In Panel A, a 0.10 increase in trailing ambivalence is associated with a 0.8–2.4 p.p. (monthly) or 0.5–1.5 p.p. (quarterly) increase in disagreement. The Panel B and Panel C patterns mirror this: a 0.10 increase in ambivalence raises the switching probability by 0.3–0.6 p.p. and subjective uncertainty by up to 0.0023 (p value < 0.01 throughout). The stronger association at longer horizons is consistent with richer recent information histories creating more scope for heterogeneous retrieval of past realizations.⁵⁰

Salient recent returns as natural cues If the most recent return reinforces the dominant sign of recent history, it may cue recall of similar past returns and compress disagreement; if it goes against the dominant sign, it may prime recall of minority returns and raise disagreement. Appendix Table A.25 examines this pattern by regressing disagreement on an indicator for a positive most-recent S&P 500 return, an indicator for whether this cue is majority-aligned, and their interaction. The interaction is negative and statistically significant across specifications (implied majority-aligned effect $0.015 - 0.046 = -0.031$, p value < 0.01 , column 1). This asymmetry mirrors the cue-treatment results in Experiment 1 and is consistent with cue-based retrieval in which salient recent returns shape which past returns households bring to mind (Jiang, Liu, Peng and Yan, 2025).

Result 5 (Ambivalence, cues, and expectations in the Michigan Survey). *In both stock price and inflation expectations, more ambivalent recent histories are associated with greater disagreement, greater within-individual instability, and higher subjective uncertainty, at all horizons and sam-*

⁵⁰ Appendix R repeats the analysis for households' one-year-ahead inflation expectations. The same patterns hold: disagreement and within-household revisions rise with trailing inflation volatility at all horizons and sampling frequencies, as does a rounding-based proxy for subjective uncertainty, used because the inflation question elicits a point forecast rather than a probability.

Table 3: STOCK PRICE EXPECTATIONS AND TRAILING STOCK-RETURN AMBIVALENCE

	Monthly			Quarterly		
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Disagreement in stock price movement expectations						
Ambivalence (1 year)	0.081*** (0.009)			0.045*** (0.005)		
Ambivalence (5 years)		0.145*** (0.014)			0.100*** (0.008)	
Ambivalence (10 years)			0.236*** (0.029)			0.147*** (0.016)
Cumulative return (1 month)	-0.111*** (0.038)	-0.101*** (0.036)	-0.096** (0.039)	-0.155*** (0.044)	-0.129*** (0.034)	-0.114*** (0.038)
Cumulative return (10 years)	-0.005*** (0.002)	-0.007*** (0.002)	-0.001 (0.002)	-0.006*** (0.002)	-0.004** (0.002)	0.004** (0.002)
<i>R-squared</i>	0.265	0.299	0.250	0.223	0.426	0.304
<i>Observations</i>	274	274	274	274	274	274
Panel B. Belief instability in stock price movement expectations						
Ambivalence (1 year)	0.046*** (0.010)			0.031*** (0.006)		
Ambivalence (5 years)		0.034** (0.015)			0.040*** (0.008)	
Ambivalence (10 years)			0.059** (0.027)			0.038*** (0.015)
Cumulative return (1 month)	-0.200*** (0.038)	-0.202*** (0.038)	-0.200*** (0.038)	-0.224*** (0.038)	-0.208*** (0.038)	-0.204*** (0.038)
Cumulative return (10 years)	-0.001 (0.002)	-0.001 (0.002)	0.000 (0.002)	-0.001 (0.002)	-0.000 (0.002)	0.002 (0.002)
Inter-interview return	-0.000 (0.013)	0.009 (0.013)	0.013 (0.013)	0.008 (0.013)	0.002 (0.013)	0.013 (0.013)
<i>R-squared</i>	0.001	0.001	0.001	0.001	0.001	0.001
<i>Observations</i>	53743	53743	53743	53743	53743	53743
Panel C. Subjective uncertainty in stock price expectations						
Ambivalence (1 year)	0.004*** (0.001)			0.003*** (0.001)		
Ambivalence (5 years)		0.014*** (0.002)			0.009*** (0.001)	
Ambivalence (10 years)			0.023*** (0.003)			0.014*** (0.002)
Cumulative return (1 month)	0.002 (0.004)	0.003 (0.004)	0.004 (0.004)	-0.001 (0.004)	0.001 (0.004)	0.002 (0.004)
Cumulative return (10 years)	-0.000 (0.000)	-0.000 (0.000)	0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	0.001*** (0.000)
Demographics	✓	✓	✓	✓	✓	✓
<i>R-squared</i>	0.027	0.027	0.027	0.027	0.028	0.027
<i>Observations</i>	135201	135201	135201	135201	135201	135201

Notes: MSC stock price expectations, June 2002–March 2025. Panel A: dependent variable is cross-sectional standard deviation of binary expectations (+1 if reported probability $\geq 50\%$, -1 otherwise). Panel B: indicator for switching direction between consecutive interviews (6 months apart). Panel C: individual uncertainty $U_{it} = p_{it}(1 - p_{it})$. Columns (1)–(3) use monthly frequency; (4)–(6) quarterly. All panels control for cumulative S&P 500 returns (1-month and 10-year); Panel B additionally controls for the inter-interview cumulative return. Robust standard errors in parentheses. Panel A results are robust to Newey–West robust standard errors; results in Panels B and C are robust to clustering standard errors by interview month. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

pling frequencies. A salient recent return is associated with lower disagreement when it aligns with the majority sign of recent history, and with higher disagreement when it runs against it.

These patterns are reduced-form and observational: trailing ambivalence may also correlate with macroeconomic volatility, news coverage, or other state variables that independently affect expectations, and the return controls absorb only part of this variation. We therefore read the MSC evidence as external validity for the experimentally identified comparative statics rather than as an independent test of the mechanism.

9 Discussion and Conclusion

Expectations are typically modeled as functions of an agent’s information set. Motivated by our preregistered evidence, this paper models them instead as functions of recalled information, which need not coincide with objective information, and identifies stochastic recall as a common source of belief disagreement, belief instability, and heterogeneity in subjective uncertainty. Once retrieval is stochastic, posterior consensus and belief stability are no longer implied by Bayesian learning, even when agents share common priors, observe the same information, store all signals, and update optimally on what they recall.

Relation to memory-based belief formation in economics Our model belongs to the memory-based belief formation literature (Mullainathan, 2002; Bordalo, Gennaioli and Shleifer, 2020a; Bordalo, Conlon, Gennaioli, Kwon and Shleifer, 2023; Gennaioli, Leva, Schoenle and Shleifer, 2024; Conlon and Kwon, 2025; Wachter and Kahana, 2024), whose central object is the probability that a stored signal is recalled. That literature studies how cues and similarity shift these probabilities and hence the level of beliefs, in effect averaging over many retrieval draws. We keep the same primitive and study the realized draws: because beliefs rest on a limited sample from memory, agents with identical memory databases recall different evidence, and the same agent recalls different evidence at different times. Disagreement, instability, and dispersion in subjective uncertainty are therefore distributional implications of this family of models.

Retrieval versus noisy processing An influential alternative locates the stochasticity in beliefs in noisy processing: the mind encodes information into an imprecise internal impression, and beliefs are an optimal but deterministic function of that impression (Khaw, Li and Woodford, 2021; Woodford, 2020; Azeredo da Silveira, Sung and Woodford, 2024). Under noisy processing

the stored record is corrupted and reading it is deterministic; under stochastic recall the record is intact and reading it is stochastic. Our evidence places an important part of the stochasticity (most of the within-group belief variance in our experiments) at retrieval: beliefs move with the measured content of recall, content-free cues shift beliefs by changing what is recalled, forgotten signals re-emerge, and confidence varies with the recalled sample. The distinction matters for intervention: if the record is intact, aids to retrieval alone can move beliefs and compress disagreement without conveying any new information, which is precisely what the effort and cue treatments do.

Composition of information and co-movement Disagreement, instability, and uncertainty are governed not only by the amount of information available but by its composition. Ambivalent histories amplify all three outcomes because stochastic recall more often generates divergent recalled samples, which explains their co-movement in the experimental and survey evidence.

Central bank communication The cue results suggest a channel through which central bank communication can shape household expectations: by altering which past realizations come to mind. Reference to inflation episodes that reinforce the dominant sign of recent history cues retrieval of similar past observations and compresses disagreement; reference to salient outliers contrary to the dominant pattern primes minority signals and can initially fragment expectations before, at higher cue strength, overturning the dominant pattern in recall. This complements existing accounts of how monetary policy communication moves expectations (Coibion, Gorodnichenko and Weber, 2022; Roth, Wiederholt and Wohlfart, 2023): alongside attention and salience, communication operates through which past data households retrieve.

Implications for measurement and information design Observed cross-sectional dispersion in expectations surveys (the Survey of Professional Forecasters, the Michigan Survey of Consumers, and household and firm expectations panels) need not reflect heterogeneous information sets across respondents; stochastic recall predicts that respondents drawing from a common information history form different beliefs simply by retrieving different stored signals. This channel offers a complementary reading of work that uses dispersion as a measure of information frictions (Mankiw, Reis and Wolfers, 2003; Coibion and Gorodnichenko, 2015) and points to retrieval-side interventions, such as simplified messages, repetition, and salient summary

statistics, as ways to compress dispersion without changing the information set.

Directions for future work Two extensions follow directly from the mechanism. First, the format in which information arrives should matter through the recall probability: Graeber, Roth and Zimmermann (2024) show that stories are retained better than statistics after a delay, and in our framework a higher recall probability compresses disagreement and instability. Whether the same facts generate different degrees of disagreement and instability when they arrive as stories rather than statistics is left for future work. Second, our re-elicitations are 24 hours apart. Re-eliciting beliefs and recall at several delays, for example after a few hours, one day, and several days, would trace how instability evolves with the interval between elicitations and how quickly a retrieval draw fades.

Concluding remarks Stochastic recall operates at a primitive stage of belief formation, the retrieval of stored signals, and combines naturally with heterogeneous information acquisition, rational inattention, and salience-based weighting.

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Online Appendix

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Appendix A Summaries of Studies and Experiments

Appendix Table A.1: OVERVIEW OF MAIN DATA COLLECTIONS

Study	Domain		Information and variation	Valid sample	Elicitations	Incentives		AEA RCT #
Field studies								
Study 1: Wave 1	Financial	mar- kets	Monthly stock returns (past 1 year)	CloudResearch Connect (606)	Recall stat.; esti- mate of mean re- turn	Estimate accu- racy		17773
	Wave 2	Financial markets	No new information; Sec- ond elicitation	CloudResearch Connect (500)	Recall stat.; esti- mate of mean re- turn	Estimate accu- racy + Partici- pation bonus		17773
Study 2: Wave 1	Macroeconomics		Annual inflation rates (past 10 years)	CloudResearch Connect (600)	Recall stat.; esti- mate of mean in- flation	Estimate accu- racy		17773
	Wave 2	Macroeconomics	No new information; Sec- ond elicitation	CloudResearch Connect (460)	Recall stat.; esti- mate of mean in- flation	Estimate accu- racy + Partici- pation bonus		17773
Study 3: Wave 1	Public policy		Tariff opinion headlines	CloudResearch Connect (600)	Recall +/- num- bers; opinion evaluation	Recall	accu- racy	17773
	Wave 2	Public policy	No new information; Sec- ond elicitation	CloudResearch Connect (522)	Recall +/- num- bers; opinion evaluation	Recall	accu- racy + Partici- pation bonus	17773
Study 4: Wave 1	Firm-level news		EV company news head- lines	CloudResearch Connect (611)	Recall +/- num- bers; news evalua- tion	Recall	accu- racy	17773
	Wave 2	Firm-level news	No new information; Sec- ond elicitation	CloudResearch Connect (522)	Recall +/- num- bers; news evalua- tion	Recall	accu- racy + Partici- pation bonus	17773
Experimental studies								
Experiment 1 (belief disagree- ment)	Asset evaluation		Binary signals; ambivalence of information; recall prob- ability; contextual cues	Prolific (4,758)	Recall; probabilis- tic and binary be- liefs	Judgment	ac- curacy	17150
Experiment 2 (Robustness; disagreement)	Asset evaluation		Binary signals; ambivalence of information; recall prob- ability	Prolific (2,039)	Recall; probabilis- tic and binary be- liefs	Judgment	ac- curacy	17671
Experiment 3 (belief instabil- ity)	Asset evaluation		No new information; Sec- ond elicitation	Prolific (1,744)	Recall; probabilis- tic and binary be- liefs	Judgment	accuracy + participation bonus	17671
Experiment 4 (AR(1) process; belief disagree- ment)	Statistic aggrega- tion		Statistical observation from AR(1) processes	CloudResearch Connect (2,301)	Recall stat.; esti- mate of quantita- tive mean	Judgment	ac- curacy	17805
Experiment 5 (AR(1) process; belief instabil- ity)	Statistic aggrega- tion		No new information; Sec- ond elicitation	CloudResearch Connect (1,867)	Recall stat.; esti- mate of mean	Judgment	accuracy + participation bonus	17805

Appendix B Additional Results for Field Studies 1 - 4

Appendix Table A.2: FIELD STUDIES 1 - 4 DEMOGRAPHIC CHARACTERISTICS

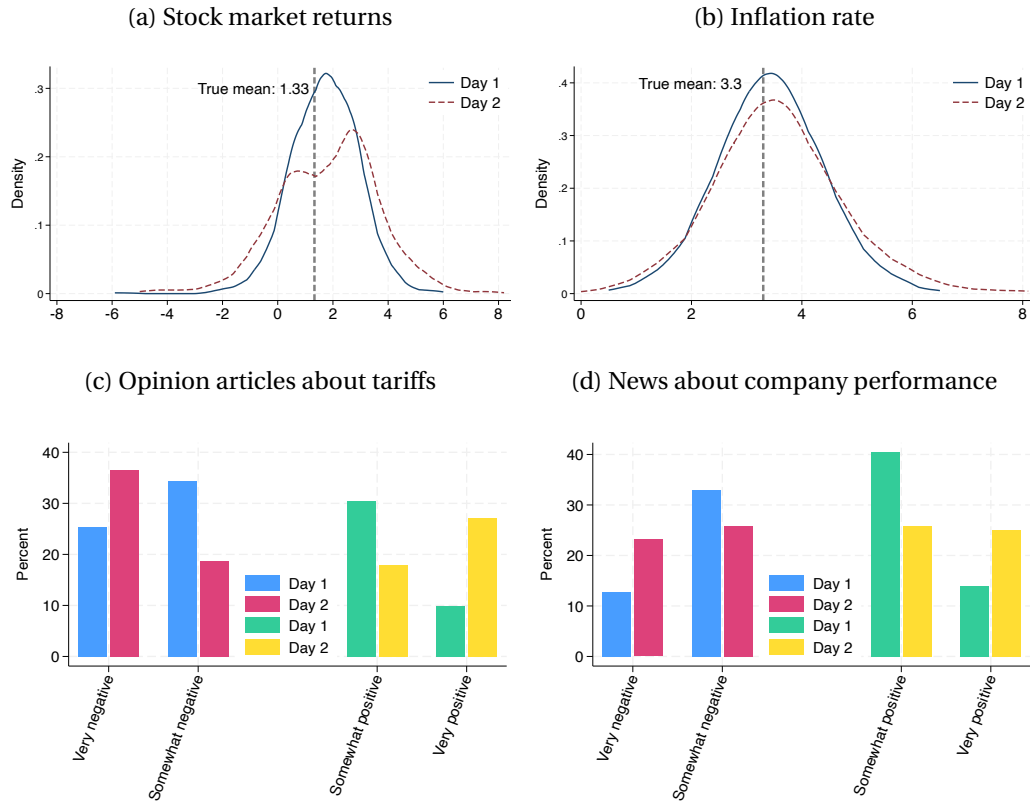
	Study 1 (Stock) (1)	Study 2 (Inflation) (2)	Study 3 (Tariff) (3)	Study 4 (EV Company) (4)
Observations	606	600	600	611
Female (%)	48.7	51.0	49.7	49.9
Age (years)	47.4 (18.8)	48.6 (18.1)	46.2 (18.0)	47.3 (18.8)
Bachelor's degree or higher (%)	29.7	28.8	32.7	29.8
Employed full-time (%)	44.2	47.2	44.3	44.0
Household income < \$60,000 (%)	57.1	57.0	57.0	57.0

Notes: The table reports baseline demographic and financial characteristics of participants in each of the four field studies. Entries are percentages for indicator variables and means for continuous variables, with standard deviations reported in parentheses. All studies were conducted using the same recruitment platform, eligibility criteria, and survey procedures. Participants were restricted to taking part in only one study.

Appendix Table A.3: ATTRITION IN THE TWO-WAVE STUDIES: DAY-2 PARTICIPATION AND DEMOGRAPHICS

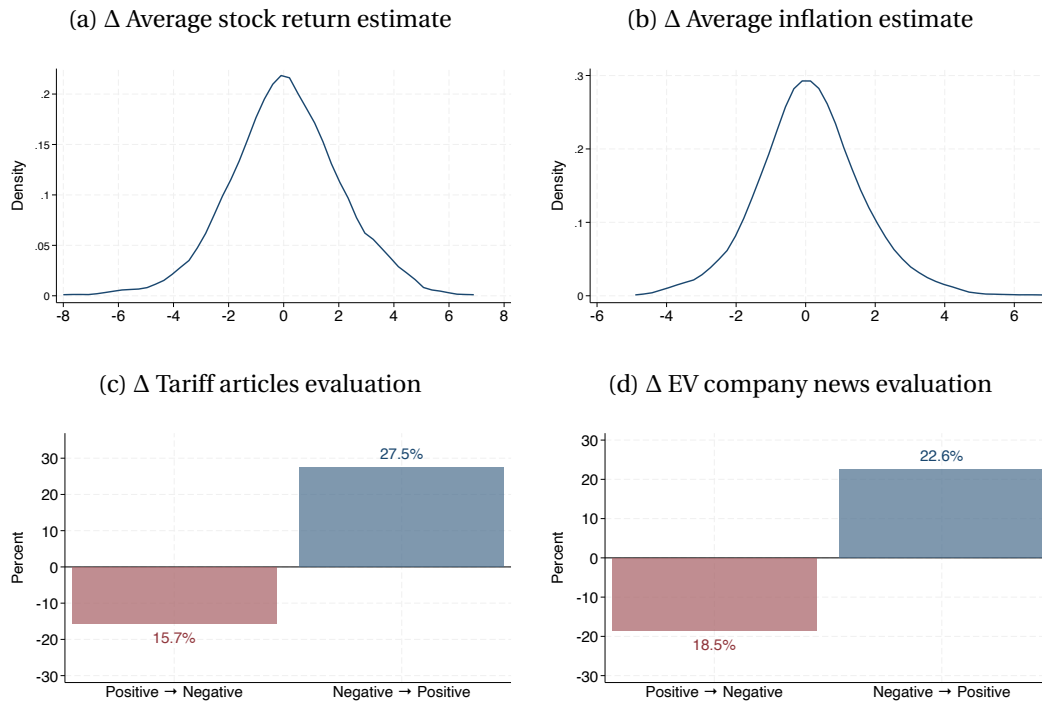
	Field studies				Experiments	
	(1) Study 1	(2) Study 2	(3) Study 3	(4) Study 4	(5) Exp. 3	(6) Exp. 5
Female	-0.020 (0.031)	0.029 (0.034)	0.007 (0.027)	0.005 (0.029)	-0.001 (0.016)	0.008 (0.016)
Age	-0.000 (0.001)	-0.001 (0.001)	-0.002* (0.001)	-0.000 (0.001)	-0.001** (0.001)	-0.000 (0.001)
Bachelor or above	0.024 (0.033)	0.048 (0.038)	0.048 (0.029)	0.022 (0.032)	0.010 (0.016)	0.050*** (0.017)
Full-time employed	0.011 (0.031)	0.016 (0.034)	-0.009 (0.029)	-0.037 (0.029)	-0.019 (0.016)	-0.015 (0.016)
Above-median income	-0.071** (0.032)	-0.027 (0.036)	-0.004 (0.030)	-0.055* (0.031)	-0.016 (0.016)	0.018 (0.017)
Stock investor	-0.042 (0.031)	0.039 (0.036)	-0.026 (0.029)	0.037 (0.032)	-0.006 (0.016)	0.002 (0.017)
Day-2 return rate	0.827	0.790	0.870	0.854	0.855	0.811
Joint <i>F</i> -statistic	1.257	0.813	1.045	1.049	1.310	2.024
Observations	606	600	600	611	2039	2301
R-squared	0.014	0.008	0.012	0.010	0.004	0.005

Notes: The table reports linear probability regressions of an indicator for Day-2 participation on Day-1 demographic characteristics, separately for every study with a second wave: the four motivating field studies (Study 1: stock returns; Study 2: inflation; Study 3: tariff headlines; Study 4: EV-company headlines), the Belief Instability Experiment (Experiment 3), and the AR(1) re-elicitation experiment (Experiment 5). Day-2 participation equals one if the respondent completed the Day-2 recall and estimate (or belief) elicitation. *Above-median income* indicates a household income above the within-study median bracket. The joint *F*-statistic row reports the test statistic for the null that all demographic coefficients are jointly zero. Robust standard errors are reported in parentheses. Statistical significance is denoted by *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$.



Appendix Figure A.1: DISTRIBUTIONS OF BELIEF AND OPINIONS ACROSS FOUR STUDIES

Notes: The figure plots the distribution of beliefs and opinion elicited immediately after information exposure (Day 1, solid lines or bars on the left hand side) and one day later (Day 2, dashed lines or bars on the right hand side), across the four field studies. The top-left panel shows the distribution of estimated mean monthly stock market rate of returns over the past 12 months; the top-right panel shows the distribution of estimated average rate of annual inflation over the past decade. The bottom-left panel shows evaluations of the sentiment of opinion articles on U.S. trade tariffs, and the bottom-right panel shows evaluations of news coverage about a major electric vehicle company. In all panels, Day 2 beliefs are elicited without any additional information.



Appendix Figure A.2: BELIEFS ARE UNSTABLE EVEN IN THE ABSENCE OF NEW INFORMATION

Notes: The figure plots the cross-sectional distribution of within-individual belief changes between Part 1 and Part 2 across the four field studies. Panels (a) and (b) show the distribution of changes in reported beliefs, defined as the difference between Part 2 and Part 1 beliefs. For stock returns, beliefs are the reported average monthly return over the past 12 months; for inflation, beliefs are the reported average annual inflation rate over the past 10 years. Panels (c) and (d) summarize discrete belief revisions for evaluative judgments, reporting the percentage of respondents who switch from positive (somewhat or very) to negative (somewhat or very) assessments, and vice versa, between elicitations.

Appendix Table A.4: CROSS-SECTIONAL RELATIONSHIP BETWEEN STATED BELIEF AND RECALLED INFORMATION

	(1) Day 1 elicitation	(2)	(3) Day 2 elicitation	(4)
Panel A (Study 1): Stock market return				
Average of recalled returns	0.895*** (0.016)	0.898*** (0.016)	0.942*** (0.017)	0.943*** (0.017)
Constant	0.170*** (0.031)	-0.401*** (0.109)	0.187*** (0.040)	-0.073 (0.165)
Demographics		✓		✓
Observations	605	605	501	501
R-squared	0.866	0.890	0.880	0.885
Panel B (Study 2): Inflation rate				
Average of recalled inflation	0.899*** (0.026)	0.903*** (0.027)	0.876*** (0.025)	0.877*** (0.026)
Constant	0.315*** (0.092)	0.230 (0.155)	0.431*** (0.092)	0.378** (0.161)
Demographics		✓		✓
Observations	583	583	474	474
R-squared	0.707	0.712	0.754	0.760
Panel C (Study 3): Tariff opinion article headlines				
Recalled positive share	2.599*** (0.118)	2.630*** (0.124)	1.933*** (0.151)	1.951*** (0.154)
Constant	-1.389*** (0.052)	-1.334*** (0.166)	-0.997*** (0.077)	-0.869*** (0.214)
Demographics		✓		✓
Observations	600	600	522	522
R-squared	0.274	0.291	0.164	0.179
Panel D (Study 4): Electric vehicle news headlines				
Recalled positive share	3.949*** (0.173)	4.211*** (0.179)	2.536*** (0.191)	2.663*** (0.198)
Constant	-2.015*** (0.088)	-1.729*** (0.187)	-1.303*** (0.105)	-1.254*** (0.230)
Demographics		✓		✓
Observations	611	611	522	522
R-squared	0.326	0.358	0.174	0.210

Notes: The table reports OLS estimates relating recalled information to contemporaneous belief or opinion measures across four studies. In Panel A, the dependent variable is respondents' estimate of the average monthly stock market return over the past 12 months. In Panel B, the dependent variable is respondents' estimate of the average annual inflation rate over the past 10 years. In Panel C, the dependent variable is a binary evaluation of the overall sentiment of opinion articles about a tariff policy, coded as +1 for positive and -1 for negative. Panel D, the dependent variable is a binary evaluation of the overall sentiment of news about an electric vehicle company, coded as +1 for positive and -1 for negative. Columns (1)–(2) use recall elicited on Day 1, while columns (3)–(4) use recall elicited on Day 2. Robust standard errors are reported in parentheses. Recall measures are expressed in percentage points. Statistical significance is denoted by *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$.

Appendix Table A.5: WITHIN-INDIVIDUAL RELATIONSHIP BETWEEN CHANGES IN BELIEF AND CHANGES IN RECALLED INFORMATION

	(1)	(2)
Panel A (Study 1): Stock market return		
Δ average of recalled returns	0.935*** (0.017)	0.933*** (0.018)
Constant	0.085*** (0.031)	0.336 (0.212)
Demographics		✓
Observations	500	500
R-squared	0.870	0.878
Panel B (Study 2): Inflation rate		
Δ average of recalled inflation	0.893*** (0.026)	0.894*** (0.027)
Constant	0.028 (0.033)	0.062 (0.165)
Demographics		✓
Observations	460	460
R-squared	0.756	0.763
Panel C (Study 3): Tariff opinion article headlines		
Δ recalled positive share	2.096*** (0.256)	2.054*** (0.262)
Constant	0.041 (0.052)	0.102 (0.267)
Demographics		✓
Observations	522	522
R-squared	0.105	0.121
Panel D (Study 4): Electric vehicle news headlines		
Δ recalled positive share	2.733*** (0.307)	2.714*** (0.319)
Constant	-0.037 (0.052)	-0.260 (0.291)
Demographics		✓
Observations	522	522
R-squared	0.116	0.139

Notes: The table reports OLS estimates relating changes in recalled information to changes in beliefs or opinions between Part 1 and Part 2 across four studies. In Panel A, the dependent variable is the change in respondents' stated beliefs about the average monthly stock market return over the past 12 months. In Panel B, the dependent variable is the change in respondents' stated beliefs about the average annual inflation rate over the past 10 years. In Panel C, the dependent variable is the change in respondents' evaluation of the overall sentiment of opinion articles about a tariff policy, coded as +1 for positive and −1 for negative. In Panel D, the dependent variable is the change in respondents' evaluation of the overall sentiment of news about an electric vehicle company, coded as +1 for positive and −1 for negative. Columns (1)–(2) use changes in recall measured between Part 1 and Part 2. Robust standard errors are reported in parentheses. Changes in recall are expressed in percentage points. Statistical significance is denoted by *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$.

Appendix C Proof of Lemma 1

As discussed in the text, when N^+ and N^- are large, X_i^+ and X_i^- can be approximated by normal distributions:

$$X_i^+ \sim \mathcal{N}(N^+ p, N^+ p(1-p))$$

$$X_i^- \sim \mathcal{N}(N^- p, N^- p(1-p))$$

Define $N = N^+ + N^-$. Let $Z_i = X_i^+ - X_i^-$, then $Z_i \sim \mathcal{N}((N^+ - N^-)p, p(1-p)N)$.

To find $P(X_i^+ < X_i^-) = P(Z_i < 0)$:

$$P(Z_i < 0) = P\left(Z_i^* < \frac{0 - (N^+ - N^-)p}{\sqrt{p(1-p)N}}\right) = P\left(Z_i^* < -\frac{(N^+ - N^-)p}{\sqrt{p(1-p)N}}\right)$$

Using the standard normal distribution function Φ , we get:

$$P(X_i^+ < X_i^-) = \Phi\left(-\frac{(N^+ - N^-)p}{\sqrt{p(1-p)N}}\right)$$

Because $\Phi(-x) = 1 - \Phi(x)$, this simplifies to:

$$P(X_i^+ < X_i^-) = 1 - \Phi\left(\frac{(N^+ - N^-)p}{\sqrt{p(1-p)N}}\right).$$

Similarly, we get that

$$q = P(X_i^+ > X_i^-) = \Phi\left(\frac{(N^+ - N^-)p}{\sqrt{p(1-p)N}}\right).$$

Note that $\hat{Y}_i = 1$ when $X_i^+ > X_i^-$. $\hat{Y}_i = -1$ when $X_i^+ \leq X_i^-$. Thus, the cross-sectional mean and variance of DMs' beliefs are

$$E(\hat{Y}_i) = 2q - 1 \tag{C.1}$$

and

$$Var(\hat{Y}_i) = 4q(1-q) \tag{C.2}$$

Remark. When N goes to infinity, q goes to 1 if $N^+ > N^-$ and to 0 if $N^+ < N^-$. In either case, disagreement $4q(1-q)$ goes to zero.

Appendix D Proof of Lemma 2

Recall that the posterior probability that the asset is good is

$$\mu_i = \left[1 + \left(\frac{1-b}{b}\right)^{X_i^+ - X_i^-}\right]^{-1}, \quad b > \frac{1}{2}.$$

Define recalled net evidence as

$$D_i \equiv X_i^+ - X_i^-.$$

Then

$$\mu_i = \left[1 + \left(\frac{1-b}{b}\right)^{D_i}\right]^{-1}.$$

Let

$$\alpha \equiv \log\left(\frac{b}{1-b}\right) > 0.$$

Using

$$\left(\frac{1-b}{b}\right)^{D_i} = e^{-\alpha D_i},$$

we can rewrite the posterior as

$$\mu_i = \frac{1}{1 + e^{-\alpha D_i}}.$$

Subjective uncertainty is defined as

$$U_i = \mu_i(1 - \mu_i).$$

Substituting the expression for μ_i ,

$$U_i = \frac{1}{1 + e^{-\alpha D_i}} \left(1 - \frac{1}{1 + e^{-\alpha D_i}}\right) = \frac{e^{-\alpha D_i}}{(1 + e^{-\alpha D_i})^2}.$$

Multiplying numerator and denominator by $e^{\alpha D_i}$,

$$U_i = \frac{1}{e^{\alpha D_i} + 2 + e^{-\alpha D_i}}.$$

Using the identity

$$\cosh(x) = \frac{e^x + e^{-x}}{2},$$

this becomes

$$U_i = \frac{1}{2(1 + \cosh(\alpha D_i))}.$$

This expression implies two properties immediately.

First, U_i depends only on $|D_i|$, because $\cosh(\alpha D_i)$ is an even function:

$$\cosh(\alpha D_i) = \cosh(\alpha |D_i|).$$

Second, U_i is strictly decreasing in $|D_i|$. To see this, define

$$g(x) \equiv \frac{1}{2(1 + \cosh(\alpha x))}, \quad x \geq 0.$$

Then

$$U_i = g(|D_i|).$$

Differentiating,

$$g'(x) = -\frac{\alpha \sinh(\alpha x)}{2(1 + \cosh(\alpha x))^2}.$$

Since $\alpha > 0$ and $\sinh(\alpha x) > 0$ for all $x > 0$, it follows that

$$g'(x) < 0 \quad \text{for all } x > 0.$$

Hence U_i is strictly decreasing in $|D_i|$.

Therefore, subjective uncertainty is maximized when $|D_i| = 0$, i.e. when $X_i^+ = X_i^-$, and decreases as the absolute recalled net evidence $|X_i^+ - X_i^-|$ increases. This proves Lemma 2.

Appendix E Proofs for Disagreement (Predictions 1 and 2)

We firstly prove the first part of the proposition. DMs' disagreement about the asset value $Var(\hat{Y}_i)$ is an increasing function of q when $q \leq \frac{1}{2}$ and a decreasing function of q when $q \geq \frac{1}{2}$.

Recall the degree of ambivalence is

$$A = 1 - \left| \frac{N^+ - N^-}{N} \right| \quad (\text{E.1})$$

If $N^+ < N^-$, $A = 1 + \frac{N^+ - N^-}{N}$, so A is increasing in $\frac{N^+ - N^-}{N}$. In this case, $q < \frac{1}{2}$ and q is increasing in $\frac{N^+ - N^-}{N}$. Thus, we have $Var(\hat{Y}_i)$ increasing in A .

If $N^+ > N^-$, $A = 1 - \frac{N^+ - N^-}{N}$, so A is decreasing in $\frac{N^+ - N^-}{N}$. In this case, $q > \frac{1}{2}$ and q is decreasing in $\frac{N^+ - N^-}{N}$. Thus, we have $Var(\hat{Y}_i)$ increasing in A .

Taken together, we have proved that the DM's belief disagreement $Var(\hat{Y}_i)$ is an increasing function of the degree of ambivalence A .

Now we prove the second part. For $N^+ > N^-$, we have $q \geq \frac{1}{2}$ and q is increasing in p . Furthermore, $Var(\hat{Y}_i)$ is decreasing in q when $q \geq \frac{1}{2}$. Thus, we have $Var(\hat{Y}_i)$ decreasing in p . For $N^+ < N^-$, we have $q \leq \frac{1}{2}$ and q is decreasing in p . Furthermore, $Var(\hat{Y}_i)$ is increasing in q when $q \leq \frac{1}{2}$. Again, we have $Var(\hat{Y}_i)$ decreasing in p . Taking the two cases together, we have $Var(\hat{Y}_i)$ decreasing in p .

Appendix F Proofs for Belief Instability (Predictions 1 and 2)

To characterize the probability $\Pi(N^+, N^-, p)$, we use the same large-sample approximation as in [Appendix C](#). Let

$$Z_\tau = X_{i,\tau}^+ - X_{i,\tau}^-.$$

As shown in [Appendix C](#), when N^+ and N^- are large and p is bounded away from zero and one,

$$Z_\tau \sim \mathcal{N}((N^+ - N^-)p, p(1 - p)N).$$

Since $\hat{Y}_\tau = 1$ if and only if $Z_\tau > 0$, it follows that the probability of a positive belief at any date τ is

$$q = \Pr(\hat{Y}_\tau = 1) = \Phi(c(N^+, N^-, p)), \quad c(N^+, N^-, p) = \frac{(N^+ - N^-)p}{\sqrt{p(1 - p)N}},$$

where $\Phi(\cdot)$ denotes the standard normal cumulative distribution function. Because recall realizations are independent across elicitation dates, $\hat{Y}_t$ and $\hat{Y}_{t+1}$ are i.i.d. Bernoulli random variables with success probability q . Therefore,

$$\Pi(N^+, N^-, p) = \Pr(\hat{Y}_t \neq \hat{Y}_{t+1}) = 2q(1 - q) = 2\Phi(c)(1 - \Phi(c)).$$

Recall the definition of informational ambivalence as

$$A = 1 - \frac{|N^+ - N^-|}{N}.$$

The result can be proved by following the argument in [Appendix E](#).

Appendix G Proof for Subjective Uncertainty (Prediction 1)

Under the normal approximation used in Lemma 1,

$$D_i = X_i^+ - X_i^- \sim \mathcal{N}(m, \sigma^2),$$

where

$$m = (N^+ - N^-)p, \quad \sigma^2 = Np(1-p).$$

By Lemma 2, subjective uncertainty is

$$U_i = h(D_i), \quad h(x) \equiv \frac{1}{2(1 + \cosh(\alpha x))},$$

where h is an even function and strictly decreasing in $|x|$.

Hence expected subjective uncertainty is approximately

$$E[U_i] \approx F(m) \equiv E[h(m + \sigma Z)], \quad Z \sim \mathcal{N}(0, 1).$$

Because h is even, $F(m) = F(-m)$. It therefore suffices to show that $F(m)$ is decreasing in $|m|$.

For $m > 0$, differentiating under the integral sign yields

$$F'(m) = \int_{-\infty}^{\infty} h'(x) \phi_{\sigma}(x - m) dx,$$

where $\phi_{\sigma}(\cdot)$ denotes the density of $\mathcal{N}(0, \sigma^2)$. Splitting the integral at zero and using that h' is odd,

$$F'(m) = \int_0^{\infty} h'(x) [\phi_{\sigma}(x - m) - \phi_{\sigma}(x + m)] dx.$$

For every $x > 0$ and $m > 0$,

$$\phi_{\sigma}(x - m) > \phi_{\sigma}(x + m),$$

while $h'(x) < 0$ for all $x > 0$. Therefore every integrand is negative, implying

$$F'(m) < 0 \quad \text{for all } m > 0.$$

Thus $F(m)$ is decreasing in $|m|$.

Finally, holding N and p fixed,

$$|m| = p|N^+ - N^-| = pN(1 - A),$$

which is decreasing in A . Since $E[U_i] \approx F(m)$ and F is decreasing in $|m|$, it follows that expected subjective uncertainty is increasing in informational ambivalence A .

Appendix H Proof for Contextual Cues (Prediction 3)

Throughout, the no-cue case is $a = 0$.

Setup Without loss of generality, let the cue raise the recall probability of positive signals from p to $\tilde{p} = p + a$, $a \in (0, 1 - p]$ (a cue on negative signals follows by relabeling); the cue is on the majority type if $N^+ \geq N^-$ and on the minority type if $N^+ < N^-$. Under the normal approximation of Lemma 1, recalled net evidence $D_i = X_i^+ - X_i^-$ has mean $v(a) = N^+(p + a) - N^-p$ and variance $\sigma^2(a) = N^+(p + a)(1 - p - a) + N^-p(1 - p)$, so the DM believes $\hat{Y}_i = 1$ with probability $\Phi(m(a))$,

where $m(a) = v(a)/\sigma(a)$, and

$$\text{Var}(\hat{Y}_i) = 4\Phi(m(a))(1 - \Phi(m(a))) \equiv V(a),$$

which is strictly decreasing in $|m(a)|$. Differentiating,

$$m'(a) = \frac{N^+ s(\tilde{p})}{\sigma^3(a)}, \quad s(\tilde{p}) = \tilde{p}\left(\frac{1}{2}N^+ - N^- p\right) + N^- p\left(\frac{3}{2} - p\right),$$

where s is affine in $\tilde{p} \in [p, 1]$, so $m'(a) > 0$ whenever $s(p) > 0$ and $s(1) > 0$. For subjective uncertainty, Lemma 2 shows that U_i is strictly decreasing in $|D_i|$; under the mean-based approximation, $E[U_i]$ is strictly decreasing in $|v(a)|$.

Item 1: cue on the majority type Let $N^+ \geq N^-$ and $d \equiv N^+ - N^- \geq 0$. Then

$$s(p) = \frac{p}{2}(4N^-(1-p) + d) > 0, \quad s(1) = \frac{N^-}{2}(1-p)(1+2p) + \frac{d}{2} > 0,$$

so $m'(a) > 0$. Since $m(0) \geq 0$, $|m(a)|$ is strictly increasing and $V(a)$ strictly decreasing in a ; in particular, $V(a) < V(0)$. Likewise, $v(a) = dp + N^+ a > 0$ is strictly increasing in a , so $E[U_i]$ is strictly decreasing in a and lies below its no-cue level.

Item 2: cue on the minority type Let $N^+ < N^-$, so $m(0) < 0$.

Comparison with the no-cue level. Direct computation gives

$$m(a)^2 - m(0)^2 = \frac{a(a - \Delta) N^+ p N^2 [\lambda(1-p) + (2\lambda - 1)^2 p]}{\sigma^2(a) \sigma^2(0)},$$

where

$$\Delta = \frac{(2\lambda - 1)^2 p(1 - 2p) - 2(2\lambda - 1)p(1 - p)}{\lambda(1 - p) + (2\lambda - 1)^2 p}, \quad \lambda = \frac{N^+}{N^-}.$$

The bracketed term is strictly positive, so for $a > 0$ the sign of $m(a)^2 - m(0)^2$ is the sign of $a - \Delta$: disagreement exceeds its no-cue level for $a < \Delta$, equals it at $a = \Delta$, and falls below it for $a > \Delta$.

Single-peakedness of disagreement. The mean $v(a) = v(0) + N^+ a$, with $v(0) < 0$, is strictly increasing in a and vanishes at

$$a = \Theta \equiv \frac{(N^- - N^+)p}{N^+} = \frac{|N^+ - N^-|p}{N^m}.$$

Let $\Theta \leq 1 - p$, equivalently $N^- p \leq N^+$. Then

$$s(p) = \frac{p}{2}(N^+ + N^-(3 - 4p)) > 0, \quad s(1) = \frac{1}{2}(N^+ + N^- p(1 - 2p)) > 0$$

(the brackets are positive for $p \leq \frac{3}{4}$ and $p \leq \frac{1}{2}$, respectively; for larger p , $N^- \leq N^+/p$ gives $s(p) \geq \frac{3}{2}N^+(1-p) > 0$ and $s(1) \geq N^+(1-p) > 0$), so $m'(a) > 0$. Hence $m(a) < 0$ on $(0, \Theta)$, $m(\Theta) = 0$, and $m(a) > 0$ on $(\Theta, 1 - p]$: $|m(a)|$ is strictly decreasing and then strictly increasing, so $V(a)$ is strictly increasing for $a < \Theta$, maximal at $a = \Theta$, and strictly decreasing for $a > \Theta$. Because $V(\Theta) = 1 > V(0)$, the no-cue level is regained only on the decreasing branch, so $\Theta < \Delta$. If instead $\Theta > 1 - p$, the peak is not feasible and disagreement is increasing throughout the feasible range. Let $q(a) = P(X_i^+ \geq X_i^-)$, so that $V(a) = 4q(a)(1 - q(a))$. Because X_i^+ is binomial with success probability $p + a$, which increases in a in the sense of first-order stochastic dominance, while X_i^- does not depend on a , $q(a)$ is nondecreasing in a (this step uses no approximation). Because

$v(a) < 0$ for all feasible a , $q(a) = \Phi(m(a)) < \frac{1}{2}$ throughout. Since $4q(1 - q)$ is increasing in q on $(0, \frac{1}{2})$, $V(a)$ is increasing in a .

Expected subjective uncertainty. When $\Theta \leq 1 - p$, $|v(a)|$ is strictly decreasing on $(0, \Theta)$ and strictly increasing thereafter, so under the mean-based approximation $E[U_i]$ is strictly increasing for $a < \Theta$, maximal at $a = \Theta$, and strictly decreasing for $a > \Theta$. If instead $\Theta > 1 - p$, then $v(a) < 0$ and $|v(a)|$ is strictly decreasing on the whole feasible range, so $E[U_i]$ is increasing in a throughout. This establishes Prediction 3.

Appendix I Posterior Consensus, Uncertainty, and Stability under Imperfect Recall

We summarize the implications of imperfect recall for three benchmark properties associated with shared information: posterior consensus (Aumann, 1976; Geanakoplos and Polemarchakis, 1982), common subjective uncertainty, and belief stability. These properties underlie many canonical models of expectation formation (Mankiw, Reis and Wolfers, 2003; Sims, 2003; Coibion and Gorodnichenko, 2015).

Theorem 1 (Posterior Consensus, Subjective Uncertainty, and Stability under Imperfect Recall). *Suppose DMs share a common prior, observe and store the same finite signal history containing at least one positive and one negative signal, and recall each signal independently with probability $p \in (0, 1)$ at each belief-formation date. DMs update optimally based on recalled information. Then:*

1. *with strictly positive probability, realized binary beliefs disagree across agents;*
2. *with strictly positive probability, subjective uncertainty differs across agents;*
3. *with strictly positive probability, a given agent's realized binary belief differs across two elicitations with no new information.*

Moreover, if the stored history has a strict majority, disagreement, uncertainty heterogeneity, and instability all converge to zero as $p \rightarrow 1$, and are identically zero when $p = 1$.

Proof intuition With both positive and negative signals present, any recall draw has strictly positive probability of yielding different realized balances of recalled positive and negative signals. Because both realized binary beliefs and subjective uncertainty depend on recalled net evidence, independent recall across agents generates disagreement and uncertainty heterogeneity; fresh recall draws across time generate belief revisions even without new information. Perfect recall is the only case in which all agents retrieve the same signal history, so realized binary beliefs coincide, subjective uncertainty is identical, and beliefs are stable over time.

Appendix J The Quantitative Model: Setup and Proofs

Environment Let $\{Z_t\}_{t \geq 1}$ follow the stationary AR(1) process

$$Z_t - \mu = \rho(Z_{t-1} - \mu) + \varepsilon_t, \quad (\text{J.1})$$

where $|\rho| < 1$ and $\varepsilon_t \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ are i.i.d. shocks, so that $\text{Var}(Z_t) = \sigma_\varepsilon^2 / (1 - \rho^2)$. The long-run mean μ is unknown. A continuum of DMs observes the same realized history $\{Z_1, \dots, Z_T\}$ but has imperfect memory; each DM must form a belief about the mean of the realizations.

Belief formation At the recall stage, each DM recalls each realization Z_k independently with probability $p \in (0, 1)$. Let $I_i^k \in \{0, 1\}$ indicate whether DM i recalls observation Z_k . At the information-processing stage, DM i forms a belief about the mean by averaging recalled observations:

$$\mu_i = \frac{\sum_{k=1}^T I_i^k Z_k}{\sum_{k=1}^T I_i^k}.^{51} \quad (\text{J.2})$$

Cross-sectional disagreement Disagreement is defined as the cross-sectional variance of beliefs, $\mathcal{D} \equiv \text{Var}(\mu_i)$.

Proposition A.1. Suppose $\{Z_t\}$ follows (J.1) and DMs recall each observation independently with probability $p \in (0, 1)$. Then, for large T ,

$$\mathcal{D} \approx \frac{1-p}{Tp} \text{Var}(Z_t) = \frac{1-p}{Tp} \cdot \frac{\sigma_\varepsilon^2}{1-\rho^2}. \quad (\text{J.3})$$

Equivalently, conditional on the realized history,

$$\text{Var}(\mu_i | Z_1, \dots, Z_T) \approx \frac{1-p}{Tp} \widehat{\text{Var}}(Z_1, \dots, Z_T),$$

where $\widehat{\text{Var}}$ denotes the sample variance.

Proof of Proposition A.1 Fix a realized history $Z = (Z_1, \dots, Z_T)$. All randomness below is taken with respect to DMs' recall indicators $\{I_i^k\}_{k=1}^T$, which are independent across k , satisfy

$$\mathbb{E}[I_i^k] = p, \quad \text{Var}(I_i^k) = p(1-p),$$

and are independent of Z .

For DM i , define

$$S_i = \sum_{k=1}^T I_i^k Z_k, \quad N_i = \sum_{k=1}^T I_i^k, \quad \mu_i = \frac{S_i}{N_i}.$$

Let

$$\bar{Z} = \frac{1}{T} \sum_{k=1}^T Z_k, \quad \delta_k = Z_k - \bar{Z}.$$

⁵¹We ignore the case in which no number is recalled, which occurs with probability $(1-p)^T$ and is negligible for large T .

Using $\sum_{k=1}^T p \delta_k = 0$,

$$\mu_i - \bar{Z} = \frac{\sum_{k=1}^T I_i^k \delta_k}{N_i} = \frac{\sum_{k=1}^T (I_i^k - p) \delta_k}{N_i}.$$

Write

$$N_i = Tp + \Delta_{N,i}, \quad \Delta_{N,i} = \sum_{k=1}^T (I_i^k - p),$$

where $\Delta_{N,i} = O_p(\sqrt{T})$. Using the Taylor expansion

$$\frac{1}{Tp + \Delta_{N,i}} = \frac{1}{Tp} - \frac{\Delta_{N,i}}{(Tp)^2} + O_p(T^{-2}),$$

and defining

$$\Delta_{S,i} = \sum_{k=1}^T (I_i^k - p) \delta_k,$$

we obtain

$$\mu_i - \bar{Z} = \frac{1}{Tp} \Delta_{S,i} - \frac{1}{(Tp)^2} \Delta_{N,i} \Delta_{S,i} + O_p(T^{-3/2}).$$

Conditional on Z , $\Delta_{S,i}$ is a sum of independent mean-zero terms, so

$$\text{Var}(\Delta_{S,i} | Z) = p(1-p) \sum_{k=1}^T \delta_k^2.$$

The remaining terms contribute $o(T^{-1})$ to the variance. Hence,

$$\text{Var}(\mu_i | Z) \approx \frac{1}{(Tp)^2} \text{Var}(\Delta_{S,i} | Z) = \frac{1-p}{Tp} \widehat{\text{Var}}(Z_1, \dots, Z_T).$$

Taking expectations over histories yields

$$\mathcal{D} \approx \frac{1-p}{Tp} \text{Var}(Z_t) = \frac{1-p}{Tp} \cdot \frac{\sigma_\varepsilon^2}{1-\rho^2}.$$

Within-individual instability Fix a realized history $\{Z_1, \dots, Z_T\}$ and consider a DM whose belief is elicited twice, with no new information arriving in between. At each elicitation $\tau \in \{1, 2\}$, the DM draws a fresh recall realization $\{I_{i,\tau}^k\}_{k=1}^T$, where $\{I_{i,\tau}^k\}_{k=1}^T \sim \text{Bernoulli}(p)$ independently across k and τ ,⁵² and reports

$$\hat{\mu}_{i,\tau} = \frac{\sum_{k=1}^T I_{i,\tau}^k Z_k}{\sum_{k=1}^T I_{i,\tau}^k}.$$

We measure belief instability as the expected squared revision,

$$\mathcal{J}(Z_1, \dots, Z_T) \equiv \mathbb{E} \left[(\hat{\mu}_{i,1} - \hat{\mu}_{i,2})^2 \mid Z_1, \dots, Z_T \right].$$

Proposition A.2. *Suppose $\{Z_t\}$ follows (J.1) and recall is independent across observations and elicitation with probability $p \in (0, 1)$. Then, for large T ,*

$$\mathcal{J}(Z_1, \dots, Z_T) \approx \frac{2(1-p)}{Tp} \widehat{\text{Var}}(Z_1, \dots, Z_T). \quad (\text{J.4})$$

⁵²Independence across elicitation is again a benchmark. Under correlated recall (each observation's second retrieval repeats the first with probability κ and is drawn fresh otherwise), all single-elicitation results, including Proposition A.1 and the estimation of Section 6, are unchanged, while instability generalizes to $\mathcal{J} \approx 2(1-\kappa) \frac{1-p}{Tp} \widehat{\text{Var}}(Z_1, \dots, Z_T)$ with $\text{Corr}(\hat{\mu}_{i,1}, \hat{\mu}_{i,2} \mid Z_1, \dots, Z_T) \approx \kappa$: instability stays proportional to the realized variance of the history, with the same comparative statics, for all $\kappa < 1$. See the Supplementary Appendix.

Unconditionally,

$$\mathbb{E}[(\hat{\mu}_{i,1} - \hat{\mu}_{i,2})^2] \approx \frac{2(1-p)}{Tp} \text{Var}(Z_t) = \frac{2(1-p)}{Tp} \cdot \frac{\sigma_\varepsilon^2}{1-\rho^2}. \quad (\text{J.5})$$

Proof of Proposition A.2 Fix a realized history $Z = (Z_1, \dots, Z_T)$. Consider a single DM i whose belief is elicited twice, at $\tau \in \{1, 2\}$, with no new information arriving between elicitations. At each elicitation, the DM draws an independent recall realization $\{I_{i,\tau}^k\}_{k=1}^T$, where $\{I_{i,\tau}^k\}_{k=1}^T \sim \text{Bernoulli}(p)$ independently across k and τ .

Define

$$S_{i,\tau} = \sum_{k=1}^T I_{i,\tau}^k Z_k, \quad N_{i,\tau} = \sum_{k=1}^T I_{i,\tau}^k, \quad \hat{\mu}_{i,\tau} = \frac{S_{i,\tau}}{N_{i,\tau}}.$$

We study

$$\mathcal{J}(Z_1, \dots, Z_T) = \mathbb{E}[(\hat{\mu}_{i,1} - \hat{\mu}_{i,2})^2 \mid Z_1, \dots, Z_T].$$

Conditional on Z , $\hat{\mu}_{i,1}$ and $\hat{\mu}_{i,2}$ are i.i.d., hence

$$\mathcal{J}(Z_1, \dots, Z_T) = 2 \text{Var}(\hat{\mu}_{i,1} \mid Z_1, \dots, Z_T).$$

From Proposition A.1,

$$\text{Var}(\hat{\mu}_{i,1} \mid Z_1, \dots, Z_T) \approx \frac{1-p}{Tp} \widehat{\text{Var}}(Z_1, \dots, Z_T).$$

Therefore,

$$\mathcal{J}(Z_1, \dots, Z_T) \approx \frac{2(1-p)}{Tp} \widehat{\text{Var}}(Z_1, \dots, Z_T),$$

which is equation (J.4).

Taking expectations over histories,

$$\mathbb{E}[(\hat{\mu}_{i,1} - \hat{\mu}_{i,2})^2] \approx \frac{2(1-p)}{Tp} \text{Var}(Z_t) = \frac{2(1-p)}{Tp} \cdot \frac{\sigma_\varepsilon^2}{1-\rho^2},$$

which is equation (J.5).

Remark. The Supplementary Appendix extends the analysis to one-step-ahead forecasts in the AR(1) environment and shows that stochastic retrieval generates forecast disagreement through both heterogeneous recall of the most recent realization and dispersion in beliefs about the mean.

Appendix K Quantitative Estimation: Simulation Details

Baseline model and simulation The baseline model features a recall probability p governing Bernoulli sampling of each stored signal. Given a treatment group g with N_g^+ positive and N_g^- negative signals, the model implies recalled counts (X_{ig}^+, X_{ig}^-) and an individual binary belief $\hat{Y}_{ig}$ determined by equation (3.1).

For any candidate parameter vector, we compute model moments by Monte Carlo simulation. Specifically, we draw $R = 500$ realizations of recalled positive and negative signals from independent binomial distributions, $X_{ig}^+ \sim \text{Bin}(N_g^+, p_g^+)$, $X_{ig}^- \sim \text{Bin}(N_g^-, p_g^-)$, construct simulated

beliefs $\hat{Y}_{ig} = 1\{X_{ig}^+ \geq X_{ig}^-\} - 1\{X_{ig}^+ < X_{ig}^-\}$, and compute the implied mean belief $M_g^{\text{mod}} = \mathbb{E}(\hat{Y}_{ig})$, which we match to the corresponding experimental moment M_g^{exp} .

Informational ambivalence (Groups 1–5) Groups 1–5 form the baseline environment, in which recall is governed by a single common probability p .⁵³ We estimate p by minimizing

$$\sum_{g=1}^5 (M_g^{\text{mod}}(p) - M_g^{\text{exp}})^2.$$

Retrieval effort (Groups 6 and 7) Groups 6 and 7 share the same signal mix as Group 3 but introduce retrieval-effort interventions. We estimate group-specific recall probabilities, p_6 and p_7 , by matching the model-implied mean belief in each group to its experimental counterpart. These estimates capture how the retrieval interventions shift effective recall probabilities relative to the Group 3 baseline.

Contextual cues (Groups 8 and 9) Groups 8 and 9 share the Group 2 signal mix but introduce contextual cues that distort recall asymmetrically. In Group 8, the positive cue raises the recall probability of positive signals only,

$$p_8^+ = \hat{p} + a^+, \quad p_8^- = \hat{p},$$

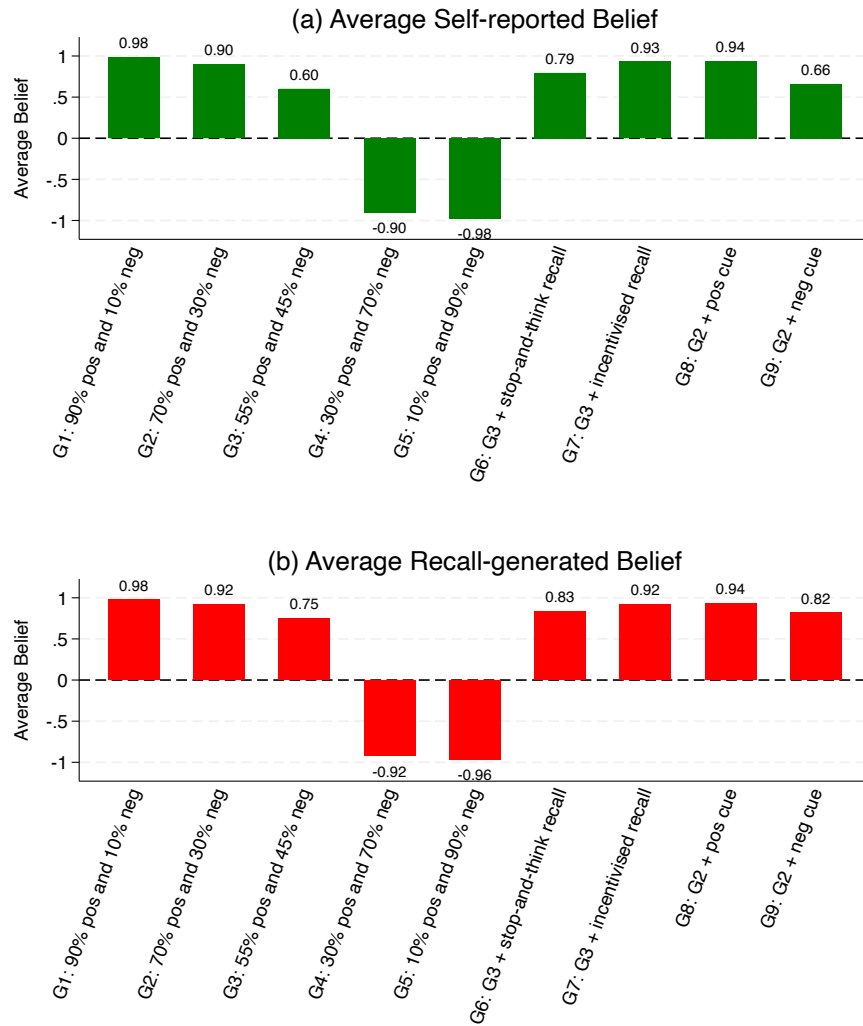
while in Group 9 the negative cue raises the recall probability of negative signals only,

$$p_9^+ = \hat{p}, \quad p_9^- = \hat{p} + a^-,$$

with $a^+, a^- \geq 0$. Holding the baseline recall probability fixed at $\hat{p}$, we estimate a^+ and a^- by matching model-implied and experimental mean beliefs in the two cue treatments.

⁵³This restriction is supported by the mechanism evidence in Section 4.3, which shows no significant differences in recall accuracy across Groups 1–5.

Appendix L Additional Belief Disagreement Experiment Results



Appendix Figure A.3: AVERAGE BELIEF ABOUT THE COMPANY STATE BY GROUP (REPORTED AND RECALL-GENERATED BELIEFS)

Notes: The figure plots the average binary belief across participants in each of the nine experimental groups. In the top panel (self-reported beliefs), beliefs take value 1 if the respondent reports that the company is “good” and -1 if the respondent reports that it is “bad”; group-level averages therefore summarize the net balance of positive versus negative beliefs within each treatment. In the bottom panel (recall-generated beliefs), beliefs are coded as 1 if the number of recalled positive signals is weakly greater than the number of recalled negative signals, and -1 otherwise.

Appendix Table A.6: BELIEF DISAGREEMENT EXPERIMENT DEMOGRAPHIC CHARACTERISTICS

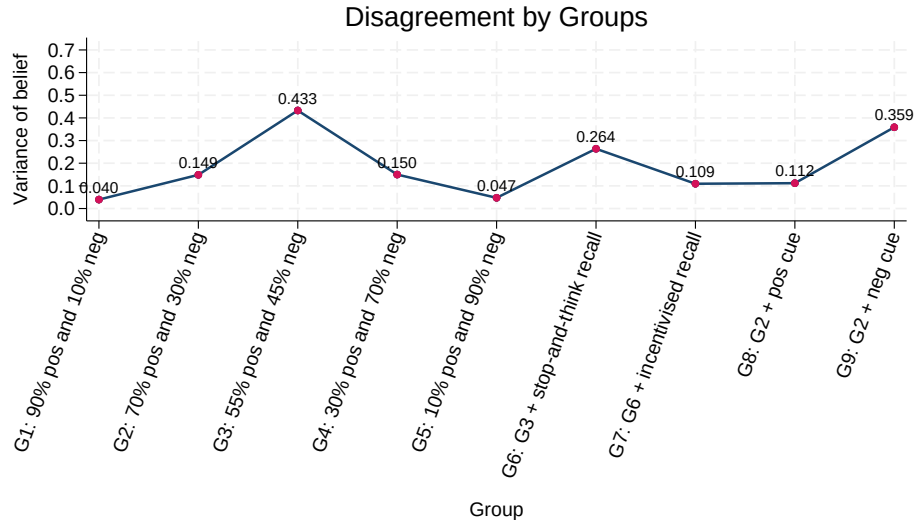
	Full sample (1)	T1 (2)	T2 (3)	T3 (4)	T4 (5)	T5 (6)	T6 (7)	T7 (8)	T8 (9)	T9 (10)	<i>p</i> -value (11)
Observations	4,758	503	554	526	571	505	506	544	525	524	
Panel A: Personal background											
Female (%)	59.06	56.86	57.22	62.74	56.22	59.41	61.07	60.29	60.38	57.63	0.37
Age (years)	44.83 (14.11)	44.16 (14.13)	44.88 (14.21)	44.19 (13.67)	44.05 (12.95)	43.26 (14.22)	45.90 (13.88)	45.67 (14.45)	44.78 (14.73)	46.58 (14.54)	0.40
Above Bachelor's degree (%)	62.61	66.60	63.00	65.59	61.65	61.39	58.30	60.85	63.43	62.79	0.21
Employed full-time (%)	55.82	57.65	54.51	53.42	55.69	58.22	56.72	52.39	57.33	56.87	0.54
Household income < \$60,000 (%)	56.89	53.88	58.30	56.84	56.74	57.03	58.30	56.07	61.14	53.63	0.39
Panel B: Financial characteristics											
Homeowner (%)	60.30	62.82	61.37	59.70	60.60	60.79	59.68	60.11	58.48	59.16	0.94
Stock investor (%)	56.56	55.27	59.75	57.22	57.09	58.42	57.31	55.15	53.71	54.96	0.61

Notes: This table reports baseline demographic characteristics by experimental treatment group. Entries are percentages for indicator variables and means for continuous variables, with standard deviations reported in parentheses. Treatment groups T1–T9 correspond to different information treatments described in the text. The final column reports *p*-values from an ANOVA test of equality of means across treatment groups.

Appendix Table A.7: STOCHASTIC RECALLS OF NEWS AND BELIEF FORMATION

	(1)	(2)	(3)	(4)	(5)	(6)
%Positive news recalled	0.037*** (0.000)	0.037*** (0.000)				
$1\{\text{Recall}_i^+ > \text{Recall}_i^-\}$			1.662*** (0.019)	1.661*** (0.019)		
Recall balance (sign)					0.941*** (0.005)	0.941*** (0.005)
Demographics		✓		✓		✓
Observations	4758	4758	4758	4758	4758	4758
R-squared	0.642	0.645	0.739	0.740	0.842	0.843

Notes: The dependent variable is the participant's self-reported binary belief about the company state, coded as 1 for "Good" and -1 for "Bad." %Positive news recalled is the share of recalled signals that are positive. $1\{\text{Recall}_i^+ > \text{Recall}_i^-\}$ is an indicator equal to one if the number of recalled positive signals is greater than the number of recalled negative signals, and zero otherwise. Recall balance (sign) is defined as $\text{sgn}(\text{Recall}_i^+ - \text{Recall}_i^-)$ and takes values 1, 0, and -1 when recalled positives exceed, equal, or fall short of recalled negatives, respectively. Robust standard errors are in parentheses. Statistical significance is denoted by *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$.

**Appendix Figure A.4: DISAGREEMENT ABOUT COMPANY TYPE BY GROUP (CONSISTENT SAMPLE)**

Notes: The figure plots disagreement in participants' reported binary beliefs across the nine experimental groups, calculated in the consistent-recall subsample. This subsample includes only respondents whose belief implied by their recalled signals is consistent with their reported belief, and we restrict to it when computing disagreement. Disagreement is measured as the cross-sectional variance of the indicator for believing the company is "good" (1) versus "bad" (-1) within each treatment. Groups 1–5 vary the degree of informational ambivalence while holding the number of signals fixed; disagreement is lowest when signals are one-sided (Groups 1 and 5) and highest when signals are most balanced (Group 3). Groups 6 and 7 keep the signal mix of Group 3 but increase recall effort: Group 6 adds Stop-and-Think and Group 7 layers a surprise accuracy incentive on top, both reducing disagreement relative to Group 3. Groups 8 and 9 keep the signal mix of Group 2 and introduce associative memory cues: cueing dominant positive signals (Group 8) lowers disagreement, whereas cueing non-dominant negative signals (Group 9) raises disagreement, consistent with the model's predictions on cue direction and strength.

Appendix Table A.8: COUNT RECALL ERRORS AND INFORMATION AMBIVALENCE ACROSS TREATMENTS

Dep. variable: Absolute count recall errors	(1) #Positive news	(2) #Negative news	(3) #Negative news	(4) #Negative news
Group 1 ($A = 0.2$)	2.195*** (0.601)	2.550*** (0.817)	2.604*** (0.512)	2.847*** (0.759)
Relative to Group 1				
Group 2 ($A = 0.6$)	-0.032 (0.908)	-0.036 (0.913)	0.385 (0.846)	0.414 (0.841)
Group 3 ($A = 0.9$)	-0.259 (0.907)	-0.283 (0.903)	0.453 (0.851)	0.433 (0.843)
Group 4 ($A = 0.6$)	-0.666 (0.773)	-0.673 (0.775)	0.308 (0.850)	0.308 (0.851)
Group 5 ($A = 0.9$)	-1.100 (0.727)	-1.096 (0.732)	1.392 (1.266)	1.395 (1.257)
Demographics		✓		✓
Observations	2659	2659	2659	2659
R-squared	0.020	0.028	0.012	0.021

Notes: This table tests whether recall accuracy varies with the degree of informational ambivalence across treatment groups. The dependent variables are the absolute recall errors for the number of positive (in Columns (1) and (2)) and negative news (in Columns (3) and (4)), respectively. Treatments groups 1-5 correspond to the informational ambivalence 0.2, 0.6, 0.9, 0.6 and 0.2, respectively; Group 1 is the omitted reference group, so coefficients are relative to Group 1. Columns (2) and (4) additionally control for employment status, housing tenure, mortgage type, household income, gender, age, college degree, and region. Clustered robust standard errors at the news-recall level are reported in parentheses. Statistical significance is denoted by *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$.

Appendix Table A.9: RECALL ACCURACY BY SIGNAL SIGN AND SIGNAL POSITION

	(1)	(2)
	Recall accuracy	
Panel A. Positive vs. negative signals		
Positive	0.510***	0.510***
	(0.007)	(0.007)
Relative to Positive		
Negative	0.009	0.009
	(0.010)	(0.010)
Demographics		✓
Observations	10636	10636
Panel B. Position / Recency		
First	0.519***	0.519***
	(0.010)	(0.010)
Relative to First		
Fifth	-0.006	-0.006
	(0.014)	(0.014)
Fifteenth	0.005	0.005
	(0.014)	(0.014)
Last	-0.015	-0.015
	(0.014)	(0.014)
Demographics		✓
Observations	10636	10636
Panel C. Retrieval-effort treatments		
Group 3 (no intervention)	0.513***	0.514***
	(0.011)	(0.011)
Relative to Group 3		
Group 6 (Stop-and-Think)	-0.012	-0.016
	(0.015)	(0.015)
Group 7 (G6 + incentivized recall)	-0.006	-0.007
	(0.015)	(0.015)
Demographics		✓
Observations	6304	6304

Notes: This table reports average marginal effects from probit regressions of recall accuracy on signal sign (Panel A), signal position (Panel B), and the retrieval-effort treatments (Panel C). The dependent variable is an indicator for correctly recalling the probed signal. In Panel A, pooling Groups 1–5, the explanatory variable is an indicator for whether the signal is negative, with positive signals as the reference category. In Panel B, pooling Groups 1–5, the explanatory variables are indicators for the 5th, 15th, and last signal positions, with the 1st signal as the reference category. In Panel C, the sample is restricted to the retrieval-effort arms (Groups 3, 6, and 7) and the explanatory variables are indicators for the two interventions, with Group 3 as the reference category. Column (1) reports estimates without controls; column (2) adds demographic controls. Standard errors clustered at the participant level are reported in parentheses. Statistical significance is denoted by *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$.

Appendix Table A.10: INSTRUMENTAL-VARIABLE ESTIMATES OF THE EFFECT OF RECALL ON BELIEFS AND UNCERTAINTY

	(1)	(2)	(3)
	Binary belief	Prob. belief	Uncertainty
Panel A. Incentive and elicitation order (Group 7 vs. Group 3)			
<i>First stage</i>			
Incentivized Recall (Group 7)	4.299*** (0.204)	4.299*** (0.204)	1.238*** (0.113)
First-stage F	443.4	443.4	120.9
<i>Reduced form</i>			
Incentivized Recall (Group 7)	0.329*** (0.038)	15.409*** (0.891)	-0.033*** (0.003)
<i>2SLS</i>			
Share pos. recalled (%)	0.077*** (0.008)	3.585*** (0.211)	
Net imbalance			-0.027*** (0.003)
Observations	1070	1070	1070
Panel B. Incentive only, elicitation order held fixed (Group 7 vs. Group 6)			
<i>First stage</i>			
Incentivized Recall (Group 7)	2.712*** (0.241)	2.712*** (0.241)	0.349** (0.163)
First-stage F	126.3	126.3	4.6
<i>Reduced form</i>			
Incentivized Recall (Group 7)	0.136*** (0.031)	5.026*** (0.846)	-0.012*** (0.003)
<i>2SLS</i>			
Share pos. recalled (%)	0.050*** (0.010)	1.853*** (0.288)	
Net imbalance			-0.034** (0.017)
Observations	1050	1050	1050

Notes: The table instruments recall with the Incentivized Recall treatment (Group 7) and reports, within each panel, the first stage, the reduced form, and 2SLS. Panel A contrasts Group 7 with Group 3, which observes the same signal history; Group 7 adds the full retrieval-effort intervention (Stop-and-Think plus a surprise monetary reward for accurate recall). Panel B contrasts Group 7 with Group 6, which shares the same elicitation order (recall before beliefs), so only the incentive differs. Outcomes are the binary belief (column 1), the reported probability that the company is good (column 2), and subjective uncertainty $U_i = p_i(1 - p_i)$ (column 3); the endogenous regressor is the recalled positive share in columns (1)–(2) and the realized net imbalance $|X_i^+ - X_i^-|$ in column (3). The bonus rewards the accuracy of reported recall against the true signal composition rather than the content of beliefs, so under a common-state interpretation it should sharpen reported recall without moving beliefs. The column (3) first stage in Panel B is weak ($F = 4.6$); we rely on the reduced form there. Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Appendix Table A.11: RECALL ACCURACY AND DEMOGRAPHIC CHARACTERISTICS

Dep. Variable: Absolute recall errors	(1) #Positive news	(2) #Negative news
Employment status		
Full-time	0.053 (0.153)	0.289 (0.247)
Part-time (for someone or self-employed)	0.239 (0.229)	0.562 (0.390)
Not in Labor force	-0.088 (0.227)	0.245 (0.445)
Student	0.254 (0.454)	0.229 (0.485)
Other	0.357 (0.389)	0.049 (0.298)
Homeownership		
Rent	-0.014 (0.101)	0.136 (0.204)
Other	0.128 (0.321)	-0.250 (0.302)
Mortgage		
Fixed-rate	0.107 (0.164)	0.080 (0.216)
Variable-rate	-0.313* (0.161)	-0.584 (0.428)
HH income	0.000 (0.022)	-0.004 (0.024)
Female	0.086 (0.142)	0.137 (0.162)
Age (grouped)	-0.025 (0.032)	-0.050 (0.054)
College degree	-0.009 (0.112)	-0.197 (0.204)
Region		
Midwest	-0.117 (0.143)	-0.335 (0.263)
South	0.099 (0.138)	-0.051 (0.222)
West	-0.048 (0.158)	-0.392 (0.256)
Constant	1.678*** (0.307)	3.138*** (0.424)
Observations	2659	2659
R-squared	0.003	0.005

Notes: This table examines whether recall accuracy within treatment groups 1-5 varies with respondent demographics. The dependent variables are the absolute recall errors for the number of positive (Column (1)) and negative news (Column (2)), respectively. Each coefficient reports the association between a demographic characteristic and recall error, estimated by OLS. Reported categories are relative to omitted reference groups for each set of indicators (employment status, homeownership, mortgage type, and region). Robust standard errors are in parentheses. Statistical significance is denoted by *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$.

Appendix Table A.12: RECALL ACCURACY WITH CUES BY DEMOGRAPHICS CHARACTERISTICS

	(1)	(2)	(3)	(4)
	Group 8		Group 9	
Dep. Variable: Absolute recall errors	#Positive news	#Negative news	#Positive news	#Negative news
Employment status				
Full-time	-1.160 (1.083)	-0.627 (0.572)	0.264 (0.420)	0.132 (0.578)
Part-time	-0.376 (1.027)	-0.543 (0.648)	0.184 (0.609)	-0.414 (0.943)
Not in the labor force	-1.064 (1.209)	-0.040 (0.872)	0.479 (0.807)	0.059 (0.926)
Student	-1.394 (1.229)	-1.255 (0.823)	-0.460 (1.041)	-0.356 (1.241)
Other	-0.999 (1.011)	-0.244 (0.438)	-0.086 (0.848)	-0.157 (0.828)
Homeownership				
Rent	0.265 (0.438)	0.064 (0.324)	-0.018 (0.271)	-0.107 (0.170)
Other	0.860 (1.082)	0.741 (1.192)	-0.166 (0.466)	-0.824 (0.707)
Mortgage				
Fixed-rate	-0.264 (0.518)	-0.174 (0.397)	0.280 (0.345)	0.118 (0.305)
Variable-rate	0.522 (0.790)	-0.290 (0.525)	-1.316* (0.682)	-1.957* (1.062)
HH income	-0.028 (0.087)	-0.111 (0.094)	0.043 (0.048)	0.001 (0.056)
Female	0.095 (0.364)	-0.012 (0.378)	-0.066 (0.324)	0.160 (0.497)
Age (grouped)	0.009 (0.077)	0.002 (0.068)	-0.079 (0.122)	-0.122 (0.192)
College degree	0.052 (0.344)	-0.033 (0.255)	-0.194 (0.315)	-0.616 (0.553)
Region				
Midwest	-0.681 (0.654)	-0.641 (0.518)	-0.450 (0.363)	-0.222 (0.416)
South	-0.443 (0.493)	-0.637 (0.515)	0.299 (0.368)	0.468 (0.477)
West	-0.237 (0.602)	-0.224 (0.477)	-0.038 (0.391)	0.272 (0.500)
Constant	3.543** (1.342)	3.173** (1.494)	3.886*** (0.962)	3.532** (1.505)
Observations	525	525	524	524
R-squared	0.018	0.021	0.016	0.021

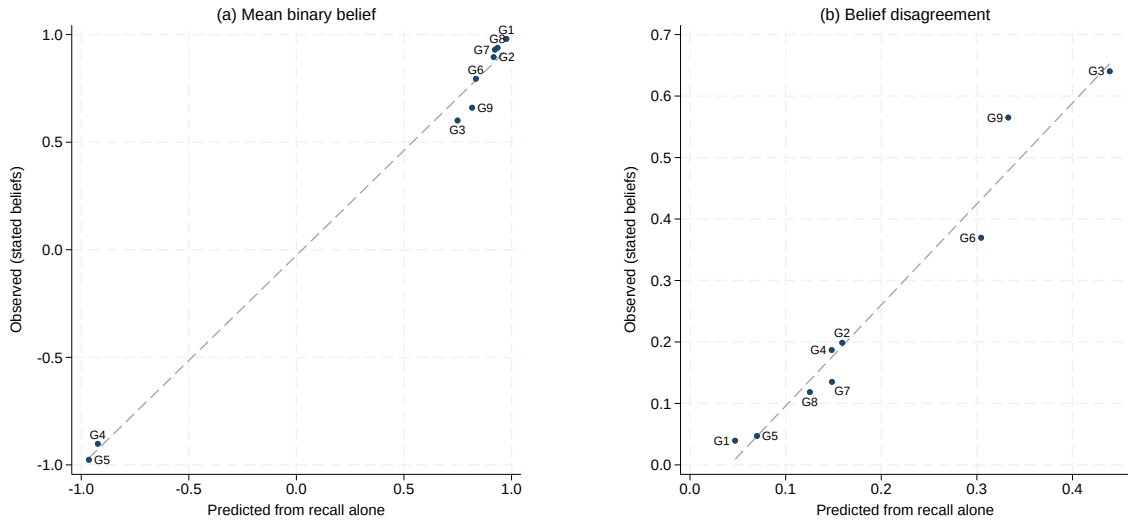
Notes: This table tests whether recall accuracy with cues varies across demographic characteristics in Groups 8 and 9. The dependent variables are absolute recall errors for positive and negative news. Columns (1)–(2) report estimates for Group 8 and columns (3)–(4) for Group 9. Each column presents OLS regressions of recall error on employment status, homeownership, mortgage type, household income, gender, age group, education, and region indicators; reported categories are relative to omitted reference groups within each set of indicators. Statistical significance is denoted by *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$.

Appendix Table A.13: TREATMENT EFFECTS ON BELIEFS CONDITIONAL ON MEASURED RECALL

	(1)	(2)	(3)	(4)
Panel A. Retrieval effort (Groups 3, 6, and 7; Group 3 omitted)				
Dominant recall	0.782*** (0.034)		0.763*** (0.034)	
Group 6 (Stop-and-Think)		0.194*** (0.044)	0.129*** (0.033)	0.107*** (0.038)
Group 7 (G6 + incentivized recall)		0.329*** (0.038)	0.197*** (0.030)	0.133*** (0.032)
Net-evidence indicators				✓
Observations	1576	1576	1576	1576
R-squared	0.463	0.047	0.480	0.534
Panel B. Contextual cues (Groups 2, 8, and 9; Group 2 omitted)				
Dominant recall	0.922*** (0.023)		0.903*** (0.023)	
Group 8 (positive cue)		0.044* (0.024)	0.027** (0.013)	0.035 (0.022)
Group 9 (negative cue)		-0.235*** (0.038)	-0.145*** (0.027)	-0.056*** (0.021)
Net-evidence indicators				✓
Observations	1603	1603	1603	1603
R-squared	0.575	0.048	0.593	0.755

Notes: The table reports OLS regressions of participants' reported binary belief about the company's type (1 = good, -1 = bad) on treatment indicators and measures of recalled information in Experiment 1. Panel A uses the retrieval-effort sample (Groups 3, 6, and 7; Group 3 omitted); Panel B uses the cue sample (Groups 2, 8, and 9; Group 2 omitted). *Dominant recall* is constructed mechanically from each participant's recalled signal counts, equal to 1 if recalled positive signals are at least as many as recalled negative signals and -1 otherwise. Column (4) replaces the dominant-recall indicator with a full set of indicators for recalled net evidence, $X^+ - X^-$, so that no functional form is imposed on how net evidence enters beliefs. Because recalled counts are a noisy, post-elicitation measure of retrieval, the residual treatment effects in columns (3) and (4) are upper bounds on channels operating outside recall. Robust standard errors are reported in parentheses. Statistical significance is denoted by *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$.

Appendix Figure A.5: PARAMETER-FREE CHECK: RECALL DATA ALONE PREDICT BELIEF MOMENTS



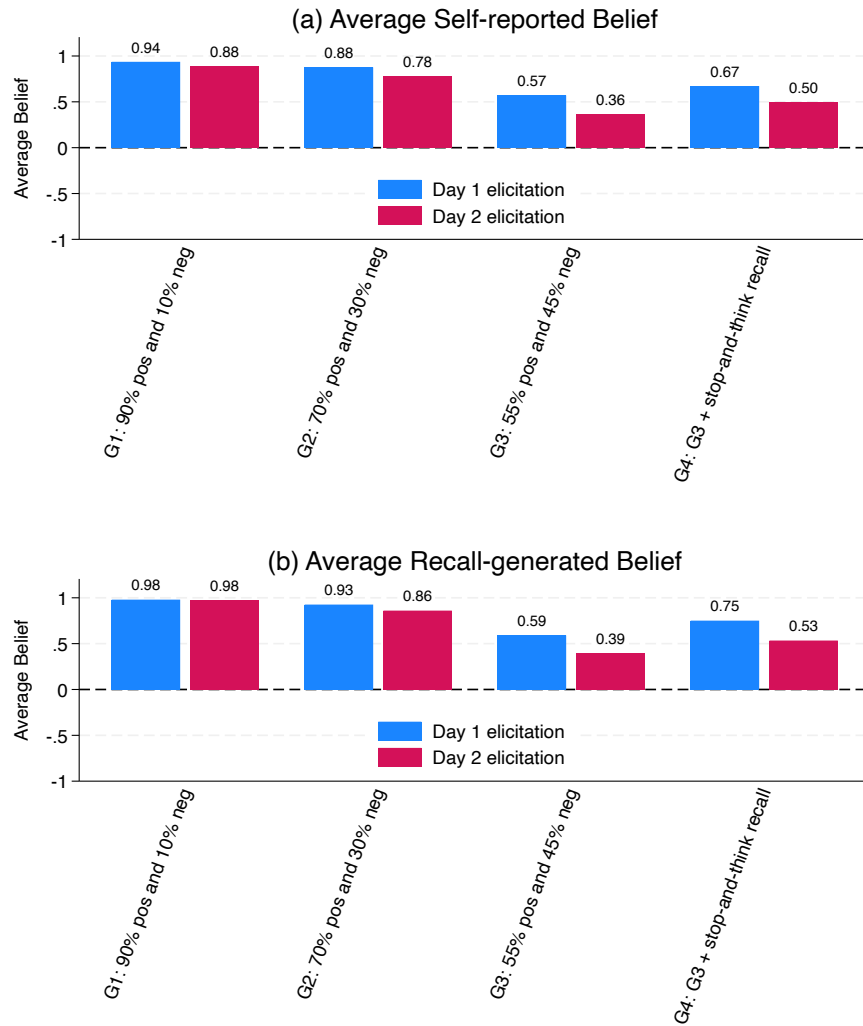
Notes: The figure compares group-level belief moments predicted from recall data alone with the observed moments of stated beliefs in Experiment 1. For each group, $\hat{q}$ denotes the share of participants whose recalled counts contain at least as many positive as negative signals; the model's belief rule then implies a mean binary belief of $2\hat{q} - 1$ (Panel (a)) and a variance of binary beliefs, disagreement, of $4\hat{q}(1 - \hat{q})$ (Panel (b)). No stated-belief data and no estimated parameters enter the predictions. Dashed lines are least-squares fits; in Panel (b), Groups 4 and 7 coincide. The visible gaps, in the most ambivalent group (G3) and under the negative cue (G9), are the groups where stated beliefs are more dispersed than the recall margin alone predicts, consistent with the residual treatment effects in Table A.13.

Appendix Table A.14: REALIZED NET IMBALANCE, CUE AND BELIEF UNCERTAINTY

	(1)	(2)
Panel A: Recalled information predicts belief uncertainty		
Realized net imbalance	-0.003*** (0.000)	-0.003*** (0.000)
Constant	0.207*** (0.001)	0.215*** (0.005)
Demographics		✓
Observations	4758	4758
R-squared	0.093	0.097
Panel B: Information ambivalence and belief uncertainty		
Group 3: 55% pos	0.217*** (0.002)	0.223*** (0.005)
Relative to Group 3		
Group 1: 90% pos	-0.053*** (0.003)	-0.052*** (0.003)
Group 2: 70% pos	-0.019*** (0.003)	-0.018*** (0.003)
Group 4: 30% pos	-0.022*** (0.003)	-0.022*** (0.004)
Group 5: 10% pos	-0.075*** (0.003)	-0.075*** (0.003)
Group 6: Group 3 + Stop-and-Think	-0.021*** (0.003)	-0.021*** (0.003)
Group 7: Group 6 + incentivized recall	-0.033*** (0.003)	-0.033*** (0.003)
Demographics		✓
Observations	3709	3709
R-squared	0.155	0.159
Panel C: Effect of cues on belief uncertainty		
Group 2	0.198*** (0.002)	0.204*** (0.008)
Relative to Group 2		
Group 8: Group 2 + pos cue	-0.037*** (0.004)	-0.037*** (0.004)
Group 9: Group 2 + neg cue	0.028*** (0.003)	0.028*** (0.003)
Demographics		✓
Observations	1603	1603
R-squared	0.200	0.209

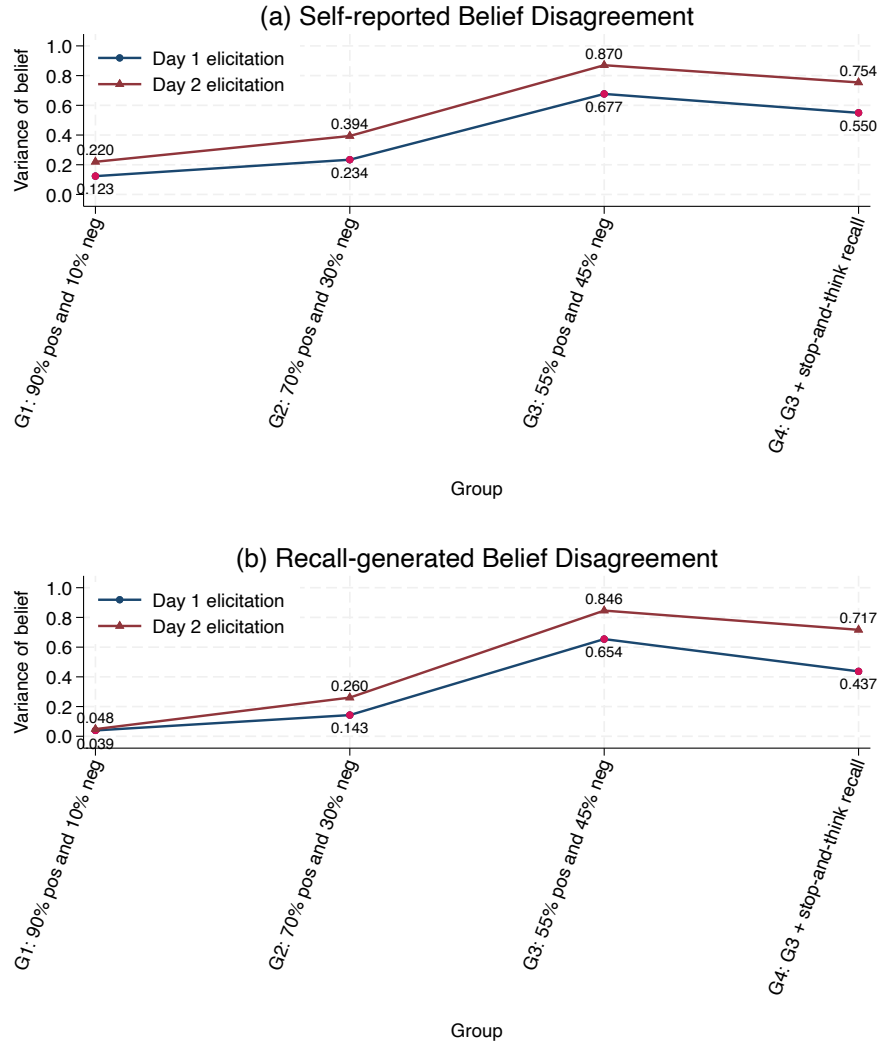
Notes: This table reports results from the Belief Disagreement Experiment on the effect of realized net imbalance recalled by respondents and cues on belief uncertainty. Probabilistic beliefs are elicited as percentage-point probabilities that the company is good. Let $p_i \in [0, 1]$ denote respondent i 's reported probability that the company is good; when beliefs are elicited as $P(\text{bad})$, we convert to the common scale by setting $p_i = 1 - P(\text{bad})$. The dependent variable is belief uncertainty, defined as $p_i(1 - p_i)$. Panel A relates belief uncertainty to *realized net imbalance*, defined as the absolute value of $X_i^+ - X_i^-$, where X_i^+ and X_i^- denote the numbers of positive and negative items recalled. Panel B relates belief uncertainty to informational ambivalence (Groups 1-5) in information provision and recall probability (Groups 3, 6 and 7) using treatment-group indicators (omitted category is Group 3). Panel C estimates the effect of cues on belief uncertainty (omitted category is Group 2). Robust standard errors are reported in parentheses. Statistical significance is denoted by *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$.

Appendix M Additional Robustness Experiment 2 Results



Appendix Figure A.6: ROBUSTNESS EXPERIMENT: AVERAGE BELIEF ABOUT THE COMPANY STATE BY GROUP (REPORTED AND RECALL-GENERATED BELIEFS)

Notes: The figure plots the average binary belief across participants in each of the four experimental groups. In the top panel (self-reported beliefs), beliefs take value 1 if the respondent reports that the company is “good” and –1 if the respondent reports that it is “bad”; group-level averages therefore summarize the net balance of positive versus negative beliefs within each treatment. In the bottom panel (recall-generated beliefs), beliefs are coded as 1 if the number of recalled positive signals is weakly greater than the number of recalled negative signals, and –1 otherwise. Groups 1–3 vary the degree of informational ambivalence while holding the number of signals fixed; Groups 4 keeps the signal composition of Group 3 but increase recall effort (via Stop-and-Think).



Appendix Figure A.7: ROBUSTNESS EXPERIMENT: DISAGREEMENT ABOUT COMPANY TYPE BY GROUP (SELF-REPORTED AND RECALL-GENERATED BELIEFS)

Notes: The figure plots disagreement in participants' reported binary beliefs across the four experimental groups. Disagreement in the top figure (self-reported beliefs) is measured as the cross-sectional variance of the indicator for believing the company is "good" (1) versus "bad" (-1) within each treatment. Disagreement in the bottom figure (recall-generated beliefs) is coded as 1 if the number of recalled positive signals is weakly greater than the number of recalled negative signals, and -1 otherwise. Groups 1–3 vary the degree of informational ambivalence while holding the number of signals fixed; Groups 4 keeps the signal composition of Group 3 but increase recall effort (via Stop-and-Think).

Appendix Table A.15: ROBUSTNESS EXPERIMENT 2 DEMOGRAPHIC CHARACTERISTICS

	Full sample (1)	T1 (2)	T2 (3)	T3 (4)	T4 (5)	<i>p</i> -value (6)
Observations	2,039	503	514	516	506	
Panel A: Personal background						
Female (%)	59.49	56.86	56.61	63.37	61.07	0.14
Age (years)	44.87 (13.99)	44.16 (14.13)	45.11 (14.22)	44.33 (13.72)	45.90 (13.88)	0.77
College (%)	87.15	87.87	87.55	88.18	84.98	0.40
Employed full-time (%)	55.37	57.65	54.09	53.10	56.72	0.42
Household income < \$60,000 (%)	56.94	53.88	58.75	56.78	58.30	0.46
Panel B: Financial characteristics						
Homeowner (%)	60.96	62.82	61.48	59.88	59.68	0.71
Stock investor (%)	57.38	55.27	59.92	56.98	57.31	0.51

Notes: This table reports baseline demographic characteristics by experimental treatment group. Entries are percentages for indicator variables and means for continuous variables, with standard deviations reported in parentheses. Treatment groups T1–T4 correspond to different information treatments described in the text. The final column reports *p*-values from an ANOVA test of equality of means across treatment groups.

Appendix N Additional Results for Experiment 3 (Belief Instability Experiment)

Appendix Table A.16: PURE-FORGETTING TEST: DAY-2 VS DAY-1 RECALL COUNTS (EXPERIMENT 3)

	T1 (90% pos)	T2 (70% pos)	T3 (55% pos)	T4 (T3 + S&T)	All
<i>Share with Day-2 counts exceeding Day-1 counts (%)</i>					
Day 2 positive > Day 1 positive	16.7	28.6	33.6	37.6	29.4
Day 2 negative > Day 1 negative	30.3	26.4	36.3	26.9	30.0
Day 2 total > Day 1 total	33.7	35.4	38.6	35.3	35.8
Any violation (positive or negative)	44.2	49.5	57.6	57.0	52.2
Observations	412	444	446	442	1744

Notes: The table tests the implication of deterministic (pure) forgetting that each individual's Day-2 recalled set is a subset of her Day-1 recalled set. Under this restriction, the Day-2 counts of recalled positive and recalled negative signals cannot exceed the corresponding Day-1 counts. Each entry reports the share of participants in Experiment 3 whose Day-2 count strictly exceeds the corresponding Day-1 count – a violation of pure forgetting. Treatment groups T1–T4 are defined as in Table 1: T1 (90% positive), T2 (70% positive), T3 (55% positive), and T4 (T3 with a Stop-and-Think recall intervention). The sample consists of all participants with valid recall counts on both days.

Appendix Table A.17: INFORMATION AMBIVALENCE AND BELIEF INSTABILITY

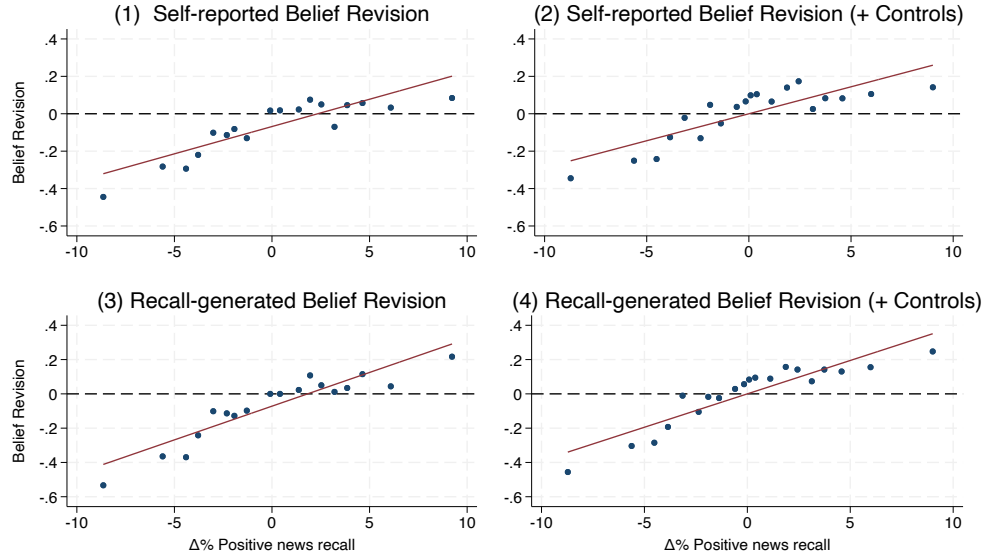
	binary belief instability		implied belief instability	
	(1)	(2)	(3)	(4)
T1: 90% pos and 10%	0.073*** (0.013)	0.071 (0.047)	0.002 (0.002)	-0.038 (0.037)
Relative to T1				
T2: 70% pos and 30%	0.040** (0.020)	0.040** (0.020)	0.040*** (0.010)	0.041*** (0.010)
T3: 55% pos and 45%	0.230*** (0.025)	0.229*** (0.025)	0.258*** (0.021)	0.258*** (0.021)
T4: T3 + stop-and-think recall	0.115*** (0.023)	0.114*** (0.023)	0.199*** (0.019)	0.202*** (0.020)
Demographics		✓		✓
Observations	1744	1744	1744	1744
R-squared	0.054	0.058	0.101	0.107

Notes: This table reports estimates from regressions of belief instability indicators on treatment-group dummies in Experiment 3. The dependent variable is an indicator equal to one if a participant's belief reverses between Part 1 and Part 2, and zero otherwise. Columns (1)–(2) use self-reported binary beliefs, while Columns (3)–(4) use recall-generated beliefs constructed from participants' remembered signals. The omitted reference group is T1 (90% positive, 10% negative signals). T2 and T3 correspond to increasingly ambivalent information environments, and T4 implements a Stop-and-Think recall intervention in the T3 signal environment. Columns (2) and (4) include demographic controls. Robust standard errors are reported in parentheses. Statistical significance is denoted by *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$.

Appendix Table A.18: BELIEF INSTABILITY AND CHANGES IN RECALLED INFORMATION

	(1)	(2)	(3)	(4)
Panel A. Self-reported binary belief instability				
Δ%Positive news recall	0.029*** (0.003)	0.029*** (0.003)		
Δ%Positive news recall(Direction)			0.114*** (0.011)	0.113*** (0.011)
Demographics		✓		✓
Observations	1744	1744	1744	1744
R-squared	0.090	0.096	0.067	0.074
Panel B. Recall-generated belief instability				
Δ%Positive news recall	0.039*** (0.002)	0.039*** (0.002)		
Δ%Positive news recall (Direction)			0.149*** (0.009)	0.148*** (0.009)
Demographics		✓		✓
Observations	1744	1744	1744	1744
R-squared	0.220	0.225	0.155	0.161
Panel C. Belief switching conditional on recalled-majority switching				
	Recalled majority switches		Recalled majority unchanged	
Share switching stated belief	0.827 (0.025)		0.074 (0.007)	
Observations	225		1,519	
Difference = 0.753 (robust s.e. = 0.026), z = 28.8				

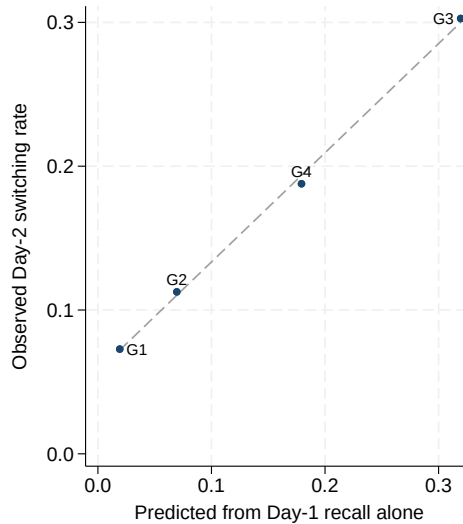
Notes: This table relates within-individual changes in recalled information to within-individual belief instability between Part 1 (Day 1) and Part 2 (Day 2). In Panel A (self-reported belief), the dependent variable is $\Delta\text{Belief}_i \in \{-1, 0, 1\}$, constructed from the respondent's self-reported binary belief on Day 1 and Day 2. It equals 0 if the respondent reports the same belief across days, 1 if they revise from "bad" to "good," and -1 if they revise from "good" to "bad." In Panel B (recall-generated belief), the dependent variable is $\Delta\text{RecallBelief}_i \in \{-1, 0, 1\}$, constructed analogously from a *recall-generated* binary belief in each part. In each part, the recall-generated belief is coded as "good" if the respondent recalls at least as many positive as negative news items, and "bad" otherwise. $\Delta\text{RecallBelief}_i$ equals 0 if this recall-generated classification is unchanged across days, 1 if it switches from "bad" to "good," and -1 if it switches from "good" to "bad." The regressor $\Delta\%$ Positive news recall $_i$ is the change in the share of recalled items that are positive, defined as $\text{share_pos}_{i,2} - \text{share_pos}_{i,1}$. Columns (3)–(4) use $\Delta\%$ Positive news recall (Direction) $_i \in \{-1, 0, 1\}$, which equals 1 if the positive recall share increases, -1 if it decreases, and 0 if it is unchanged. Columns with a check mark include demographic controls. All models are estimated by OLS with heteroskedasticity-robust standard errors in parentheses. Panel C reports the share of participants who switch their self-reported binary belief between Part 1 and Part 2, separately for participants whose recall-generated classification (the recalled majority) switches and for those whose classification is unchanged; standard errors of the proportions are in parentheses, and the difference is tested with a two-sample proportion test. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.



Appendix Figure A.8: ROBUSTNESS EXPERIMENT 2: BELIEF INSTABILITY AND CHANGES IN RECALLED INFORMATION

Notes: The figure plots the average binary belief across participants in each of the four experimental groups. In the top panel (self-reported beliefs), beliefs take value 1 if the respondent reports that the company is “good” and -1 if the respondent reports that it is “bad”; group-level averages therefore summarize the net balance of positive versus negative beliefs within each treatment. In the bottom panel (recall-generated beliefs), beliefs are coded as 1 if the number of recalled positive signals is weakly greater than the number of recalled negative signals, and -1 otherwise. Groups 1–3 vary the degree of informational ambivalence while holding the number of signals fixed; Groups 4 keeps the signal composition of Group 3 but increase recall effort (via Stop-and-Think).

Appendix Figure A.9: PARAMETER-FREE CHECK: DAY-1 RECALL PREDICTS DAY-2 BELIEF SWITCHING



Notes: For each group of Experiment 3, $\hat{q}$ is the share of returning participants whose Day-1 recalled counts contain at least as many positive as negative signals; under independent retrieval draws, the model implies a Day-2 switching probability of $2\hat{q}(1 - \hat{q})$. No stated-belief data and no estimated parameters enter the predictions. The dashed line is the least-squares fit. Group 4 implements the Stop-and-Think intervention.

Appendix Table A.19: INSTRUMENTAL VARIABLE ESTIMATES OF THE EFFECT OF RECALL ON BELIEFS AND UNCERTAINTY (EXPERIMENT 3)

	(1) Binary belief	(2) Prob. belief good	(3) Uncertainty U_i
Panel A. First stage			
	Day 2 share pos.	Day 2 share pos.	Day 2 net imbalance
Day 1 recall measure	1.003*** (0.008)	1.003*** (0.008)	0.933*** (0.007)
First-stage F	14,488	14,488	20,024
Panel B. 2SLS			
Day 2 share pos. recalled (%)	0.022*** (0.001)	0.506*** (0.031)	
Day 2 net imbalance			-0.002*** (0.000)
Observations	1,744	1,744	1,744

Notes: This table reports estimates from Experiment 3, in which the same participant is observed on Day 1 and Day 2 with no new information arriving between elicitations. Following [Gillen, Snowberg and Yariv \(2019\)](#), we use the Day 1 recall as an instrument for the Day 2 recall in regressions of Day 2 outcomes on Day 2 recall. The two-stage least squares model is

$$R_i^{(2)} = \pi_0 + \pi_1 R_i^{(1)} + v_i \quad (\text{first stage, Panel A})$$

$$Y_i^{(2)} = \alpha + \beta \hat{R}_i^{(2)} + \varepsilon_i \quad (\text{second stage, Panel B}).$$

The outcome $Y_i^{(2)}$ is the Day 2 binary belief $\hat{Y}_i$ in column (1), the Day 2 reported probability p_i that the company is good (in percent) in column (2), and the Day 2 subjective uncertainty $U_i = p_i(1 - p_i)$ in column (3). The endogenous regressor $R_i^{(2)}$ is the Day 2 share of recalled signals that are positive in columns (1) and (2), and the Day 2 realized net imbalance $|X_i^+ - X_i^-|$ in column (3); the instrument $R_i^{(1)}$ is the corresponding Day 1 measure. Under classical measurement error in recall, OLS would attenuate the recall-belief slope toward zero; the instrumental variable estimator recovers the slope free of this attenuation. The first-stage F -statistics (14,488 and 20,024) are far above conventional weak-instrument thresholds, reflecting the high within-individual persistence of recall measures across the two days; the persistence is not an artifact of pooling treatment groups, as the first stage remains far above conventional thresholds with group fixed effects ($F = 2,821$ for the recalled positive share). The 2SLS estimates are statistically significant in every column, with the theoretically predicted signs: a higher recalled positive share predicts a more positive belief, and a higher realized net imbalance predicts lower subjective uncertainty. Robust standard errors are reported in parentheses. Statistical significance is denoted by *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$.

Appendix O Other Alternative Explanations

This appendix addresses five further alternative explanations for the recall–belief link, complementing the principal threats addressed in Section 5.3.

Experimenter demand Respondents might report an “expected” answer rather than their own belief. Several features make this unlikely. Beliefs are incentivized for accuracy (a bonus for correctly identifying the company’s type), so conforming at the expense of accuracy is costly, and the hypothetical company carries no signaled “right” answer. The comparative statics, moreover, track the signal composition we vary exogenously and persist when beliefs are built mechanically from recalled signals, neither of which a demand effect on stated answers can produce. The instability experiment adds a further check: Experiment 3 re-elicits the same participants a day later under identical materials, so a demand-driven respondent would simply repeat the earlier answer, yet many revise (Result 3). The cue treatments are the most exposed, since a cue could be read as a hint about the “right” answer. But participants are told the cue carries no information and must pass a comprehension check confirming this to enter the study, and beliefs remain incentivized for accuracy, so deliberately following the cue against one’s own reading would be costly rather than rewarded. The cue nonetheless moves beliefs, and it does so by shifting the *recalled composition* itself (Groups 8 and 9), which points to a retrieval channel rather than a demand response.

Cognitive defaults Under cognitive uncertainty, respondents may default to a focal answer to economize on effort and reuse it for both the recall and belief tasks, mechanically inducing a recall–belief association (Enke and Graeber, 2023). Two features of Experiment 1 argue against this. First, recalled signal composition tracks the experimentally varied true composition across Groups 1–5 and responds to the incentivized-recall treatment (Group 7), indicating genuine engagement rather than a fixed default. Second, to provide additional evidence against cognitive defaults as the driver of our results, we follow Jiang, Liu, Peng and Yan (2025) and recompute the disagreement and subjective-uncertainty patterns on the subsample of participants who correctly recalled at least three of four probed signals, indicating clear effort and suggesting they did not rely on cognitive defaults; within this subsample the results are, if anything, sharper: belief disagreement still rises more than twelvefold from the one-sided environments (variance 0.05 in Groups 1 and 5) to the most ambivalent (variance 0.63 in Group 3), and subjective uncertainty still peaks under balanced evidence (Group 3, 0.22). Cognitive defaulting therefore cannot account for the results.

External recording and perfect recall of previous beliefs External note-taking would imply near-perfect accuracy, but fewer than 0.5 percent of participants in any Experiment 1 group answer all recall questions correctly. Perfect recall of previous beliefs combined with non-

Bayesian updating would imply that cues and retrieval-effort interventions cannot move beliefs, since these manipulations leave the previously reported belief intact; instead, both substantially shift disagreement, supporting imperfect recall of signals rather than perfect recall of previous beliefs. Most directly, if participants could recall or had externally recorded their previously reported belief, they would simply reproduce it upon re-elicitation and their beliefs would remain stable; instead, a substantial share revise their binary belief in the 24-hour re-elicitation under unchanged information (30.3 percent in the most ambivalent group; Result 3), precisely the instability that recall of a prior belief would preclude.

Pure forgetting A natural alternative is deterministic forgetting: each signal, once dropped from memory, cannot return. Under this restriction, each Experiment 3 respondent's Day-2 recalled set must be a subset of her Day-1 recalled set, so Day-2 counts of recalled positive and negative signals cannot exceed the corresponding Day-1 counts. Appendix Table A.16 sharply rejects this alternative: 52.2 percent of the 1,744 two-day participants recall more positives or more negatives on Day 2 than on Day 1, 35.8 percent recall a strictly larger total on Day 2 than Day 1, and the violation rate rises monotonically with informational ambivalence (from 44.2 percent in Group 1 to 57.6 and 57.0 percent in Groups 3 and 4). Signals re-emerge across days at scale, consistent with stochastic recall but not with deterministic memory decay.⁵⁴

Stochastic choice and noisy aggregation Two models could match our comparative statics without stochastic recall: a noisy signal-aggregation model, in which beliefs are a weighted sum of all signals plus noise, and stochastic-choice models such as random utility or probability matching, in which a fixed belief is turned into a binary answer by a fresh random draw. Both produce more disagreement, instability, and uncertainty when evidence is more balanced. The difference is where the variability sits: both leave it unanchored to what the respondent retrieves, whereas stochastic recall ties it to the measured recalled content. Three findings favor the latter. First, within-person belief revisions co-move with revisions in recalled information roughly one for one (Result 1), which content-free noise has no reason to do.⁵⁵ Second, raising retrieval effort enlarges the recalled subsample and so must shrink the dispersion of the recalled mean. The effort treatments do this without adding information: they reduce disagreement by up to 80 percent and lower uncertainty (Groups 6 and 7), whereas reporting noise has no

⁵⁴One might worry that this Day-2 re-emergence reflects guessing rather than genuine retrieval. The qualitative task records only the numbers of positive and negative signals recalled, so we cannot verify the content of re-emerged items here. The quantitative experiment (Section 6), which elicits the exact values recalled, allows this check directly: most Day-2 values absent from a participant's Day-1 recall are genuine data points that were actually shown rather than noise (Appendix Table A.21).

⁵⁵The Quantitative Belief Experiment (Section 6), which records the exact values recalled, provides a sharper version: the data points appearing in a respondent's Day-2 recall but absent on Day-1 are predominantly genuine values that were in fact shown rather than noise (Appendix Table A.21), which a fixed belief with noisy reporting cannot generate.

retrieval channel and predicts no effect. Third, an uninformative cue moves beliefs by shifting the measured recalled composition (Groups 8 and 9): it gives an aggregation rule no signal to re-weight and leaves a fixed belief untouched, so neither standard model has a channel for the cue effect we observe. The only variant that does respond re-weights toward the cued signal type, but then its weights must equal the recall we measure, which is stochastic recall under another name. We do not dispute that survey responses are partly stochastic; we locate the channel at retrieval.

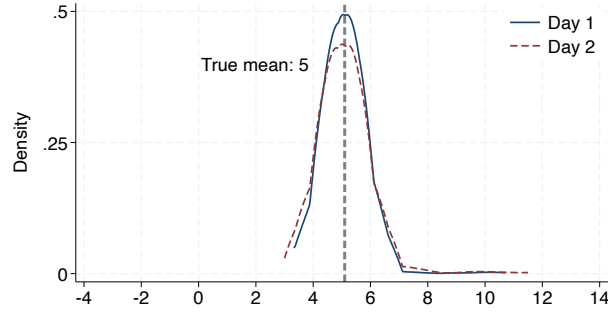
Appendix P Additional Results for Experiment 4 (Quantitative Belief Disagreement)

Appendix Table A.20: EXPERIMENTS 4-5: DEMOGRAPHIC CHARACTERISTICS

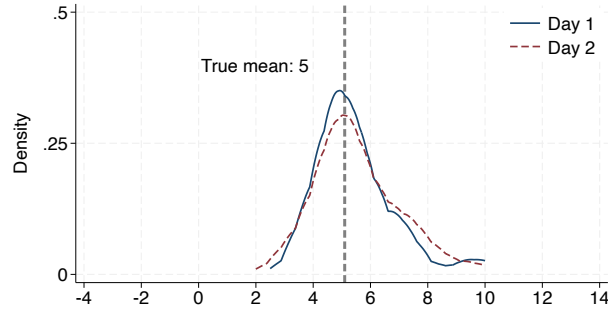
	Full sample	Group 1 ($\rho = 0.1, \sigma_\epsilon = 0.5$)	Group 2 ($\rho = 0.9, \sigma_\epsilon = 0.5$)	Group 3 ($\rho = 0.9, \sigma_\epsilon = 5$)	p -value (ANOVA)
Observations	2,301	816	771	714	
<i>Panel A: Personal background</i>					
Female (%)	51.11	50.98	51.23	51.12	0.995
Age (years)	48.94 (15.34)	49.41 (15.47)	49.13 (15.19)	48.20 (15.36)	0.280
Bachelor's degree or higher (%)	28.55	28.80	29.83	26.89	0.577
Employed full-time (%)	47.02	47.18	46.56	47.34	0.953
Household income < \$60,000 (%)	56.84	57.11	56.42	57.00	0.962
<i>Panel B: Financial characteristics</i>					
Stock investor (%)	44.02	44.12	44.36	43.56	0.951

Notes: This table reports summary statistics by experimental group in the quantitative belief experiments. Groups differ in the persistence ρ and innovation standard deviation σ_ϵ of the AR(1) data-generating process, while the long-run mean is held fixed at $\mu = 5$. Female, Bachelor's degree or higher, employed full-time, household income below \$60,000, and stock investor are indicator variables and are reported as percentages. Age is reported as the sample mean, with standard deviation in parentheses. The final column reports p -values from one-way ANOVA tests of equality across the three groups. For indicator variables, the ANOVA tests correspond to tests of equality of group means. All variables are measured in Part 1 of the experiment.

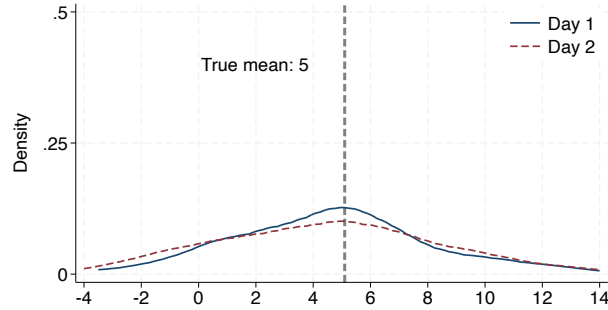
(a) Group 1: Low persistence, low volatility



(b) Group 2: High persistence, low volatility

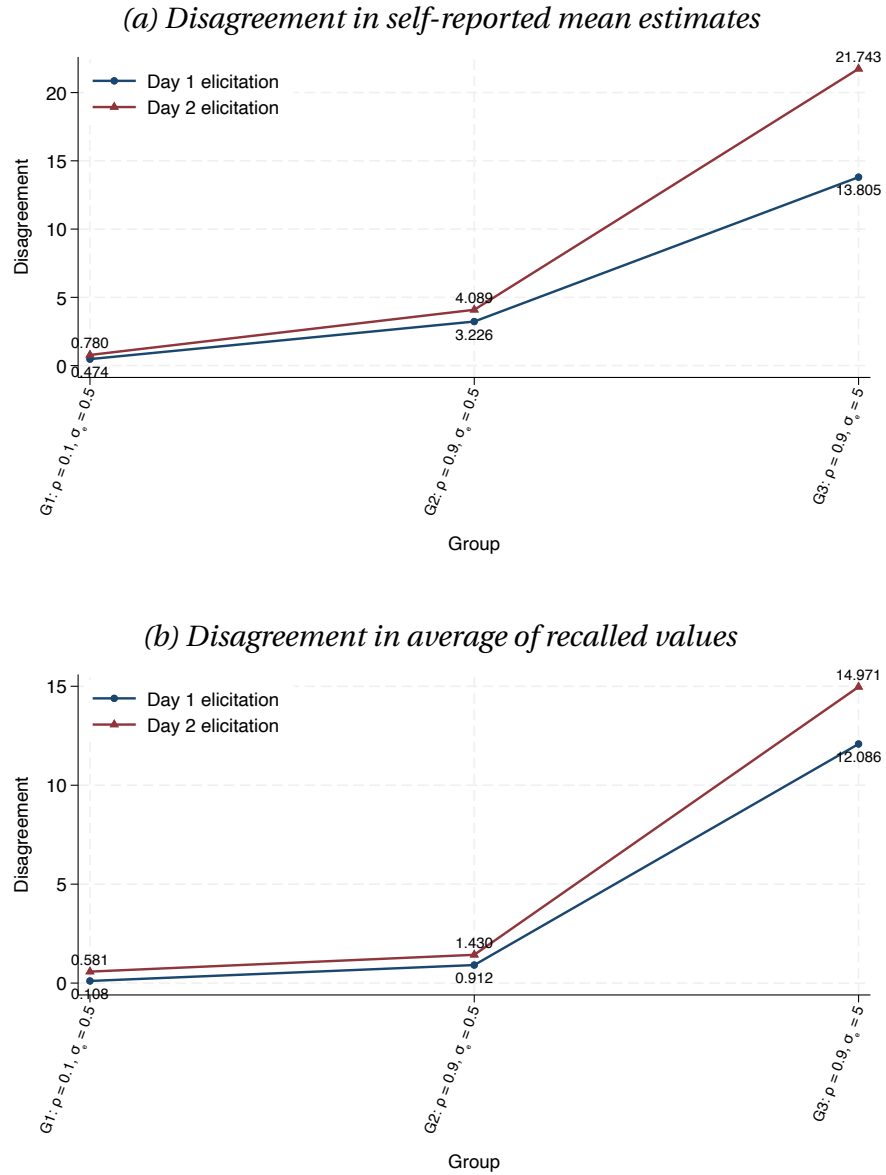


(c) Group 3: High persistence, high volatility



Appendix Figure A.10: QUANTITATIVE BELIEF EXPERIMENT: DISTRIBUTION OF ESTIMATED AVERAGE ACROSS GROUPS AND ACROSS DAYS

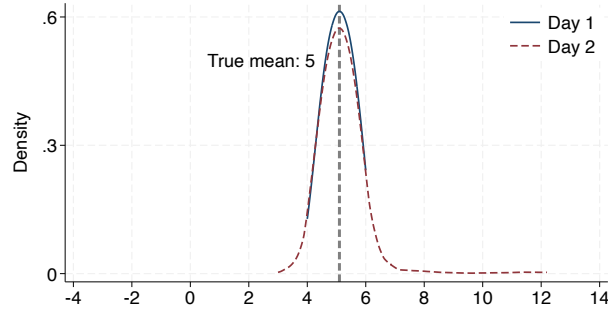
Notes: The figure plots the distribution of reported estimates of the series average on Day 1 and Day 2 for three data-generating processes with a common mean $\mu = 5$ and differing persistence and volatility. Group 1 observes data from a low-persistence, low-volatility process ($\rho = 0.1, \sigma_\epsilon = 0.5$). Group 2 observes data from a high-persistence, low-volatility process ($\rho = 0.9, \sigma_\epsilon = 0.5$). Group 3 observes data from a high-persistence, high-volatility process ($\rho = 0.9, \sigma_\epsilon = 5$). The vertical dashed line indicates the true mean.



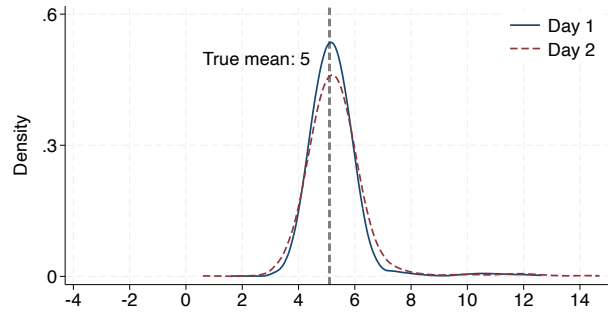
Appendix Figure A.11: QUANTITATIVE BELIEF EXPERIMENT: DISAGREEMENT IN MEAN ESTIMATES BY GROUP

Notes: The figure plots disagreement in participants' estimates of the series mean across the three AR(1) treatment groups. In panel (a), disagreement is measured as the cross-sectional variance of participants' self-reported estimates of the series average. In Panel B, disagreement is measured as the cross-sectional variance of recall-generated estimates, computed as the average of the values each participant recalls. Groups vary persistence ρ and innovation standard deviation σ_η while holding the long-run mean fixed at $\mu = 5$. Day 2 beliefs are elicited without any new information.

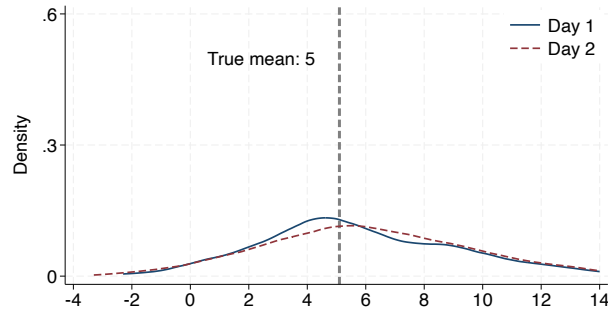
(a) Group 1: Low persistence, low volatility



(b) Group 2: High persistence, low volatility



(c) Group 3: High persistence, high volatility



Appendix Figure A.12: QUANTITATIVE BELIEF EXPERIMENT: DISTRIBUTION OF AVERAGE RECALLED VALUES ACROSS GROUPS AND ACROSS DAYS

Notes: The figure plots the cross-sectional distribution of average recalled values on Day 1 and Day 2 for three data-generating processes with a common mean $\mu = 5$ and differing persistence and volatility. Group 1 observes data from a low-persistence, low-volatility process ($\rho = 0.1, \sigma_\epsilon = 0.5$). Group 2 observes data from a high-persistence, low-volatility process ($\rho = 0.9, \sigma_\epsilon = 0.5$). Group 3 observes data from a high-persistence, high-volatility process ($\rho = 0.9, \sigma_\epsilon = 5$). The vertical dashed line indicates the true mean.

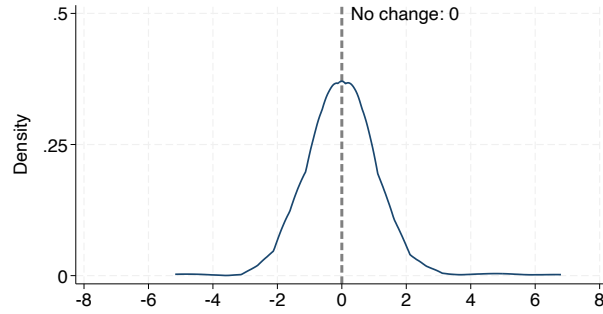
Appendix Q Additional Results for Experiment 5 (Quantitative Belief Instability)

Appendix Table A.21: PURE-FORGETTING TEST IN THE AR(1) QUANTITATIVE EXPERIMENT: DAY-2 VALUES WITHOUT A DAY-1 COUNTERPART

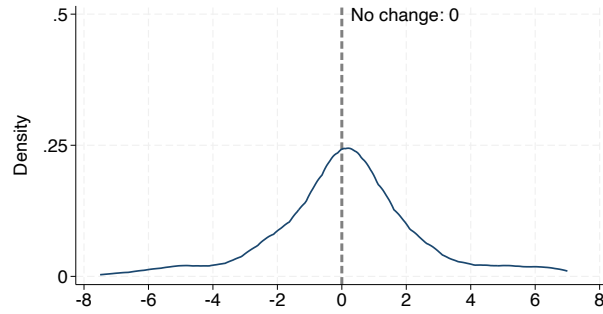
Matching tolerance	Share of participants with ...		
	≥ 1 new value	≥ 2 new values	≥ 3 new values
<i>Panel A: All groups (Observations = 1,867)</i>			
Exact match	94.3	67.1	37.0
$\pm 1\%$	93.4	65.7	35.6
$\pm 2.5\%$	89.7	60.0	30.7
$\pm 5\%$	82.9	47.9	22.0
$\pm 10\%$	65.8	30.2	13.5
<i>Panel B: By group, $\pm 2.5\%$ matching tolerance</i>			
Group 1 ($\rho = 0.1, \sigma = 0.5$; Observations = 606)	78.5	38.8	12.7
Group 2 ($\rho = 0.9, \sigma = 0.5$; Observations = 617)	91.2	60.8	31.8
Group 3 ($\rho = 0.9, \sigma = 5.0$; Observations = 644)	98.6	79.3	46.6

Notes: The table tests the value-level implication of deterministic (pure) forgetting in the AR(1) two-day quantitative experiment. Under pure forgetting, every numerical value a participant recalls on Day 2 must also have appeared in her Day-1 recall. A Day-2 value v is treated as having a Day-1 counterpart if some Day-1 value s satisfies $|v - s| \leq \tau |v|$ for the matching tolerance $\tau \in \{0, 1\%, 2.5\%, 5\%, 10\%\}$. Reported entries are the share of participants (in percent) whose Day-2 recalled set contains at least $k \in \{1, 2, 3\}$ values without any Day-1 counterpart under the given tolerance. Panel A pools all groups; Panel B reports group-level shares at $\pm 2.5\%$ tolerance. The three groups vary the persistence ρ and innovation standard deviation σ of the data-generating process while holding the long-run mean fixed at $\mu = 5$.

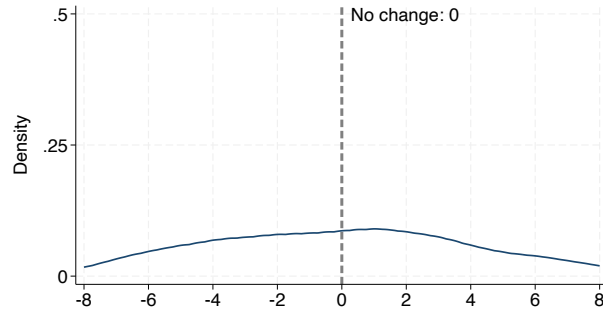
(a) Group 1: Low persistence, low volatility



(b) Group 2: High persistence, low volatility



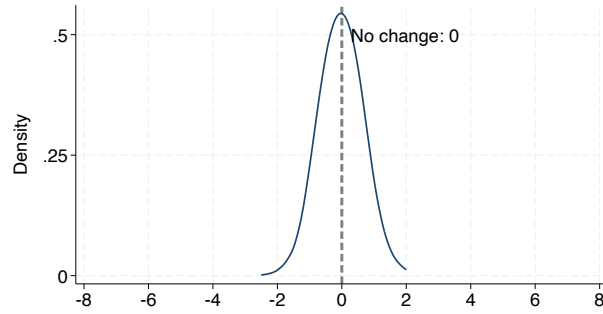
(c) Group 3: High persistence, high volatility



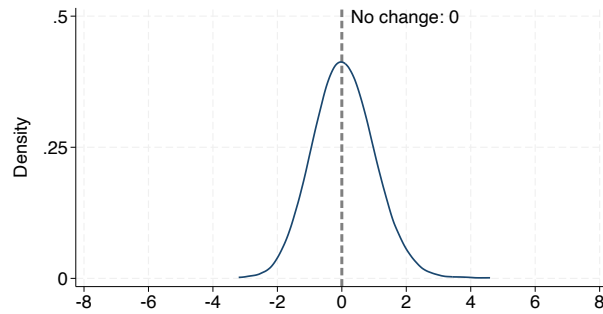
Appendix Figure A.13: QUANTITATIVE BELIEF EXPERIMENT: DISTRIBUTION OF BELIEF CHANGES ACROSS GROUPS

Notes: The figure plots the cross-sectional distribution of belief changes in the estimated series mean between Day 1 and Day 2 (Day 2 minus Day 1) for three data-generating processes with a common mean $\mu = 5$ and differing persistence and volatility. Group 1 observes data from a low-persistence, low-volatility process ($\rho = 0.1, \sigma_\epsilon = 0.5$). Group 2 observes data from a high-persistence, low-volatility process ($\rho = 0.9, \sigma_\epsilon = 0.5$). Group 3 observes data from a high-persistence, high-volatility process ($\rho = 0.9, \sigma_\epsilon = 5$). The vertical dashed line indicates zero change.

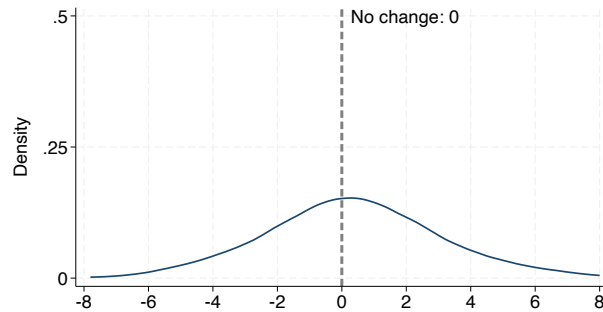
(a) Group 1: Low persistence, low volatility



(b) Group 2: High persistence, low volatility



(c) Group 3: High persistence, high volatility



Appendix Figure A.14: QUANTITATIVE BELIEF EXPERIMENT: DISTRIBUTION OF RECALL CHANGES ACROSS GROUPS

Notes: The figure plots the cross-sectional distribution of changes in mean recalled values between Day 1 and Day 2 (Day 2 minus Day 1) for three data-generating processes with a common mean $\mu = 5$ and differing persistence and volatility. Group 1 observes data from a low-persistence, low-volatility process ($\rho = 0.1, \sigma_\epsilon = 0.5$). Group 2 observes data from a high-persistence, low-volatility process ($\rho = 0.9, \sigma_\epsilon = 0.5$). Group 3 observes data from a high-persistence, high-volatility process ($\rho = 0.9, \sigma_\epsilon = 5$). The vertical dashed line indicates zero change.

Appendix Table A.22: STATED BELIEFS AND RECALLED INFORMATION IN THE QUANTITATIVE BELIEF EXPERIMENT

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Day 1 cross-sectional relationship between recall and beliefs (estimate of the series average)								
Average recall	0.784*** (0.029)	0.782*** (0.029)	0.951*** (0.073)	0.935*** (0.073)	0.845*** (0.056)	0.799*** (0.057)	0.824*** (0.029)	0.833*** (0.028)
Constant	0.968*** (0.152)	0.477* (0.280)	0.239 (0.367)	0.381 (0.394)	1.277*** (0.296)	0.280 (0.436)	-0.110 (0.173)	-0.207 (0.605)
Demographics		✓		✓		✓		✓
Observations	2238	2238	777	777	771	771	690	690
Panel B: Day 2 cross-sectional relationship between recall and beliefs (estimate of the series average)								
Average recall	0.708*** (0.048)	0.703*** (0.049)	0.985*** (0.110)	0.977*** (0.107)	0.857*** (0.109)	0.816*** (0.109)	0.760*** (0.052)	0.759*** (0.054)
Constant	1.332*** (0.261)	0.835* (0.450)	0.070 (0.547)	0.158 (0.623)	1.357** (0.570)	0.219 (0.693)	0.094 (0.333)	-0.090 (1.000)
Demographics		✓		✓		✓		✓
Observations	1867	1867	606	606	617	617	644	644
Panel C: Within-individual relationship between changes in recall and changes in beliefs								
Δ average recall	0.915*** (0.055)	0.918*** (0.055)	0.924*** (0.089)	0.922*** (0.090)	0.762*** (0.108)	0.748*** (0.108)	0.937*** (0.060)	0.938*** (0.062)
Constant	-0.048 (0.069)	-0.002 (0.443)	0.020 (0.040)	0.027 (0.240)	0.103 (0.090)	-0.076 (0.608)	-0.259 (0.178)	-0.243 (1.147)
Demographics		✓		✓		✓		✓
Observations	1867	1867	606	606	617	617	644	644

Notes: This table reports regressions relating participants' beliefs about the mean of a stochastic process to recalled information in the quantitative belief experiment. Panel A reports cross-sectional regressions for Day 1, relating each participant's self-reported estimate of the series average (dependent variable) to the average value of observations they recall from the sequence (independent variable). Panel B reports analogous cross-sectional regressions for Day 2, in which beliefs are re-elicited without any new information. Panel C reports within-individual regressions relating changes in beliefs between Day 1 and Day 2 to changes in the average of recalled values over the same period. Specifications with demographic controls include age, gender, education, household income, employment status, state of residence, and indicators for stock market participation. All regressions are estimated separately by treatment group, which varies the persistence ρ and innovation standard deviation σ_η of the data-generating process while holding the long-run mean fixed at $\mu = 5$. Robust standard errors are reported in parentheses. Changes in recall are expressed in percentage points. Statistical significance is denoted by *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$.

Appendix Table A.23: INSTRUMENTAL VARIABLE ESTIMATES OF THE EFFECT OF RECALL ON BELIEFS (EXPERIMENTS 4–5)

	(1) Pooled	(2) Pooled, group FE
Day 2 recall (2SLS)	0.460*** (0.065)	0.509*** (0.067)
Group fixed effects		✓
Observations	1,867	1,867
First-stage F	608	223

Notes: This table applies the instrumental variable approach of [Gillen, Snowberg and Yariv \(2019\)](#) to Experiments 4 and 5, in which the same participant is observed on Day 1 and Day 2 with no new information arriving between elicitations. We instrument the Day 2 recall (the mean of recalled data points) with the Day 1 recall in regressions of the Day 2 stated mean estimate on the Day 2 recall. Column (1) pools all three groups; column (2) adds group fixed effects. The first-stage F -statistics in both columns are well above conventional weak-instrument thresholds, and the 2SLS estimates of the recall-belief slope are statistically significant at the 1% level, indicating that the recall-belief link is not an artifact of classical measurement error in reported recall. Robust standard errors are reported in parentheses. Statistical significance is denoted by *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$.

Appendix R Inflation Expectations from the Michigan Survey of Consumers

This appendix presents results for household inflation expectations from the Michigan Survey of Consumers, paralleling the stock price analysis in Section 8.

Empirical setup We study households' quantitative expectations about inflation over the next year, using MSC data from June 1980 to March 2025. Disagreement is measured as the monthly cross-sectional standard deviation of reported expected inflation rates. The MSC inflation question elicits a point forecast rather than a probability, so no direct analogue of the uncertainty measure $U_{it} = p_{it}(1 - p_{it})$ is available; following [Binder \(2017\)](#), we instead proxy subjective uncertainty with an indicator for round-number responses (expected inflation a multiple of five percent). Because inflation expectations are quantitative rather than binary, we use realized inflation volatility as the empirical analogue of an ambivalent recent information environment. For horizon $h \in \{1Y, 5Y, 10Y\}$ and frequency $f \in \{\text{Monthly}, \text{Quarterly}\}$, we define inflation ambivalence as

$$A^f(h) = \sqrt{\frac{1}{T_h} \sum_{t'=1}^{T_h} \left(\pi_{t-t'}^f - \bar{\pi}_h^f \right)^2},$$

where π_t^f denotes realized inflation and $\bar{\pi}_h^f$ is the average realized inflation rate over the corresponding trailing window.

Results Table [A.24](#) presents the results. Panel A shows that inflation disagreement rises with trailing inflation volatility across horizons and sampling frequencies. The effect is stronger at longer horizons: in the monthly specifications, the coefficients increase from 0.015 at the one-year horizon to 0.072–0.086 at the five- and ten-year horizons, conditional on the contemporaneous level of inflation.

Panel B relates inflation ambivalence to within-individual instability, measured as the absolute change in a respondent's one-year-ahead expected inflation rate between consecutive interviews. More volatile inflation histories are associated with larger within-household revisions: the coefficients range from 0.017 to 0.066 and remain significant after controlling for the contemporaneous level of inflation.

Panel C examines the rounding-based uncertainty proxy. More volatile inflation histories are associated with a higher probability of round-number responses at all horizons and sampling frequencies (coefficients 0.003–0.007, p values < 0.01), controlling for the contemporaneous level of inflation, and the pattern is unchanged when zero responses, which are focal answers, are excluded. Ambivalent inflation histories are thus associated with more disagreement, larger revisions, and greater subjective uncertainty, paralleling the stock-price results despite a different

empirical proxy for ambivalence.

Appendix Table A.24: INFLATION EXPECTATIONS AND TRAILING INFLATION AMBIVALENCE

	Monthly			Quarterly		
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Disagreement in inflation expectations						
Ambivalence (1 year)	0.015*** (0.005)			0.015*** (0.005)		
Ambivalence (5 years)		0.072*** (0.010)			0.102*** (0.012)	
Ambivalence (10 years)			0.086*** (0.010)			0.109*** (0.011)
Current inflation	0.159*** (0.030)	0.155*** (0.029)	0.147*** (0.030)	0.238*** (0.044)	0.224*** (0.042)	0.217*** (0.043)
<i>R-squared</i>	0.078	0.145	0.135	0.115	0.209	0.181
<i>Observations</i>	538	538	538	538	538	538
Panel B. Belief instability in inflation expectations						
Ambivalence (1 year)	0.020*** (0.001)			0.017*** (0.001)		
Ambivalence (5 years)		0.054*** (0.002)			0.066*** (0.003)	
Ambivalence (10 years)			0.051*** (0.002)			0.050*** (0.003)
Current inflation	0.116*** (0.005)	0.112*** (0.005)	0.107*** (0.005)	0.162*** (0.007)	0.154*** (0.007)	0.155*** (0.007)
<i>R-squared</i>	0.007	0.011	0.008	0.009	0.013	0.010
<i>Observations</i>	105253	105253	105253	105253	105253	105253
Panel C. Belief uncertainty in inflation expectations (proxied by round-number responses)						
Ambivalence (1 year)	0.003*** (0.000)			0.003*** (0.000)		
Ambivalence (5 years)		0.006*** (0.000)			0.007*** (0.000)	
Ambivalence (10 years)			0.003*** (0.000)			0.003*** (0.000)
Current inflation	0.004*** (0.000)	0.003*** (0.000)	0.003*** (0.000)	0.005*** (0.000)	0.004*** (0.000)	0.005*** (0.000)
<i>R-squared</i>	0.005	0.009	0.002	0.005	0.010	0.002
<i>Observations</i>	276243	276243	276243	276243	276243	276243

Notes: Panel A: monthly cross-sectional standard deviation of one-year-ahead expected inflation. Panel B: absolute change in an individual's expectation between consecutive interviews (about six months apart). Panel C: the uncertainty proxy. Ambivalence is the standard deviation of realized inflation over a trailing window of length $h \in \{1Y, 5Y, 10Y\}$, as defined in the text. Columns (1)–(3) use monthly frequency; columns (4)–(6) quarterly. All specifications control for the contemporaneous level of inflation. Robust standard errors in parentheses. Panel A results are robust to Newey–West robust errors; results in Panels B and C are robust to clustering standard errors by interview month. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Appendix S Salient Recent Returns as Cues in Stock-Price Expectations

This appendix reports the cue analysis for stock-price expectations discussed in Section 8: whether a salient most-recent S&P 500 return that reinforces or runs against the dominant sign of recent history shifts cross-sectional disagreement, mirroring the cue comparative static in Experiment 1.

Appendix Table A.25: EFFECT OF CUES: STOCK PRICE MOVEMENT EXPECTATION

	Monthly		Quarterly	
	(1)	(2)	(3)	(4)
	Past 5Y	Past 10Y	Past 5Y	Past 10Y
Cue $\times$ Majority aligned	-0.046*** (0.010)	-0.050*** (0.010)	-0.068*** (0.010)	-0.028*** (0.008)
Cue	0.015** (0.007)	0.013** (0.006)	0.023*** (0.006)	-0.001 (0.006)
Majority aligned	0.018*** (0.006)	0.023*** (0.009)	0.026*** (0.006)	0.012* (0.006)
Ambivalence	0.049*** (0.012)	0.076*** (0.009)	0.015** (0.007)	0.041*** (0.005)
Short-term return	✓	✓	✓	✓
Long-term return	✓	✓	✓	✓
<i>R-squared</i>	0.331	0.315	0.375	0.279
Observations	274	274	274	274

Notes: This table reports OLS estimates relating stock-price-expectation disagreement in the Michigan Surveys of Consumers to recent return “cues” and the sign composition of past returns. The sample covers June 2002 through March 2025 at the monthly level (274 months). The dependent variable is the cross-sectional standard deviation of a binary stock-price-movement belief in month t . Each respondent reports the probability (0–100) that stock prices will be higher over the next 12 months; individual beliefs are coded as $b_{it} = +1$ if the reported probability is at least 50 and $b_{it} = -1$ otherwise, and disagreement is computed as the standard deviation, $sd_t(b_{it})$ within month t . *Cue* is an indicator for whether the most recent lagged one-month S&P 500 return is positive. *Majority aligned* is an indicator for whether this cue aligns with the majority sign of realized returns in a trailing window of length h , constructed from the same return series used to form the ambivalence measure; *Cue $\times$ Majority aligned* is their interaction. *Ambivalence* is a trailing sign-imbalance index, $Ambivalence(h) = 1 - \frac{|\#Neg(h) - \#Pos(h)|}{\#Neg(h) + \#Pos(h)}$, where $\#Pos(h)$ and $\#Neg(h)$ count positive and negative realized returns over the trailing window of length h . Columns (1)–(2) construct $\#Pos(h)$ and $\#Neg(h)$ using monthly return signs with horizons $h \in \{1Y, 5Y\}$; columns (3)–(4) use quarterly return signs with horizons $h \in \{4Q, 20Q\}$. All specifications control for cumulative S&P 500 returns over the previous 1 month and 10 years. The marginal effect of *Cue* equals β_{Cue} when the cue is a minority sign in the trailing window and $\beta_{Cue} + \beta_{Cue \times Maj}$ when it matches the trailing majority sign; the negative interaction therefore implies attenuation when the cue aligns with the majority. Robust standard errors are reported in parentheses. The *Cue $\times$ Majority aligned* interaction in columns (1)–(3) is robust to Newey–West standard errors (12 lags). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.